

# Fiscal Inflation with Incomplete Information

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*We quantify fiscal inflation through the lens of incomplete information between funded and unfunded fiscal shocks. Information friction naturally breaks the Ricardian equivalence for deficits to be inflationary. We connect the unfunded shock to a pair of hidden time-varying short-term debt and inflation targets. Incomplete information alters the effects of both monetary and fiscal shocks on inflation. We use survey data (SPF and MSC) to discipline the magnitude of information friction and find that both fiscal stimulus and supply shocks contribute significantly to COVID inflation. An unfunded tax cut can increase output and reduce debt burden, at the cost of higher lingering inflation.*

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## I. Introduction

The COVID-19 pandemic and the subsequent inflationary episode have sparked a renewed debate about the role of fiscal policy and its impact on inflation. The pandemic caused substantial disruptions to the economy, including supply chain shortages, shifts in preferences toward consumption and labor supply, and the unprecedented fiscal response, among others. Although there is a consensus that all of these shocks can contribute to the increase in prices in theory, there is still much controversy about the decomposition of supply- and demand-driven inflation in practice (see Giannone and Primiceri (2024) and Bernanke and Blanchard (2023)).

An important debate centers on the relative importance of fiscal inflation. The prompt fiscal response to the COVID crisis has been the largest in peacetime. From March 2020 to March 2021, the US government issued three rounds of fiscal stimulus (See Figure 1). In total, more than \$5 trillion in federal tax cuts,

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spending increases, and stimulus checks have been spent to increase aggregate demand. These stimulus payments were coupled with large scale purchase of government debt by the central bank. These measures likely put the economy on a path to recovery, but the effects on inflation are unclear. Are fiscal expansions inflationary? If so, how large? Using similarly estimated medium-scale DSGE models, Bianchi, Faccini and Melosi (2023) find that most of the COVID inflation is due to fiscal stimulus. In contrast, Smets and Wouters (2024) claim that supply shocks, not fiscal expansion, are the primary drivers of post-pandemic inflation.

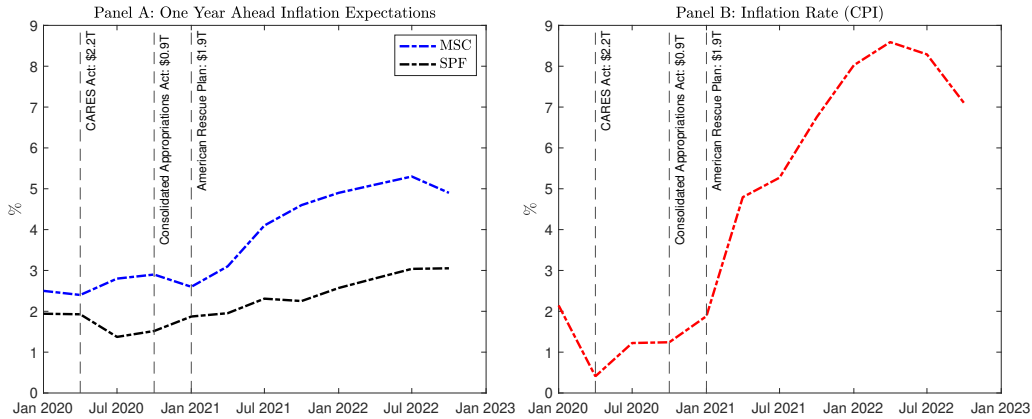


FIGURE 1. INFLATION EXPECTATIONS AND CPI INFLATION DURING COVID

**Notes:** Panel A shows the quarterly one-year ahead inflation expectations from Michigan Survey of Consumer Sentiments (MSC) and Survey of Professional Forecasters (SPF). Panel B shows the quarterly inflation rate as measured by CPI during the period of 2020:Q1 to 2022:Q4. Horizontal dotted lines mark the three fiscal stimulus payments enacted in response to COVID along with amount of disbursements in trillions of dollars.

Although there is a quantitative debate, the aforementioned literature has largely settled on the origin of potential fiscal inflation. According to the Fiscal Theory of Price Level (FTPL), as prices need to adjust to ensure that the real value of government debt equals the present value of future surpluses, fiscal policy will have a direct impact on inflation. The early influential work of Leeper (1991) proposes two separate regimes of policy behavior. In the monetary-led regime, the central bank controls inflation by adjusting the nominal interest rate more than one for one in response to inflation deviations from its target (i.e., the Taylor principle), and the fiscal authority ensures that it adjusts primary surpluses (i.e., raising taxes and cutting expenditure) properly to stabilize government debt. In contrast, in the fiscal-led regime, the fiscal authority does not adjust primary surpluses sufficiently. With an accommodating central bank that does not satisfy the Taylor principle, fiscal inflation can arise to reduce the real value of that debt. Such a debt-eroding process is central to the FTPL and can only happen in the fiscal-led regime. For the past three decades, the literature on FTPL has focused

mainly on identifying fiscal/monetary policy regimes to infer the magnitude of fiscal inflation (Leeper, Traum and Walker, 2017; Bianchi and Melosi, 2017).

The recent literature (Bianchi, Faccini and Melosi, 2023; Smets and Wouters, 2024) has moved away from policy regimes and has taken an alternative approach to model fiscal inflation. In particular, Bianchi, Faccini and Melosi (2023) design an environment with both funded and unfunded fiscal shocks. The funded shock is backed by future primary surpluses and is similar to the policy behavior in the monetary-led regime. However, the unfunded shock is unbacked, mimicking the policy behavior in the fiscal-led regime. Consequently, while the former shock satisfies the Ricardian equivalence (Barro, 1974) and is noninflationary, the latter shock can generate substantial fiscal inflation once fully accommodated by the central bank. The coexistence of both shocks allows researchers to confront a quantitative DSGE model directly using linear Bayesian estimation techniques.

Both Bianchi, Faccini and Melosi (2023) and Smets and Wouters (2024) impose the assumption of Full Information Rational Expectations (FIRE) of all economic agents. In the face of unprecedented government transfers and monetary accommodation, one immediate implication of the FIRE model is that economic agents should predict high inflation in the future, resulting in higher current inflation. However, as Figure 1 shows, neither the CPI inflation rate nor the expectations of one-year ahead inflation rate in two of the widely referenced surveys, the Survey of Professional Forecasters (SPF) and the Michigan Survey of Consumers (MSC), spiked right after the implementation of stimulus payments.<sup>1</sup> The delayed inflation response raises concerns about the validity of the full-information assumption in explaining the COVID inflation dynamics.

Moreover, the coexistence of both funded and unfunded fiscal shocks naturally introduces an information friction that has not been explored. Intuitively, since the two shocks rely on different monetary accommodations in the present and distinct fiscal backings in the future, it is likely households and firms cannot fully tell whether it is a funded shock, an unfunded shock, or a combination of both shocks when a fiscal change occurs today. This is particularly true for U.S. fiscal and monetary policy, as (i) politicians often are unable to clarify the future fiscal backings and (ii) the Federal Reserve rarely admits explicitly it is accommodating the fiscal authority. To capture this salient feature, we model the friction in the source of fiscal expansion through incomplete information. To connect directly to Bianchi, Faccini and Melosi (2023) and Smets and Wouters (2024), we work within the rational expectations paradigm.

The modeling of the funded fiscal shock is standard. To model the unfunded shock, we utilize a pair of time-varying short-term inflation and debt targets that are intertwined. Although the interpretation is slightly different, our approach is in full agreement with the original design of Bianchi, Faccini and Melosi (2023).

<sup>1</sup>The failure to anticipate the COVID-19 inflation in early 2020 was a widespread phenomenon, with economists and central bankers initially predicting a period of disinflation or deflation. For example, when inflation did start to rise in 2021, the Federal Reserve viewed it as transitory, a temporary result of supply shortages that would soon resolve.

For example, we map an analytical model to the simple endowment economy in Bianchi, Faccini and Melosi (2023) one-to-one. We show that the unfunded fiscal shock is essentially an exogenous shock to the hidden debt target. When an unfunded shock increases the debt target, it also allows the government to reduce or delay adjustments of future primary surpluses. Moreover, the unfunded shock also needs to drive the inflation target so that the central bank is being “accommodating”. We specify a persistent AR(1) debt target process. To discipline central bank accommodation, Bianchi, Faccini and Melosi (2023) consider a shadow economy in which the only shock is the unfunded fiscal shock. We show how the shadow economy can translate into an endogenous AR(1) inflation target in closed-form solutions.

Incomplete information relies on the fact that both the debt and inflation targets are hidden and cannot be perfectly observed. These targets reflect how tolerant policy makers are to short-term deviations of debt and inflation from their long-term goals. Although the U.S. has never formally adopted a numerical long-term debt target, the debt-to-GDP ratio has been rising steadily well before the COVID-19 pandemic. This trend indicates that the federal government is reluctant to stabilize the high debt burden, especially in the short term (Han, 2021). At the same time, while the Federal Reserve adopts a long-term inflation target of 2%, the central bank does not follow the strict inflation targeting rule in the short run (Ireland, 2007). Historically, the central bank considered a broad range of inflation indicators when setting the federal funds rate, and had adopted a deliberately vague “average inflation target” framework during COVID. Empirically, we find strong evidence supporting the presence of both hidden and time-varying policy targets.

Given the combination of a hidden target and a standard policy shock (either monetary or fiscal), the private sector (i.e. households and firms) faces a signal extraction problem. Since the unfunded shock enters the debt and inflation targets simultaneously, there is a subsequent informational interaction of monetary and fiscal policy. We follow the standard practice in the literature on incomplete information models (Blanchard, L’Huillier and Lorenzoni, 2013) and let households and firms use the optimal algorithm, the Kalman filter, to solve the linearized signal extraction problem.

We first illustrate the transmission mechanism in a simple endowment economy. We use analytical solutions to shed light on intuition and numerical calculations to highlight the importance of fiscal financing. Incomplete information leads to an initial under-reaction of inflation to the unfunded shock, as economic agents perceive that the shock is partially funded at the beginning. Households gradually figure out the nature of the shock, leading to a hump-shaped impulse response to inflation. Similarly, since households can also believe that a funded shock is partially unfunded, its effect on inflation is nonzero (i.e., Ricardian equivalence breaks down) and can be persistent. Lastly, as households confuse between an exogenous monetary policy shock and a changing inflation target driven by the

unfunded shock, a transitory monetary policy shock can also generate some persistent inflation.

We then estimate a medium-scale DSGE model to quantify fiscal inflation with incomplete information. To give both the demand and supply shocks a fair chance in explaining inflation *ex ante*, the information friction in the DSGE model is twofold. First, agents cannot perfectly distinguish unfunded fiscal shocks from funded fiscal or monetary policy shocks. Second, they also cannot perfectly disentangle persistent supply shocks from transitory shocks in real time. To discipline the magnitude of information friction, a novelty of the paper is that we confront the model directly to inflation expectations in SPF and MSC. To the best of our knowledge, the theoretical mechanism of incomplete information between funded and unfunded fiscal shocks, and the accompanying empirical discipline for the information friction, are both new contributions to the literature.

Under incomplete information, an unfunded fiscal shock produces a smaller immediate inflation response but more persistent inflation dynamics, as households gradually infer that unfunded shocks are not backed by future fiscal adjustments. Quantitatively, fiscal inflation arising from unfunded shocks under incomplete information is approximately 65% to 75% lower than under full information, at the estimated 95th and 5th percentiles. Despite the presence of information friction, funded transfers and tax cuts generate negligible inflation. The decomposition of shock contributions to inflation under incomplete information indicates that, at its peak of 8.7% in 2022:Q2, inflation attributable to unfunded shocks was approximately 3%, while supply shocks resulted in an additional inflation of 2.8%. Although unfunded fiscal shocks remain one of the primary drivers of post-COVID inflation in our model, their quantitative contribution is substantially smaller than the findings in Bianchi, Faccini and Melosi (2023), which attribute almost 100% of observed inflation between 2021 and 2022 to unfunded shocks.

The final sections of the paper examine the contributions of monetary policy and unfunded tax cuts to inflation. As illustrated in the simple model, interest rate cuts become more inflationary under incomplete information because agents are unable to distinguish exogenous monetary policy shocks from endogenous changes in short-term inflation target. Quantitatively, we find that information friction can generate up to a 20% larger inflation response to an exogenous rate cut relative to a full-information benchmark. Thus, while incomplete information dampens the inflationary contribution of unfunded fiscal shocks, it simultaneously amplifies the non-neutrality of monetary policy as agents attempt to infer the nature of observed signals. Lastly, incomplete information does not generate meaningful inflationary responses to funded tax shocks. Although funded labor and capital tax cuts stimulate output at the cost of higher debt, neither produces a significant effect on inflation. An unfunded tax cut, while being more stimulative and reducing the debt burden, can generate very persistent inflation that lasts years.

**Related Literature** This paper belongs to the vast literature on monetary-

fiscal policy interactions (see Sargent and Wallace (1981), Leeper (1991), Sims (1994), and Woodford (2001)). We propose an informational interaction of monetary and fiscal policy and quantify fiscal inflation through the lens of incomplete information between funded and unfunded shocks. These shocks resemble the different monetary-fiscal policy behaviors in the regime-switching literature on FTPL (see Davig, Leeper and Walker (2010) and Bianchi and Melosi (2017)). Smets and Wouters (2024) estimate a medium-scale DSGE model, which allows for a time-invariant partial fiscal backing between funded and unfunded shocks. They find that on average 80% of fiscal shocks are funded. We instead consider a dynamic signal extraction problem such that the degree of partial fiscal backing always depends on the underlying monetary-fiscal policy actions and the private sector's expectations. Consequently, the degree of fiscal backing in our model is always state-dependent and time-varying.

In addition to incomplete information, other information channels or frictions can also play a significant role in generating fiscal inflation. We share the spirit of Eusepi and Preston (2018), who propose a theory of the fiscal foundations of inflation based on imperfect knowledge and recursive least square learning. Bassetto and Miller (2025) use rational inattention and shows that when bond holders are more concerned with the possibility of a fiscally led regime, sudden inflation can occur. Angeletos, Lian and Wolf (2024) connect to the FTPL by considering a Heterogeneous Agent New Keynesian (HANK) model with a sufficiently slow fiscal adjustment. Since households are non-Ricardian in HANK models, fiscal deficits drive aggregate demand, and thus inflation. Interestingly, we arrive at a quantitatively similar result that fiscal inflation is reduced by about half compared to the full-information FTPL case.

## II. A Simple Endowment Economy

We use a simple endowment economy (see Leeper (1991)) to highlight the key information friction. The one-period nominal bond  $B_t$ , issued by the government, is sold for  $Q_t$ . The gross nominal interest rate  $R_t$  is given by the inverse of  $Q_t$ , i.e.,  $R_t = 1/Q_t$ . The steady-state gross nominal interest rate is  $1/\beta$ , where  $\beta \in (0, 1)$  is the household's time discount factor. In logarithmic linearized form, the Fisher equation is

$$(1) \quad i_t = \mathbb{E}_t^{HH} \pi_{t+1},$$

where  $i_t$  is the net nominal interest rate and  $\pi_t$  is inflation. The rational expectations operator  $\mathbb{E}_t^{HH} = \mathbb{E}_t(\cdot | I_t^{HH})$  is conditional on the household's information set  $I_t^{HH}$ , which will be specified below.

Monetary policy follows a simple rule.

$$(2) \quad i_t = \phi_\pi(\pi_t - \pi_t^*),$$

where  $\pi_t^*$  stands for the central bank's short-term inflation target and can be

varying over time (see Ireland (2007)).<sup>2</sup> The term  $\pi_t - \pi_t^*$  defines the inflation gap. The parameter  $\phi_\pi$  controls the strength with which the central bank reacts to its inflation gap and satisfies the Taylor principle (that is,  $\phi_\pi > 1$ ). To focus solely on fiscal inflation, we do not introduce an exogenous monetary policy shock at this time. We also leave the task of specifying the inflation target  $\pi_t^*$  after introducing the fiscal block.

The government budget constraint is

$$Q_t B_t + P_t S_t = B_{t-1},$$

where  $S_t$  stands for the primary surplus and  $Q_t B_t / P_t$  defines the real market value of the debt. We normalize the endowment economy's output to one. The steady-state surplus-to-output ratio  $S^* > 0$  determines the steady-state real market debt  $s_b^*$  that the government can finance in the long run. To see the point, imposing steady-state values in the government budget constraint leads to  $s_b^* + S^* = s_b^* / \beta$ . Consequently,  $s_b^* = (\beta S^*) / (1 - \beta) > 0$ , where a larger steady-state primary surplus  $S^*$  supports a higher steady-state debt  $s_b^*$ .

Let  $s_t$  and  $s_{b,t}$  denote the logarithmic deviations of the two variables from their corresponding steady-state values. The linearized government budget constraint (GBC) is

$$(3) \quad s_{b,t} = \beta^{-1} [s_{b,t-1} + i_{t-1} - \pi_t - (1 - \beta)s_t].$$

Recently, elevated public debt levels have been at the center of fiscal discussions. We consider a fiscal rule that incorporates a hidden time-varying short-term debt target (see also Bianchi and Melosi (2017) and Han (2021)). Let us first consider a log-linearized surplus rule

$$s_t = \gamma(s_{b,t-1} - s_{b,t}^*) + \eta_t,$$

where  $s_{b,t}^*$  is the logarithmic deviation of the contemporaneous debt target from its steady-state value. The parameter  $\gamma > 1$  controls the strength with which the government reacts to movements of realized debt  $s_{b,t-1}$  from its time-varying target  $s_{b,t}^*$ . The  $\eta_t$  represents an exogenous fiscal shock.

We still need to specify the law of motion for the time-varying debt target  $s_{b,t}^*$ . To achieve this goal, we draw on recent literature on funded and unfunded fiscal shocks. In particular, Bianchi, Faccini and Melosi (2023) considers the following log-linearized fiscal rule.

$$s_t = \gamma(s_{b,t-1} - s_{b,t-1}^*) + \gamma_b s_{b,t-1}^* + (\varepsilon_t^U + \varepsilon_t^F),$$

with  $0 < \gamma_b < 1 < \gamma$ ,  $\varepsilon_t^F \sim N(0, \sigma_F^2)$ , and  $\varepsilon_t^U \sim N(0, \sigma_U^2)$ . For reasons illustrated

<sup>2</sup>The central bank's long-term inflation target  $\pi^*$  can still be time-invariant. For example, the Federal Reserve adopts a 2% long-term inflation target.

below, Bianchi, Faccini and Melosi (2023) labels  $\varepsilon_t^F$  the “funded” shock, and  $\varepsilon_t^U$  the “unfunded” fiscal shock. Equating the above two fiscal rules and imposing the restriction that  $\eta_t \equiv \varepsilon_t^F$  give the law of motion for the time-varying debt target

$$(4) \quad s_{b,t}^* = (1 - \gamma_b/\gamma)s_{b,t-1}^* - \gamma_\tau^{-1}\varepsilon_t^U, \quad \varepsilon_t^U \sim N(0, \sigma_U^2);$$

which is a stationary AR(1) process with persistence  $1 - \gamma_b/\gamma \in (0, 1)$ .

The crucial distinction between “funded” and “unfunded” fiscal shocks is that while future primary surpluses back the former and therefore are not inflationary (that is,  $\varepsilon_t^F$  satisfies Ricardian equivalence), the unfunded shock  $\varepsilon_t^U$  is unbacked by necessary future fiscal adjustments and must be inflationary. In Bianchi, Faccini and Melosi (2023), all economic agents have perfect knowledge of  $\varepsilon_t^F$  and  $\varepsilon_t^U$ . Furthermore, the central bank is willing to accommodate all fiscal inflation by adjusting its  $\pi_t^*$ , which can only arise from the unfunded shock  $\varepsilon_t^U$ .

We now specify the time-varying inflation target,  $\pi_t^*$ . Consistent with Bianchi, Faccini and Melosi (2023), we assume that the unfunded fiscal shock completely drives  $\pi_t^*$ ,

$$(5) \quad \pi_t^* = \mathcal{P}(L)\varepsilon_t^U.$$

For now, we do not impose any restrictions on the serial correlations of  $\pi_t^*$  and only assume that it is a covariance stationary process, which is equivalent to not imposing any functional forms of  $\mathcal{P}(L)$  and only requiring  $\sum_{j=0}^{\infty} \mathcal{P}_j^2 < \infty$ . Technically,  $\mathcal{P}(L)$  is an analytical function.

#### A. Making connections to Leeper (1991) and Bianchi, Faccini and Melosi (2023)

Before introducing incomplete information, it is worth connecting to two papers in the literature which also consider similar endowment economies. Setting

$$(6) \quad \gamma_b = \gamma, \quad \mathcal{P}(L) = 0 \Rightarrow \text{Leeper (1991)}$$

eliminates the time-varying debt and inflation targets. In Leeper (1991), two separate parameter spaces deliver the existence and uniqueness of a rational expectation equilibrium (REE). The first equilibrium is in the monetary-led regime, where nominal interest rates  $i_t$  respond more than one-to-one to deviations in inflation from its target ( $\phi_\pi > 1$ ), and primary surpluses  $s_t$  respond strongly to deviations of debt to keep it on a stable path ( $\gamma > 1$ ). The fiscal shock is funded in the first regime. The other equilibrium is in the fiscal-led regime, where  $i_t$  responds less than one to one to inflation ( $0 \leq \phi_\pi < 1$ ), and  $s_t$  responds weakly to debt ( $0 \leq \gamma < 1$ ). The fiscal shock in the second regime is unfunded. Leeper (1991) labels the two regimes as Active Money-Passive Fiscal (AMPF) and Passive Money-Active Fiscal (PMAF). Appendix A derives both the AMPF and PMAF solutions. As debt stability is achieved with sufficient fiscal adjustments, a distinct feature of the monetary-led regime is that fiscal policy is Ricardian and



inflation is independent of the funded fiscal shocks. In the fiscal-led regime, since fiscal authority responds weakly to debt, inflation must adjust to surplus shocks to stabilize government debt. By design, there is only one type of fiscal shock, funded or unfunded, in either regime.

We share the same surplus and debt target rules as Bianchi, Faccini and Melosi (2023). In its simple model section, Bianchi, Faccini and Melosi (2023) considers the following policy rule in which the monetary authority reacts differently to funded and unfunded shocks

$$(7) \quad i_t = \phi_\pi(\pi_t - \pi_t^F) + \phi_F \pi_t^F.$$

In particular, the additional parameter  $\phi_F$  satisfies  $0 \leq \phi_F < 1$ , and  $\pi_t^F$  is the inflation that would arise in a shadow economy in which the fiscal-led regime is always in place. The derivation in Appendix A shows that

$$\pi_t^F = \frac{\beta - 1}{1 - \phi_F L} \varepsilon_t^U,$$

which is an AR(1) process. Equating the two rules (7) and (2) implies  $\pi_t^* = (1 - \phi_F/\phi_\pi) \pi_t^F$ . The setting of the inflation target process is the following

$$(8) \quad \mathcal{P}(L) = \left(1 - \frac{\phi_F}{\phi_\pi}\right) \frac{\beta - 1}{1 - \phi_F L} \Rightarrow \text{Bianchi, Faccini and Melosi (2023)}$$

leads to the simple model in Bianchi, Faccini and Melosi (2023).

Both papers assume a Full Information Rational Expectations (FIRE) environment. However, the immediate implication of the FIRE assumption suggests that the private sector should largely predict the COVID-19 inflation that would arise from large-scale fiscal stimuli. It contradicts the movements of bond yields in 2021, as few market participants anticipated the subsequent surge in inflation.

### B. Introducing incomplete information

Let  $\mathcal{M}$  denote the model structure which includes all the structural parameters and the equilibrium conditions. As in all rational expectations models, we first assume  $\mathcal{M}$  is common knowledge for the household. Consequently, our information friction is different from the imperfect information channel considered in learning models (e.g. Eusepi and Preston (2018)).

Full-information models assume that economic agents can observe or learn all shocks (i.e.,  $\varepsilon_t^F$  and  $\varepsilon_t^U$ ) perfectly. As a benchmark, we first define the information set for the FIRE case as the following

$$(9) \quad I_t^{FI} = \{\varepsilon_{t-k}^F, \varepsilon_{t-k}^U, \mathcal{M} | k \geq 0\}.$$

The superscript “FI” denotes full information. The coexistence of funded and

unfunded shocks, along with their vague clarifications from policymakers, naturally introduces incomplete information. We now formally introduce our concept of information friction while preserving the assumption of rational expectations.

Rewrite the two policy rules and define the monetary and fiscal signals  $s_{m,t}$  and  $s_{f,t}$  as

$$(10) \quad s_{m,t} = \pi_t - i_t / \phi_\pi = \pi_t^*$$

$$(11) \quad s_{f,t} = s_t - \gamma s_{b,t-1} = -\gamma s_{b,t}^* + \varepsilon_t^F.$$

We assume that households make  $\mathbb{E}_t^{HH} \pi_{t+1}$  after observing the entire history of the nominal interest rate  $\{i_{t-k} | k \geq 0\}$  and inflation  $\{\pi_{t-k} | k \geq 0\}$ , they can also observe the entire history of surpluses  $\{s_{t-k} | k \geq 0\}$  and realized real market debt  $\{s_{b,t-k} | k \geq 0\}$ . It follows that we can define the incomplete information set as

$$(12) \quad I_t^{II} = \{s_{f,t-k}, s_{m,t-k}, \mathcal{M} | k \geq 0\},$$

where the superscript “II” stands for incomplete information.

### C. An analytical case without any monetary policy shocks

So far, we have not introduced exogenous monetary policy shocks. Although it is certainly a simplification, such a modeling choice can deliver powerful insight when coupled with an appropriate assumption. To see the point, comparing the observables in  $I_t^{II}$  and  $I_t^{FI}$  implies that we can establish the household’s signal extraction problem as

$$(13) \quad \underbrace{\begin{bmatrix} s_{f,t} \\ s_{m,t} \end{bmatrix}}_{\mathbf{s}_t} = \underbrace{\begin{bmatrix} 1 & \frac{1}{1-\rho L} \\ 0 & \mathcal{P}(L) \end{bmatrix}}_{M(L)} \begin{bmatrix} \varepsilon_t^F \\ \varepsilon_t^U \end{bmatrix}$$

where  $M(L)$  is a mapping in the lag operator  $L$  that links the household signals  $\mathbf{s}_t$  to the underlying shocks  $\varepsilon_t^F$  and  $\varepsilon_t^U$ . For ease of notation, we define  $\rho = 1 - \gamma_b / \gamma$ .

The signal extraction (13) makes it clear that if the mapping  $M(L)$  is invertible, then households can perfectly learn the two types of fiscal shocks and  $I_t^{II} = I_t^{FI}$ . Consequently, the model dynamics become FIRE. The Riesz-Fisher Theorem (see Sargent (1987)) establishes a necessary and sufficient condition for the invertibility of the mapping  $M(L)$ . For invertibility to hold, we must have a nonzero determinant of the mapping (i.e.  $\det M(z) = \mathcal{P}(z) \neq 0$  for all  $z \in (-1, 1)$ .) For example, when  $\pi_t^*$  follows the stationary AR(1) process (8), there is no incomplete information between funded and unfunded fiscal shocks, even if the household’s information set is given by (12).

In other words, to preserve incomplete information,  $\mathcal{P}(L)$  must be non-invertible. A non-invertible  $\mathcal{P}(L)$  requires at least one root  $z \in (-1, 1)$  such that  $\mathcal{P}(z) = 0$ . For simplicity, we assume that there is only one such root, denoted by  $\lambda$ . As-

sumption 1 below formalizes the condition.

ASSUMPTION 1: Assume there is a unique  $\lambda \in (-1, 1)$ , where  $\lambda \neq 0$  such that  $\mathcal{P}(z) = 0$ . For all  $z \in (-1, 1)$  and  $z \neq \lambda$ ,  $\mathcal{P}(z) \neq 0$ .

Given Assumption 1 and the signal extraction problem (13), we can solve for the equilibrium inflation process analytically using the frequency domain techniques (see Kasa, Walker and Whiteman (2014) and Han, Ma and Mao (2022)). Appendix B shows that the incomplete-information inflation process follows

$$(14) \quad \pi_t^{II} = \frac{\Pi_{F,0}}{1 - \lambda L} \varepsilon_t^F + \left[ \frac{\mathcal{P}(\phi_\pi^{-1}) - \phi_\pi L \mathcal{P}(L)}{1 - \phi_\pi L} + \frac{\Pi_{U,0} - \mathcal{P}(\phi_\pi^{-1})}{1 - \lambda L} \right] \varepsilon_t^U.$$

where  $\Pi_{F,0}$  and  $\Pi_{U,0}$  are two endogenously determined constants. In contrast, the full-information inflation follows

$$(15) \quad \pi_t^{FI} = \frac{\mathcal{P}(\phi_\pi^{-1}) - \phi_\pi L \mathcal{P}(L)}{1 - \phi_\pi L} \varepsilon_t^U.$$

We make several remarks about  $\pi_t^{II}$  and  $\pi_t^{FI}$ . While  $\pi_t^{FI}$  makes it clear that the funded shock  $\varepsilon_t^F$  satisfies the Ricardian equivalence and is non-inflationary in  $\pi_t^{FI}$ , it introduces an AR(1) term,  $\Pi_{F,0}/(1 - \lambda L)\varepsilon_t^F$ , in  $\pi_t^{II}$ . The persistence of the AR(1) process is given by  $\lambda$ . Under incomplete information, even a transitory, funded fiscal shock can generate persistent inflation. The breakdown of Ricardian equivalence under incomplete information should not come as a surprise, as households cannot perfectly distinguish a funded shock from an unfunded fiscal shock. If the latter can generate some inflation, the former must also be inflationary with incomplete information. Second, compared to  $\pi_t^{FI}$ , which is fully driven by the unfunded shock  $\varepsilon_t^U$ , the same shock introduces an additional AR(1) term in  $\pi_t^{II}$ , which is given by  $\frac{\Pi_{U,0} - \mathcal{P}(\phi_\pi^{-1})}{1 - \lambda L} \varepsilon_t^U$ . The additional term allows for a more complicated fiscal inflation process.

#### D. An illustrative case with exogenous monetary policy shock

The analytical result described above relies on a technical assumption (i.e., the non-invertibility of  $\mathcal{P}(L)$ ) to preserve incomplete information. We now argue that information friction can arise naturally if there is an exogenous monetary policy shock  $e_t$ . Let the monetary policy rule be

$$(16) \quad i_t = \phi_\pi(\pi_t - \pi_t^*) + e_t, \quad e_t \sim N(0, \sigma_e^2).$$

Households can observe both the histories of  $i_t$  and  $\pi_t$ , but cannot separate the exogenous shock  $e_t$  from the endogenous inflation target  $\pi_t^*$ . All else being equal, we can establish the household's signal extraction problem between signals and

the underlying shocks as

$$\begin{bmatrix} s_{f,t} \\ s_{m,t} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \frac{1}{1-\rho L} & 0 \\ 0 & \mathcal{P}(L) & -1/\phi_\pi \end{bmatrix}}_{M_2(L)} \begin{bmatrix} \varepsilon_t^F \\ \varepsilon_t^U \\ e_t \end{bmatrix}$$

Since there are three shocks  $\{\varepsilon_t^F, \varepsilon_t^U, e_t\}$  and the private sector can observe only two signals  $\{s_{f,t}, s_{m,t}\}$ , the above non-square mapping  $M_2(L)$  must be non-invertible and can preserve incomplete information naturally.

#### E. A numerical case with a meaningful fiscal financing scheme

So far, we have concentrated fiscal discussions on primary surpluses  $S_t$ . We now introduce a more realistic fiscal financing scheme to the simple endowment economy and show how incomplete information can alter the policy analysis numerically. The fiscal financing scheme also reveals the root of the problem of elevated public debt levels.

Consider  $S_t = T_t - Z_t$  where  $T_t$  and  $Z_t$  are lump-sum taxes and transfers. The slow-moving increase in public debt begins long before the pandemic, mainly due to an aging population and higher mandatory payments of the Social Security, Medicaid, and Medicare programs (i.e., higher transfers  $Z_t$ ). For this reason, we impose constant tax revenues  $T_t \equiv T^*$ . The steady-state transfers-to-output ratio satisfies  $Z^* < T^*$ , leading to a steady-state surplus-to-output ratio  $S^* = T^* - Z^* > 0$ . The log-linearized transfers rule follows

$$(17) \quad z_t = -\gamma_z(s_{b,t-1} - s_{b,t}^*) + \varepsilon_t^F, \quad \varepsilon_t^F \sim N(0, \sigma_F^2),$$

where  $s_{b,t}^*$  is the debt target and  $\gamma_z > 0$  is the debt response parameter.<sup>3</sup> The  $\varepsilon_t^F$  stands for the “funded” transfers shock. Given a predetermined  $s_{b,t-1}$  and a zero funded shock, a higher  $s_{b,t}^*$  increases transfers  $z_t$ . The shock of “unfunded” transfers,  $\varepsilon_t^U$ , drives a very persistent debt target  $s_{b,t}^*$  as

$$(18) \quad s_{b,t}^* = \rho s_{b,t-1}^* + \varepsilon_t^U, \quad \varepsilon_t^U \sim N(0, \sigma_U^2),$$

with  $\rho \in (0, 1)$  and  $\rho \approx 1$ .

We close the model by specifying the inflation target process  $\mathcal{P}(L)$ . Consistent with Bianchi, Faccini and Melosi (2023), we assume that there is an underlying shadow economy in which only the unfunded shock  $\varepsilon_t^U$  can drive the inflation target  $\pi_t^*$ . Appendix C shows that the inflation target implied by the shadow

<sup>3</sup>Given the fiscal financing scheme, the primary surplus now follows  $s_t = -\frac{Z^*}{S^*} z_t$ . It follows that the restriction  $\gamma > 1$  translates to  $\gamma_z > S^*/Z^*$ .

economy is an AR(1) process given by

$$(19) \quad \pi_t^* = \mathcal{P}(L)\varepsilon_t^U = \frac{\Phi}{1 - \phi_F L} \varepsilon_t^U,$$

where

$$(20) \quad \Phi = \left(1 - \frac{\phi_F}{\phi_\pi}\right) (1 - \beta) \frac{Z^*}{S^*}.$$

Compared to the inflation target process (8), the term  $\beta - 1$  changes to  $1 - \beta$  in (19) as we switch from a fiscal income (i.e., primary surplus) shock to a fiscal expenditure (i.e., transfers) shock. More importantly, the fiscal financing scheme introduces an additional term  $Z^*/S^*$  in  $\Phi$ . The parameter  $\Phi$  controls the magnitude of the increase in the inflation target to an unfunded shock, indicating the required strength of central bank accommodation to fiscal inflation. As the steady-state primary surplus  $S^* = T^* - Z^*$  approaches zero, it greatly amplifies the central bank accommodation  $\Phi$  implied by the shadow economy.

Throughout the paper, we focus on incomplete information on the private sector (i.e., households and firms). It remains to ask: Does the central bank have perfect information on the unfunded shock? We think it is unlikely that the answer is yes. The question provides another reason for introducing  $e_t$  in the monetary rule (16). In addition to the usual interpretation of  $e_t$  as a policy shock, we could also interpret  $e_t$  as a measurement or implementation error of the short-term inflation target, since the central bank may also be unable to observe  $\varepsilon_t^U$  perfectly.

We redefine the fiscal signal  $s_{f,t} = z_t + \gamma_z s_{b,t-1} = \gamma_z s_{b,t}^* + \varepsilon_t^F$  so that

$$(21) \quad \begin{bmatrix} s_{f,t} \\ s_{m,t} \end{bmatrix} = \begin{bmatrix} 1 & \frac{\gamma_z}{1-\rho_L} & 0 \\ 0 & \frac{\Phi}{1-\phi_F} & -1/\phi_\pi \end{bmatrix} \begin{bmatrix} \varepsilon_t^F \\ \varepsilon_t^U \\ e_t \end{bmatrix}.$$

Given the non-square signal extraction problem (21), there is generally no analytical solution. We solve the model following the algorithm in Blanchard, L'Huillier and Lorenzoni (2013). Let  $\mathbf{y}_t$  denote the vector of endogenous state variables and  $\mathbf{s}_t$  the exogenous state variables. The equilibrium conditions can be written as

$$(22) \quad \mathbf{F}\mathbb{E}_t^{\mathbf{HH}}\mathbf{y}_{t+1} + \mathbf{G}\mathbf{y}_t + \mathbf{H}\mathbf{y}_{t-1} + \mathbf{M}\mathbf{s}_t + \mathbf{N}\mathbb{E}_t^{\mathbf{HH}}\mathbf{s}_{t+1} = \mathbf{0},$$

where  $\mathbf{F}, \mathbf{G}, \mathbf{H}, \mathbf{M}, \mathbf{N}$  are coefficient matrices. Solving (22) gives a law of motion

$$(23) \quad \mathbf{y}_t = \mathbf{P}\mathbf{y}_{t-1} + \mathbf{Q}\mathbf{s}_t + \mathbf{R}\mathbf{s}_{t|t}.$$

The  $\mathbf{s}_{t|t}$  denotes the perceived exogenous state variables of the households, which can be obtained from the Kalman recursion once we establish the household signal

extraction problem (21) as a state-space model.<sup>4</sup>

For illustration, we fix a set of parameters as a benchmark. The time discount factor  $\beta$  is 0.99, suggesting a quarterly model. Empirically, the low frequency movements in transfers are very persistent. We set the persistence of the time-varying debt target  $\rho = 0.99$ .<sup>5</sup> The other parameters are as follows:

$$\begin{aligned}\phi_\pi &= 1.5, & \gamma_z &= 0.2, & Z^* &= 0.2, & S^* &= 0.01, \\ \phi_F &= 0.95, & \sigma_e &= 0.25, & \sigma_U &= 1, & \sigma_F &= 1;\end{aligned}$$

The steady-state surplus-to-output ratio is 1%, leading to a steady-state debt-to-output ratio  $s_b^* = (\beta S^*)/(1 - \beta) = 0.99$ . The implied  $\Phi \approx 0.07$ , indicating that the central bank increases its inflation target by 7 basis points for a 1% unfunded transfer shock. The magnitudes of the monetary and fiscal shocks are chosen to be in line with existing estimates (see Smets and Wouters (2007) and Leeper, Plante and Traum (2010)).

Figure 2 plots the impulse responses of inflation (quarterly, non-annualized), nominal interest rate, real debt, and transfers to monetary and fiscal shocks (both funded and unfunded). With full information, a negative monetary policy shock  $e_1 = -\sigma_e$  generates a one-time inflation. Since inflation is transitory and  $\mathbb{E}_t^{HH} \pi_{t+1} = 0$ , the nominal interest rates remain the same. Higher inflation causes the real market value of debt to decrease, leading to higher transfers. Due to Ricardian equivalence, the funded fiscal shock  $\varepsilon_1^F = \sigma_F$  generates trivial (that is, identically zero) inflation responses. Only the unfunded fiscal shock  $\varepsilon_1^U = \sigma_U$  can generate persistent inflation with full information.

All shocks can generate non-trivial and persistent inflation in the incomplete information model. This result is intuitive: When a transitory shock (either a monetary policy or a funded fiscal) hits the economy, incomplete-information households rationally attribute the movements in their observables  $\{s_{m,t}, s_{f,t}\}$  to a linear combination of three shocks. The non-zero weight households assign to their *perceived* unfunded fiscal shock is the direct cause of the persistent inflation impulse responses. Consequently, the impulse responses of the nominal interest rate are also persistent to all three types of shock as it tracks  $\mathbb{E}_t^{HH} \pi_{t+1}$  in the Fisherian model.

Similarly, when a persistent unfunded fiscal shock ( $\varepsilon_1^U = \sigma_U$ ) hits the economy, households rationally believe that some of the observed changes in their signals  $\{s_{m,t}, s_{f,t}\}$  are due to funded fiscal and monetary policy shocks. Since both perceived funded fiscal and monetary shocks cannot generate much inflation, the initial impact is small compared to the full-information model. Over time, households gradually determine the intrinsic nature of the underlying shocks, and the impulse responses of inflation and nominal interest rates converge to their full-

<sup>4</sup>For technical details of the Kalman recursion, see Hamilton (1994). Guo and Han (2025) use the same solution algorithm and studies how time-varying fiscal foresight uncertainty impacts government spending multipliers in an incomplete information setting.

<sup>5</sup>Bianchi, Faccini and Melosi (2023) set  $\gamma_b = 0$  in (4) so that the debt target is a unit root process.

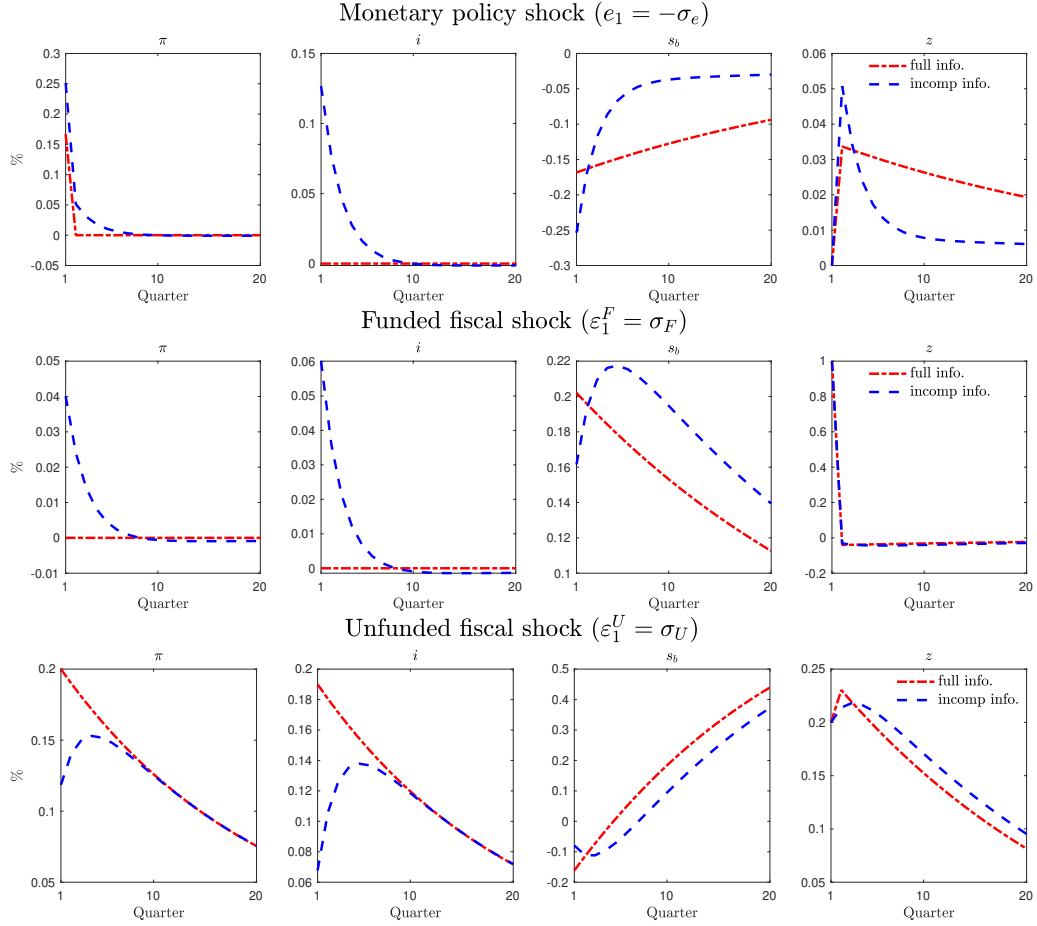


FIGURE 2. IMPULSE RESPONSES OF INFLATION (QUARTERLY, NON-ANNUALIZED), NOMINAL INTEREST RATE, REAL DEBT, AND LUMP-SUM SURPLUSES.

information counterparts within two years. The two properties force inflation and nominal interest rates to display hump-shaped impulse responses to the unfunded fiscal shock. The hump-shaped pattern has become a hallmark of incomplete-information rational expectation models.

Next, we conduct a sensitivity analysis to demonstrate how key parameters affect inflation dynamics. Figure 3 plots the impulse responses of inflation when we vary the standard deviations of the shocks  $\{\sigma_e, \sigma_F, \sigma_U\}$  one at a time. The other parameters are fixed at their benchmark values. These parameters enter the model dynamics multiplicatively only through their corresponding shocks in full-information models. With incomplete information, these standard deviations play an additional role as they enter the signal extraction problem of the household (21) and determine the weights that the household assigns to their perceived shocks.

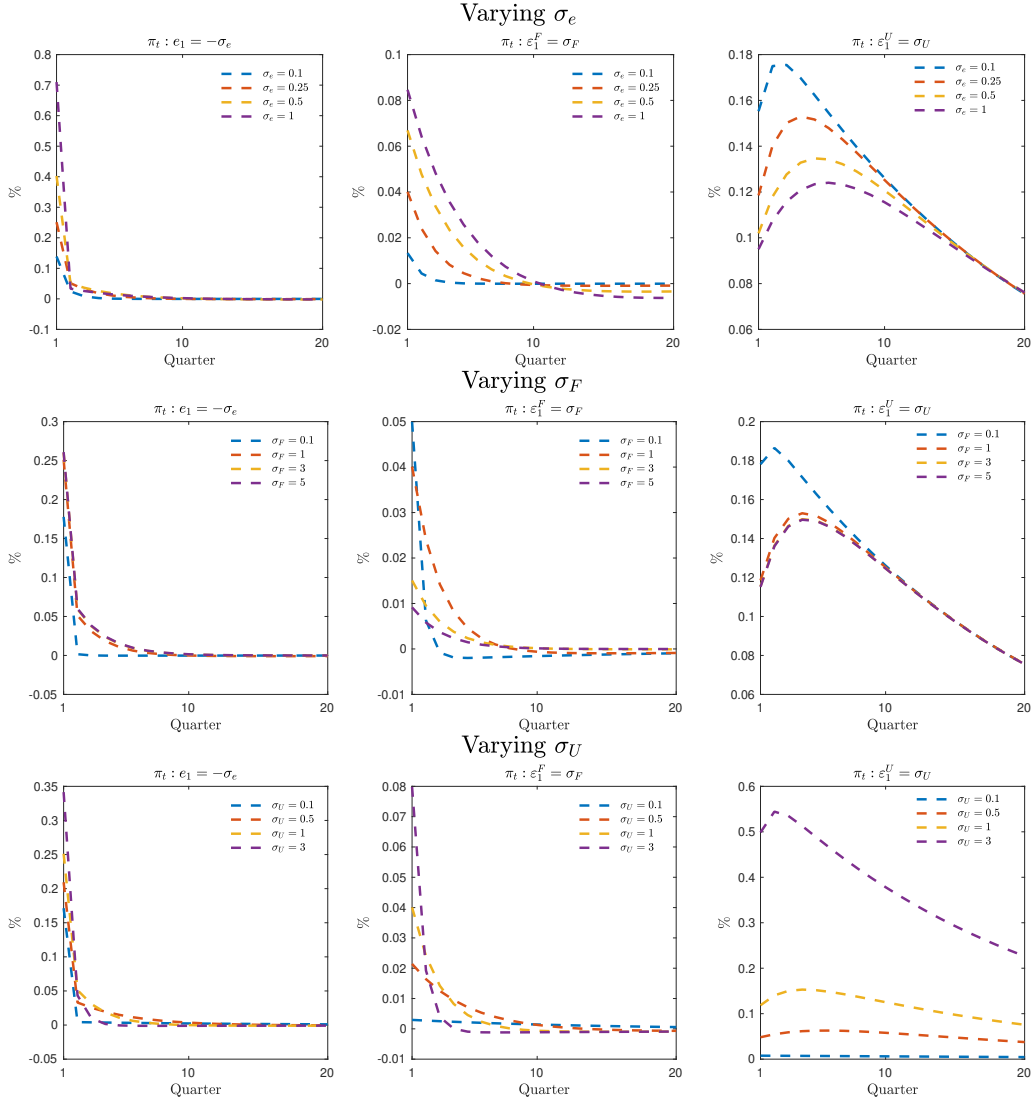


FIGURE 3. IMPULSE RESPONSES OF INFLATION (QUARTERLY, NON-ANNUALIZED) WHEN VARYING  $\{\sigma_e, \sigma_F, \sigma_U\}$ .

For example, when  $\sigma_e$  and  $\sigma_F$  are relatively small compared to  $\sigma_U$ , households will assign more weight to their perceived unfunded shock when the actual shock is  $\epsilon_1^U = \sigma_U$ , with larger inflationary impulse responses (that is, the third column of Figure 3).

Figures 4 and 5 plot the impulse responses of inflation when varying the policy parameters  $\{\phi_\pi, \phi_F, \gamma_z, S^*\}$  one at a time. As  $\phi_\pi$  increases so that the central bank becomes more hawkish, a monetary cut or a funded transfer shock generates



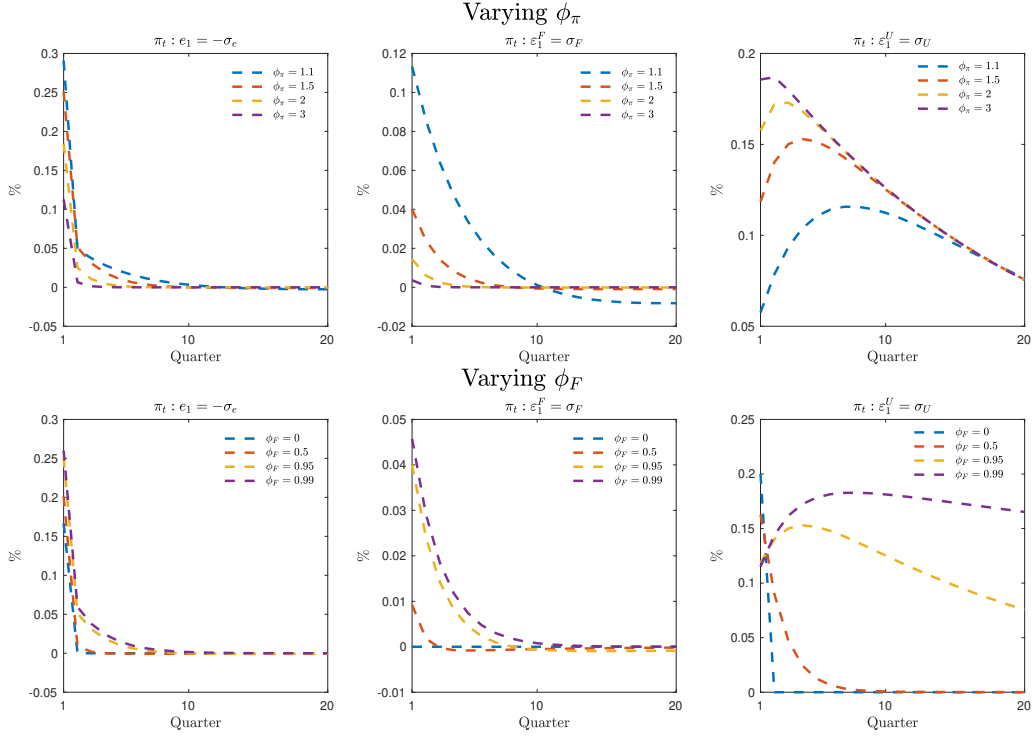


FIGURE 4. IMPULSE RESPONSES OF INFLATION (QUARTERLY, NON-ANNUALIZED) WHEN VARYING MONETARY POLICY PARAMETERS  $\{\phi_\pi, \phi_F\}$ .

smaller inflationary effects. However, a larger  $\phi_\pi$  also indicates a stronger central bank accommodation  $\Phi$  of the unfunded shock, leading to higher inflationary effect of the unfunded shock. The parameter  $\phi_F$  mainly controls the persistence of fiscal inflation. When  $\phi_F = 0$ , the central bank only allows transitory fiscal inflation by allowing  $\pi_t^*$  to temporarily deviate from its long-term target. As  $\phi_F$  approaches one, the fiscal inflation driven by the unfunded shock becomes more persistent. On the other hand,  $\Phi$  represents the degree to which the central bank accommodates the unfunded fiscal shock  $\varepsilon_t^U$  by raising its short-term inflation target  $\pi_t^*$  and controls the magnitude of fiscal inflation. As  $S^*$  decreases from 1% to 0.1% so that  $\Phi$  increases from 0.07 to 0.7, it greatly amplifies the impulse responses of inflation to unfunded shocks.

### III. A Quantitative Model

We now embed the information structure developed in the simple economy into a medium-scale New Keynesian DSGE model. Following Leeper, Traum and Walker (2017) and Bianchi, Faccini and Melosi (2023), our model provides a realistic environment with nominal rigidities, capital accumulation, hand-to-mouth

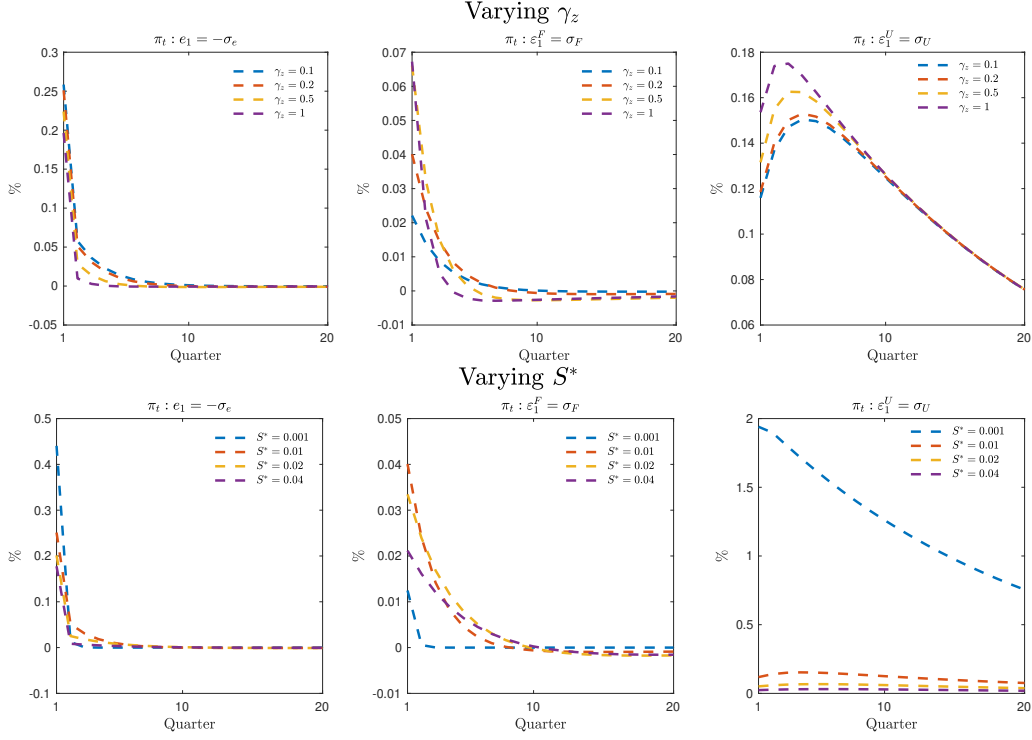


FIGURE 5. IMPULSE RESPONSES OF INFLATION (QUARTERLY, NON-ANNUALIZED) WHEN VARYING FISCAL POLICY PARAMETERS  $\{\gamma_z, S^*\}$ .

households, long-term debt, and a rich fiscal block, allowing the model to match U.S. business-cycle and fiscal dynamics. Meanwhile, it introduces incomplete information on monetary and fiscal policy in a way that directly affects inflation expectations. Agents observe policy outcomes—interest rates, inflation, debt, spending, transfers, and tax rates—but cannot perfectly disentangle funded fiscal shocks, unfunded fiscal shocks, monetary policy shocks, and persistent cost-push disturbances. The resulting signal-extraction problem is the key departure from the full-information benchmark. We describe here its main ingredients and defer the details and log-linearization to the Appendix D.

There are two types of households in the economy: savers and hand-to-mouth consumers. Both households are subject to external habit formation and a discount factor shock. They receive wage income by providing labor to firms and are subject to various taxes. They also receive lump-sum transfers from the government. Savers have access to short- and long-term government bonds and can also accumulate capital, subject to variable capacity utilization and adjustment costs in investment. Hand-to-mouth households consume all of their disposable, after-tax income and do not save. Both saver (S) and Non-saver (N) households

derive utility from consumption of the composite good and disutility from the supply of labor services:

$$(24) \quad U_t^i(j) = u_t^d \left( \ln(C_t^{*i}(j) - hC_{t-1}^{*i}) - \frac{L_t^i(j)^{1+\chi}}{1+\chi} \right), \quad i \in \{S, N\}$$

where  $u_t^d$  is a shock from the discount factor, and  $1/\chi$  is the Frisch elasticity of the labor supply.

Both households receive after-tax nominal income and lump sum transfers from the government ( $Z_t$ ). In addition, households can save by investing in a one-period government bond ( $B_t$ ), a long-term nominal government bond ( $B_t^m$ ), invest in physical capital ( $I_t$ ), and receive dividends from firms ( $D_t$ ). The budget constraint of saver households can be written as:

$$(25) \quad \begin{aligned} & P_t(1 + \tau_{C,t})C_t^S + P_t I_t^S + P_t^m B_t^m + R_{n,t}^{-1} B_t \\ &= (1 + \rho P_t^m) B_{t-1}^m + B_{t-1} + (1 - \tau_{L,t}) \int_0^1 W_t(l) L_t^S dl \\ &+ (1 - \tau_{K,t}) R_{K,t} \nu_t \bar{K}_{t-1}^S - \Psi(\nu_t) \bar{K}_{t-1}^S + P_t Z_t^S + D_t, \end{aligned}$$

where  $\tau_{C,t}$  and  $\tau_{L,t}$  denote the tax rates on consumption and labor income, respectively.  $P_t$  and  $P_t^m$  are the prices of one-year and long-term bonds. The arbitrage condition and the law of capital accumulation are described in Appendix D in more detail. The household maximizes the expected utility  $\mathbb{E}_0^{HH} \sum_{t=0}^{\infty} \beta^t U_t^S$  subject to the sequence of budget constraints in equation (25) and the law of motion of capital accumulation.

Intermediate goods firms are subject to the Calvo pricing and price indexation. To allow for balanced growth, labor-augmenting technology  $A_t$  follows an exogenous process that is stationary in the growth rate. There is a perfectly competitive sector of final good firms that produce the final consumption good  $Y_t$  by combining a unit measure of intermediate differentiated inputs. Intermediate firms produce goods according to the production function:

$$(26) \quad Y_t(i) = K_t(i)^\alpha (A_t L_t(i))^{1-\alpha} - A_t \Omega,$$

where  $\Omega$  is a fixed production cost that increases with the rate of technological advancement that increases labor  $A_t$ , and  $\alpha \in [0, 1]$  is the capital share. When setting prices, firms face Calvo-style price rigidity. Wages set by firms are subject to both sticky wages and wage indexation. The resulting log-linearized Phillips curve is given by:

$$(27) \quad \hat{\pi}_t = \frac{\beta}{1 + \xi_p \beta} \mathbb{E}_t \hat{\pi}_{t+1} + \frac{\xi_p}{1 + \xi_p \beta} \hat{\pi}_{t-1} + \kappa_p \hat{m}_t + \kappa_p \hat{p}_t,$$

where  $\kappa_p = [(1 - \beta\omega_p)(1 - \omega_p)] / [\omega_p(1 + \beta\xi_p)]$ . The total cost-push shock process

$\hat{p}_t$  is the sum of the persistent ( $\hat{u}_t^p$ ) and transitory ( $\hat{\eta}_t^p$ ) shock components.

The monetary and fiscal policy blocks in our model differ from the canonical DSGE models in introducing both time-varying inflation and debt targets. First, monetary policy follows a generalized Taylor rule

$$(28) \quad \hat{r}_t^n = \rho_r \hat{r}_{t-1}^n + (1 - \rho_r) [\phi_\pi (\hat{\pi}_t - \hat{\pi}_t^*) + \phi_y \hat{y}_t] + u_t^m,$$

which embeds interest rate inertia and satisfies the Taylor principle (that is,  $\phi_\pi > 1$ ). It also responds to the deviation of the inflation gap  $\hat{\pi}_t - \hat{\pi}_t^*$  and the output  $\hat{y}_t$ . The existence of a persistent (that is, AR (1)) monetary policy shock,  $u_t^m$ , prevents households from learning the inflation target  $\hat{\pi}_t^*$  perfectly from the monetary policy rule (28).

Fiscal policy consists of a set of expenditure, transfer, and tax rules. Let  $s_{b,t}$  denote the real market debt-to-output ratio. The fiscal authority adjusts government spending  $\hat{g}_t$ , transfers  $\hat{z}_t$ , and tax rates on capital income  $\hat{\tau}_k$  and labor income  $\hat{\tau}_l$  as follows:

$$(29) \quad \hat{g}_t = \rho_G \hat{g}_{t-1} - (1 - \rho_G) [\gamma_G (\hat{s}_{b,t-1} - \hat{s}_{b,t}^*) + \phi_{g,y} \hat{y}_t] + \varepsilon_t^g,$$

$$(30) \quad \hat{z}_t = \rho_Z \hat{z}_{t-1} - (1 - \rho_Z) [\gamma_Z (\hat{s}_{b,t-1} - \hat{s}_{b,t}^*) + \phi_{z,y} \hat{y}_t] + \varepsilon_t^z,$$

$$(31) \quad \hat{\tau}_{J,t} = \rho_J \hat{\tau}_{J,t-1} + (1 - \rho_J) \gamma_J [\hat{s}_{b,t-1} - \hat{s}_{b,t}^*] + \varepsilon_t^J, \text{ for } J \in \{k, l\};$$

where  $\gamma_G$ ,  $\gamma_Z$ , and  $\gamma_J > 0$  are large enough so that the debt remains on a stable path. We abstract from the consumption tax  $\hat{\tau}_c$  by fixing its value. As in the Fisherian economy, the time-varying debt target,  $s_{b,t}^*$ , follows a stationary AR(1) process:

$$\hat{s}_{b,t}^* = \rho_s \hat{s}_{b,t-1}^* + \varepsilon_t^U, \quad \varepsilon_t^U \sim N(0, \sigma_U^2),$$

with  $\rho_s \in (0, 1)$  and  $\varepsilon_t^U$  represents the unfunded fiscal shock. It should be noted that we do not let the tax rates respond to the output  $\hat{y}_t$ . This model choice follows Leeper, Traum and Walker (2017) and Bianchi, Faccini and Melosi (2023). It does not affect our main results.

To specify the inflation target  $\pi_t^*$ , Bianchi, Faccini and Melosi (2023) considers the shadow economy in which there is only an unfunded fiscal shock and policymakers follow a fiscal-led policy mix. All other model equations are identical across the actual and shadow economies. Due to the scale and complexity of the medium-scale DSGE model, Bianchi, Faccini and Melosi (2023) solve  $\pi_t^*$  numerically. However, for our purpose, a numerical  $\pi_t^*$  will not work as we have to put the household's signal extraction into a linear state-space form. For this purpose, we assume the shadow economy is the benchmark three-equation New Keynesian model for the production economy. We augment the model with a government budget constraint and a simple fiscal rule. We then solve the implied  $\pi_t^*$  analytically. Appendix E shows that, as in the Fisherian model, the inflation target is

still an AR(1) process, given by

$$(32) \quad \hat{\pi}_t^* = \phi_F \hat{\pi}_{t-1}^* + \Phi \cdot \varepsilon_t^U$$

with  $0 < \phi_F < 1$  and  $\Phi > 0$ .<sup>6</sup> While  $\phi_F$  governs the persistence of fiscal inflation, the other parameter  $\Phi$  controls the strength with which the central bank adjusts its short-term inflation target  $\pi_t^*$  to accommodate the unfunded shock  $\varepsilon_t^U$ .

All economic agents form rational expectations, subject to incomplete information on monetary and fiscal policy shocks, and cost-push shocks. We do not introduce information friction for any other shocks in the economy to isolate incomplete-information fiscal inflation.

#### A. Empirical Analysis

The model is estimated using Bayesian techniques to match the following 12 observables for the US economy: real per capita GDP growth, real per capita consumption growth, real per capita investment growth, a measure of the hours gap, the effective federal funds rate, the growth of average weekly earnings, price inflation based on the GDP deflator, the growth of real government transfers, the growth of government expenditure, the government debt-to-GDP ratio, labor tax revenue, and capital tax revenue. To better discipline the information frictions introduced in the model, we incorporate one-quarter-ahead inflation expectations from the Survey of Professional Forecasters (SPF) in the incomplete information model estimation. We do not take the SPF to the full information model, as it is well known in the literature that the FIRE model cannot explain certain features of the survey data (see Coibion and Gorodnichenko (2015)). In Section IV.D, we compare our baseline estimation results with the estimation results with inflation expectations one-year ahead from Michigan Survey of Consumer Sentiments (MSC) and without short-term inflation expectations. The data construction is explained in more detail in the Appendix H.

Our sample period spans from 1968:Q1 to 2022:Q3 to align with the availability of SPF forecast data. The estimation strategy to deal with the zero lower bound period after the financial crisis relies on two sub-samples, the first sample from 1968:Q1 to 2007:Q4 and the second subsample from 2008:Q1 to 2022:Q3. The measurement equations linking observables to model variables are shown in the Appendix H.H1. Consistent with Bianchi, Faccini and Melosi (2023) and Campbell et al. (2012), we discipline the second sample estimation by allowing agents' expectations of future interest rates to be informed by market forecasts. The latter subsample incorporates additional observables using overnight index swaps to measure the forecast of the federal funds rate one to ten quarters ahead. Each

<sup>6</sup>Considering a more complicated shadow economy may lead to a higher-order ARMA(p,q) inflation target process with  $p > 1, q > 0$ . We find that additional complexity may not be necessary. First, one could argue if it is indeed possible for the central bank to follow an almost perfect shadow economy like in Bianchi, Faccini and Melosi (2023). Second, we find that the AR(1) inflation target yields very similar estimation results to Bianchi, Faccini and Melosi (2023) under full information.

sample is estimated in two steps; first we calculate the likelihood of our model and elicit the mode of posterior distribution. Second, we use MCMC with 100,000 simulations with burn-in of 50% to obtain the entire posterior distribution.

The economic agents in the model form rational expectations of inflation, real interest rate, bond price, investment, marginal utility of the household of the savers, capital rental rate, capital tax rate, marginal utility of investment and real wages. We first estimate our model under the assumption that agents have full information about the shocks underlying the economy, which gives us our FIRE estimates.

We introduce incomplete information by assuming that all households, intermediate firms, and final good firms share the same information set, denoted by  $\mathcal{I}_t^{HH}$ . We assume that the agents can observe the entire history of the nominal interest rate  $\{r_{t-k}^n | k \geq 0\}$ , inflation  $\{\pi_{t-k} | k \geq 0\}$ , and output  $\{y_{t-k} | k \geq 0\}$ . The households can also observe the history of real-market debt  $\{s_{b,t-k} | k \geq 0\}$ , government spending and transfers  $\{g_{t-k}, z_{t-k} | k \geq 0\}$ , and the history of labor and capital tax rates  $\{\tau_{k,t-k}, \tau_{l,t-k} | k \geq 0\}$ . However, households cannot distinguish between exogenous shocks to fiscal and monetary policies and shocks to time-varying debt and inflation targets. In addition, households cannot perfectly distinguish between transitory and persistent components of the cost-push shock.

There are more policy shocks than signals. Thus, incomplete information arises naturally in our model. Rewriting the monetary rule (28) as

$$\hat{r}_t^n - \rho_r \hat{r}_{t-1}^n - (1 - \rho_r) (\phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t) = \underbrace{-(1 - \rho_r) \phi_\pi \pi_t^* + u_t^m}_{s_t^m}$$

indicates that the history of the variables on the right side,  $s_t^m$ , is also known to households. Similarly, rewriting the fiscal rules (29) and (30) as follows

$$\begin{aligned} \hat{g}_t - \rho_G \hat{g}_{t-1} + (1 - \rho_G) [\gamma_G \hat{s}_{b,t-1} + \phi_{g,y} \hat{y}_t] &= \underbrace{(1 - \rho_G) \gamma_G \hat{s}_{b,t}^* + \varepsilon_t^g}_{s_t^g} \\ \hat{z}_t - \rho_Z \hat{z}_{t-1} + (1 - \rho_Z) [\gamma_Z \hat{s}_{b,t-1} + \phi_{z,y} \hat{y}_t] &= \underbrace{(1 - \rho_Z) \gamma_Z \hat{s}_{b,t}^* + \varepsilon_t^z}_{s_t^z} \end{aligned}$$

suggests that the two right-hand variables,  $s_t^g$  and  $s_t^z$ , are also in the household information set. The tax rules (31) can also be rewritten with signals where  $s_t^J$  is in the household information set. That is,

$$\hat{\tau}_{J,t} - \rho_J \hat{\tau}_{J,t-1} - (1 - \rho_J) \gamma_J \hat{s}_{b,t-1} = \underbrace{-(1 - \rho_J) \gamma_J \hat{s}_{b,t}^* + \varepsilon_t^J}_{s_t^J} \quad \forall J \in \{k, l\}$$

Finally, we introduce the cost-push shock signal as a composite of transitory and

persistent components:

$$\hat{t}p_t = \underbrace{\hat{u}_t^p + \hat{\eta}_t^p}_{s_t^p}$$

We do not introduce incomplete information about any other shocks in the model and assume that they can be observed perfectly by households. Although incomplete information certainly can exist among these non-policy shocks, we know from Chahrour and Ulbricht (2023) that there is an equivalence result between the incomplete information model and a full-information model, as long as these non-policy shocks do not interact with policy and cost-push shocks. These exogenous shocks are intended to improve the empirical fit of the medium-scale DSGE model. Formally, we define the incomplete information set  $\mathcal{I}_t^{HH}$  as

$$\mathcal{I}_t^{HH} = \{s_{t-k}^m, s_{t-k}^g, s_{t-k}^z, s_{t-k}^k, s_{t-k}^l, s_{t-k}^p, \mathcal{M} | k \geq 0\}.$$

The model is then re-estimated under the incomplete information specification. Appendix F establishes the solution of the incomplete information rational expectations (IIRE) model as a state-space representation used in estimation.

#### B. Calibration and Prior Distributions

The priors for the parameters are shown in Table 1. Our choice of priors for all non-policy structural parameters and exogenous parameters follows Bianchi, Faccini and Melosi (2023) and Leeper, Traum and Walker (2017). The response of fiscal variables to debt follows a normal distribution with mean 0.1 and standard deviation 0.1. Our fiscal rule allows government spending and transfers to respond to fluctuations in output. The associated policy parameters are restricted to be positive using the Gamma distribution with mean 0.2 and standard deviation 0.05. The monetary policy parameters for inflation and output response follow a normal distribution with mean 1.5 and 0.5 respectively, and standard deviation 0.1. The AR(1) coefficients in the monetary and fiscal policy rules follow a beta distribution with mean 0.5 and standard deviation 0.1.

The prior parameter measuring the response of the short-term inflation target to the unfunded fiscal shock  $\Phi$  is defined as a Gamma distribution with mean 0.2 and standard deviation at 0.1. Our prior imposes the restriction that  $\Phi$  must be greater than 0. Consistent with other policy rules, the AR(1) coefficient for the inflation target shock follows a beta distribution with mean 0.5 and standard deviation 0.1. The AR(1) coefficient for the time-varying debt target  $\rho_s$  is calibrated at 0.99, allowing for a very high persistence of the unfunded fiscal shock. The autocorrelation parameter  $\rho$  for all persistent shocks (except cost push) follows a Beta distribution with mean 0.5 and standard deviation 0.1.

We calibrate the remaining parameters and the steady state in the model to sample averages consistent with Bianchi, Faccini and Melosi (2023). The discount factor  $\beta$  is set at 0.99, the share of capital in the production function  $\alpha$  at 0.33, the depreciation rate  $\delta$  at 2.5%, the substitution elasticity between labor and

between intermediate goods at 0.14 and the share of agents from hand to mouth  $\mu$  at 0.11. The steady state of government expenditure to GDP ratio is set at 0.11, and the steady state of labor, capital, and consumption tax rates are set at 0.186, 0.218, and 0.023, respectively. The average duration of government maturity is calibrated at 24 quarters.

#### IV. Estimation Results

We now present the results of our model estimation under the assumption of Full Information Rational Expectations (FIRE) and Incomplete Information Rational Expectations (IIRE). The impulse response functions are reported for the IIRE model estimated with and without short-run expectations as observables to illustrate how expectations data disciplines the information frictions in the model. The posterior medians along with the 90%-credible intervals for all parameters from the first sample estimation (1968:Q3 to 2007:Q4) are presented in Table 1.<sup>7</sup> The incomplete information model performs better than the full information model in the estimation, measured by a higher logarithmic likelihood, -2946.57 for the FIRE estimate versus -2915.56 for the IIRE estimate.

##### A. FIRE versus IIRE: Non-policy shocks

Table 1 provides a direct comparison between the estimation results under FIRE and IIRE. Estimates are largely similar for most structural parameters and shock processes, including policy response coefficients. We find that the central bank is slightly more aggressive in its response to inflation deviations from its long-term steady state (measured by  $\phi_\pi$ ) under full information. The responses of fiscal variables to the debt target are higher under the FIRE estimation for government expenditure, but lower for tax rules.

It should be noted that, under both FIRE and IIRE models, the data point to a very persistent inflation target  $\pi_t^*$  driven by  $\varepsilon_t^U$ . The estimated  $\phi_F$  is close to 0.98 in both models. Since  $\phi_F$  largely governs the persistence of fiscal inflation, the following impulse responses show that inflation would not converge to zero in response to an unfunded fiscal shock, even after 20 years. The finding confirms one of the main results in Bianchi, Faccini and Melosi (2023), which also shows a very persistent fiscal inflation process.

Similar estimated posterior distributions for non-policy parameters across the two frameworks imply that the impulse responses of macroeconomic variables should be quantitatively similar for all non-policy shocks which are not subject to incomplete information. Appendix G presents the impulse responses of six key macro variables, output, inflation, nominal interest rate, real interest rate, expected inflation, and real marginal cost, to all non-policy shocks in the model. We present impulse response for three models- FIRE with no expectations in red,

<sup>7</sup>The results from IIRE estimation without short-term expectations and second sample estimation for all IIRE models (2008:Q1 to 2022:Q3) are reported in Appendix G.



| Parameter                                                   | Type | Prior |      | FIRE   |        |        |       | IIRE (with SPF) |        |        |       |
|-------------------------------------------------------------|------|-------|------|--------|--------|--------|-------|-----------------|--------|--------|-------|
|                                                             |      | Mean  | Std  | Mode   | Median | 5%     | 95%   | Mode            | Median | 5%     | 95%   |
| Debt to GDP $s_b$                                           | N    | 2.40  | 0.05 | 2.414  | 2.411  | 2.334  | 2.494 | 2.407           | 2.415  | 2.343  | 2.487 |
| SS growth $100\gamma$                                       | N    | 0.5   | 0.05 | 0.382  | 0.371  | 0.306  | 0.436 | 0.437           | 0.365  | 0.328  | 0.430 |
| SS inflation $100\Pi$                                       | N    | 0.5   | 0.05 | 0.519  | 0.457  | 0.421  | 0.549 | 0.531           | 0.491  | 0.437  | 0.520 |
| Inverse Frisch $\xi$                                        | G    | 2     | 0.25 | 2.463  | 2.537  | 2.504  | 2.590 | 2.306           | 2.156  | 2.098  | 2.243 |
| Non-savers $\mu$                                            | B    | 0.11  | 0.01 | 0.091  | 0.090  | 0.077  | 0.102 | 0.090           | 0.090  | 0.078  | 0.106 |
| Wage Calvo $\omega_w$                                       | B    | 0.5   | 0.1  | 0.692  | 0.667  | 0.644  | 0.693 | 0.708           | 0.721  | 0.701  | 0.741 |
| Price Calvo $\omega_p$                                      | B    | 0.5   | 0.1  | 0.712  | 0.716  | 0.695  | 0.737 | 0.734           | 0.738  | 0.712  | 0.759 |
| Cap util cost $\psi$                                        | B    | 0.5   | 0.1  | 0.788  | 0.706  | 0.678  | 0.764 | 0.676           | 0.664  | 0.631  | 0.705 |
| Invest adj cost                                             | N    | 6     | 0.5  | 5.900  | 5.963  | 5.884  | 6.044 | 5.363           | 5.295  | 5.221  | 5.371 |
| Wage index $\chi_w$                                         | B    | 0.5   | 0.2  | 0.031  | 0.023  | 0.013  | 0.031 | 0.054           | 0.033  | 0.026  | 0.039 |
| Price index $\chi_p$                                        | B    | 0.5   | 0.2  | 0.109  | 0.105  | 0.100  | 0.118 | 0.026           | 0.015  | 0.005  | 0.029 |
| Habits $\theta$                                             | B    | 0.5   | 0.2  | 0.927  | 0.925  | 0.915  | 0.932 | 0.913           | 0.915  | 0.905  | 0.924 |
| $C/G$ sub $\alpha_G$                                        | N    | 0     | 0.1  | -0.007 | -0.017 | -0.078 | 0.061 | -0.001          | -0.001 | -0.050 | 0.057 |
| Taylor rule coeff $\phi_\pi$                                | N    | 1.5   | 0.1  | 1.728  | 1.755  | 1.660  | 1.850 | 1.784           | 1.708  | 1.583  | 1.783 |
| TR coeff on $Y$ $\phi_y$                                    | N    | 0.5   | 0.1  | 0.002  | 0.002  | 0.000  | 0.007 | 0.001           | 0.001  | 0.000  | 0.002 |
| Debt response $\gamma_G$                                    | N    | 0.1   | 0.1  | 0.325  | 0.314  | 0.267  | 0.339 | 0.269           | 0.272  | 0.249  | 0.318 |
| Debt response $\gamma_{TK}$                                 | N    | 0.1   | 0.1  | 0.002  | 0.002  | 0.000  | 0.004 | 0.003           | 0.002  | 0.000  | 0.006 |
| Debt response $\gamma_{TL}$                                 | N    | 0.1   | 0.1  | 0.001  | 0.002  | 0.000  | 0.006 | 0.019           | 0.012  | 0.002  | 0.023 |
| Debt response $\gamma_z$                                    | N    | 0.1   | 0.1  | 0.001  | 0.002  | 0.000  | 0.006 | 0.001           | 0.002  | 0.000  | 0.004 |
| Output response $\phi_{z,y}$                                | G    | 0.2   | 0.05 | 0.191  | 0.181  | 0.153  | 0.234 | 0.178           | 0.141  | 0.127  | 0.149 |
| Output response $\phi_{g,y}$                                | G    | 0.2   | 0.05 | 0.186  | 0.193  | 0.184  | 0.198 | 0.196           | 0.196  | 0.185  | 0.206 |
| Infl. target strength $\Phi$                                | G    | 0.2   | 0.1  | 0.131  | 0.145  | 0.128  | 0.173 | 0.078           | 0.075  | 0.063  | 0.089 |
| <b>AR(1) Coefficients and Standard Deviations of Shocks</b> |      |       |      |        |        |        |       |                 |        |        |       |
| AR(1) MP-rule $\rho_r$                                      | B    | 0.5   | 0.1  | 0.739  | 0.739  | 0.712  | 0.768 | 0.695           | 0.688  | 0.651  | 0.717 |
| AR(1) G-rule $\rho_g$                                       | B    | 0.5   | 0.1  | 0.973  | 0.973  | 0.965  | 0.979 | 0.967           | 0.967  | 0.954  | 0.975 |
| AR(1) Z-rule $\rho_z$                                       | B    | 0.5   | 0.1  | 0.941  | 0.942  | 0.922  | 0.959 | 0.965           | 0.963  | 0.944  | 0.975 |
| AR(1) TK-rule $\rho_{TK}$                                   | B    | 0.5   | 0.1  | 0.930  | 0.936  | 0.923  | 0.959 | 0.928           | 0.926  | 0.909  | 0.943 |
| AR(1) TL-rule $\rho_{TL}$                                   | B    | 0.5   | 0.1  | 0.933  | 0.930  | 0.911  | 0.945 | 0.938           | 0.942  | 0.924  | 0.958 |
| AR(1) Infl. target $\phi_F$                                 | B    | 0.9   | 0.05 | 0.986  | 0.982  | 0.974  | 0.992 | 0.994           | 0.994  | 0.991  | 0.997 |
| Technology $\rho_a$                                         | B    | 0.5   | 0.1  | 0.297  | 0.295  | 0.263  | 0.325 | 0.155           | 0.167  | 0.146  | 0.190 |
| Preference $\rho_b$                                         | B    | 0.5   | 0.1  | 0.432  | 0.428  | 0.376  | 0.464 | 0.705           | 0.675  | 0.617  | 0.719 |
| Monetary $\rho_m$                                           | B    | 0.5   | 0.1  | 0.368  | 0.400  | 0.363  | 0.439 | 0.374           | 0.338  | 0.314  | 0.387 |
| Investment $\rho_i$                                         | B    | 0.5   | 0.1  | 0.849  | 0.814  | 0.791  | 0.839 | 0.857           | 0.871  | 0.852  | 0.889 |
| Risk Prem. $\rho_{rp}$                                      | B    | 0.5   | 0.1  | 0.882  | 0.872  | 0.854  | 0.889 | 0.808           | 0.812  | 0.786  | 0.831 |
| Cost-push $\rho_{cp}$                                       | B    | 0.9   | 0.05 | 0.997  | 0.997  | 0.994  | 0.999 | 0.998           | 0.997  | 0.994  | 0.999 |
| Gov spending $\sigma_g$                                     | I    | 0.5   | 0.2  | 1.685  | 1.736  | 1.572  | 1.799 | 1.680           | 1.752  | 1.601  | 1.813 |
| Technology $\sigma_a$                                       | I    | 0.5   | 0.2  | 1.093  | 1.135  | 1.069  | 1.199 | 1.065           | 1.045  | 0.993  | 1.108 |
| Preference $\sigma_b$                                       | I    | 0.5   | 0.2  | 5.000  | 4.998  | 4.996  | 5.000 | 4.990           | 4.992  | 4.980  | 4.999 |
| Monetary $\sigma_m$                                         | I    | 0.5   | 0.2  | 0.275  | 0.283  | 0.262  | 0.306 | 0.285           | 0.278  | 0.258  | 0.298 |
| Investment $\sigma_i$                                       | I    | 0.5   | 0.2  | 0.603  | 0.700  | 0.646  | 0.743 | 0.629           | 0.601  | 0.544  | 0.668 |
| Wage $\sigma_w$                                             | I    | 0.5   | 0.2  | 0.324  | 0.324  | 0.292  | 0.359 | 0.342           | 0.337  | 0.313  | 0.361 |
| Price markup $\sigma_p$                                     | I    | 0.5   | 0.2  | 0.190  | 0.192  | 0.165  | 0.213 | 0.180           | 0.172  | 0.154  | 0.217 |
| Risk prem. $\sigma_{rp}$                                    | I    | 0.5   | 0.2  | 0.411  | 0.413  | 0.374  | 0.492 | 0.507           | 0.490  | 0.459  | 0.548 |
| Cost-push $\sigma_{cp}$                                     | I    | 0.5   | 0.2  | 0.155  | 0.149  | 0.132  | 0.166 | 0.125           | 0.128  | 0.113  | 0.144 |
| Transfer $\sigma_z$                                         | I    | 0.5   | 0.2  | 3.016  | 2.911  | 2.807  | 2.977 | 3.090           | 3.196  | 3.137  | 3.273 |
| Unfunded $\sigma_U$                                         | I    | 0.5   | 0.2  | 0.375  | 0.363  | 0.336  | 0.409 | 0.422           | 0.362  | 0.333  | 0.392 |
| Capital tax $\sigma_{tk}$                                   | I    | 0.5   | 0.2  | 4.957  | 4.993  | 4.976  | 4.999 | 4.995           | 4.936  | 4.905  | 4.965 |
| Labor tax $\sigma_{tl}$                                     | I    | 0.5   | 0.2  | 2.669  | 2.813  | 2.762  | 2.877 | 2.688           | 2.562  | 2.511  | 2.645 |
| GDP M.E. $\sigma_y^{ME}$                                    | I    | 0.5   | 0.2  | 0.429  | 0.438  | 0.402  | 0.482 | 0.440           | 0.447  | 0.407  | 0.519 |
| Debt to GDP M.E. $\sigma_b^{ME}$                            | I    | 0.5   | 0.2  | 0.342  | 0.330  | 0.309  | 0.351 | 0.340           | 0.380  | 0.346  | 0.421 |
| Inf expectation M.E. $\pi_{ME}^e$                           | I    | 0.5   | 0.2  | 0.000  | 0.000  | 0.000  | 0.000 | 0.163           | 0.168  | 0.145  | 0.196 |

TABLE 1—PARAMETER ESTIMATES COMPARISON: FIRE vs. IIRE

**Notes:** Table reports the mode, median, and confidence intervals from first sample estimation under FIRE and IIRE. Prior Type indicate the prior density function where N, G, B, and I stand for Normal, Gamma, Beta, and Inverse-Gamma respectively. The posterior median and confidence intervals are computed using MCMC with 100,000 simulations and 50% burn-in.

IIRE with no expectations in blue, and IIRE with SPF expectations in black. Despite the information frictions in distinguishing between transitory price markup shock and persistent cost-push shock, the response of macro variables is largely similar under FIRE and IIRE with the exception of output. A transitory markup shock results in significantly higher output in the short-run in the incomplete information model. Since incomplete information is not introduced for any other shocks, responses to other shocks reveal the effects from differences in estimated structural parameters between FIRE and IIRE models. For these non-policy shocks, the effects on macro variables are nearly identical under both frameworks. A notable difference can be seen in the response to preference shock, which is estimated to be twice as persistent under IIRE (0.377 vs. 0.677), leading to a larger impact on inflation, expected inflation, and the real interest rate compared to FIRE.

#### B. FIRE versus IIRE: Funded and unfunded fiscal shocks

The main interest of our analysis lies with the behavior of the economy in response to an unfunded fiscal shock under FIRE and IIRE. Before discussing the impulse responses we highlight the key parameters that influence the response of inflation to an unfunded shock. The estimated Taylor rule coefficient on inflation,  $\phi_\pi$ , is largely similar across models but the persistence of interest rate,  $\rho_r$  is largest under FIRE and lowest under the IIRE model with SPF expectations. As  $\rho_r$  affects the composition of monetary policy signal received by agents, it lowers the response of inflation to both unfunded and monetary policy shock. Incorporating short run expectations using SPF also reduces the estimated level of accommodation of unfunded shock by the central bank. The estimate for coefficient  $\Phi$  is largest under FIRE at 0.145 relative to 0.09 under IIRE model without short-run expectations, and 0.075 under IIRE model with SPF expectations.

Figure 6 presents the impulse response for output, inflation, and the real interest rate, under a one standard deviation unfunded fiscal shock  $\varepsilon_t^U$  or a one standard deviation funded transfer shock  $\varepsilon_t^Z$ . The magnitude of the responses are percentage deviations from the steady state. Funded and unfunded fiscal shocks are clearly identified in the model, with the funded transfers shock having quantitatively negligible effects (less than 0.02%) on inflation and real interest rate under FIRE and both IIRE models.

The real interest rate decreases in response to the unfunded shock, raising both inflation and output under both FIRE and IIRE estimation. On impact, output responds similarly to both estimations, reaching the maximum response in about four years. In contrast, the response to inflation is large and strikingly different between the two information frameworks. On impact, the annualized inflation response is negligible under incomplete information. Under full information, inflation increases immediately, reaching a maximum of 0.47 percentage points above the steady-state in only two quarters. These differences emerge despite similar posterior distributions for all policy parameters under FIRE and

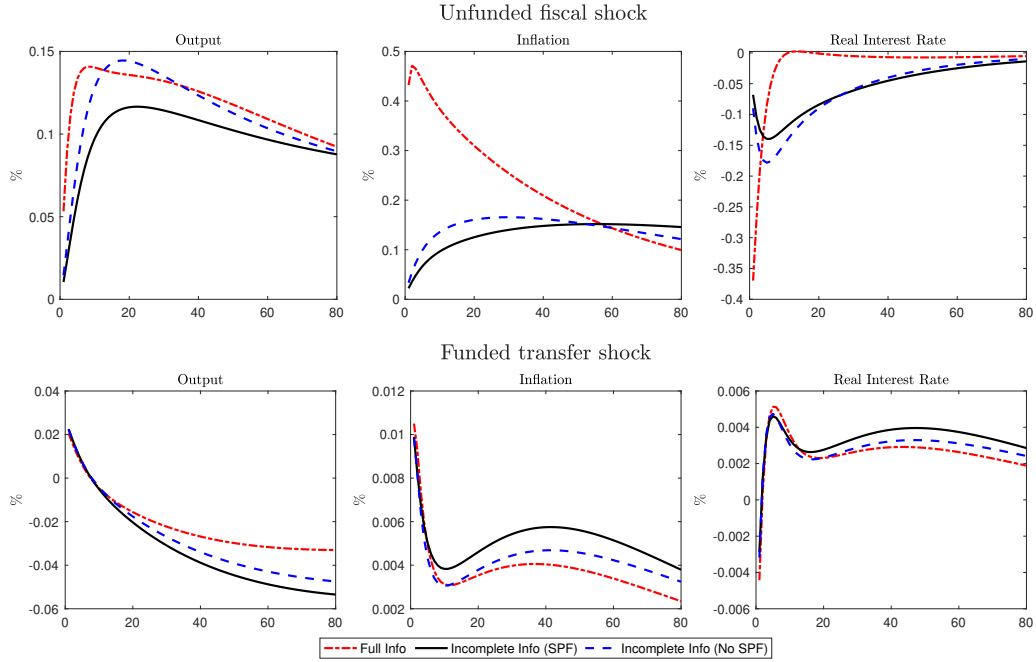


FIGURE 6. IMPULSE RESPONSES TO EXPANSIONARY FUNDED AND UNFUNDED FISCAL SHOCKS

**Notes:** Impulse response to unfunded fiscal shock (top panel) and funded transfer shock (bottom panel) for output, inflation (annualized), and real interest rate (annualized) under full information and incomplete information with and without survey expectations. The size of shocks is fixed at one standard deviation expansionary shock, as estimated in the first sample. The units of response are percentage deviations from steady state.

IIRE. Including expectations data in the estimation further lowers the response of inflation, output, and real interest rate to unfunded shocks in the model. In the long run, both output and inflation remain persistently above the steady state level.

Quantitatively, the response of inflation on impact under IIRE is near zero while inflation jumps sharply under the FIRE model where agents are able to perfectly observe the unfunded shock. As the nature of the shock gradually becomes clear under IIRE, inflation in response to an unfunded fiscal shock reaches a maximum of 0.15% and 0.17% with and without incorporating SPF expectations. Across parameter estimates at the 95% and 5% confidence intervals, the peak inflation response in the IIRE model (with expectations) remains between 65% and 75% lower than in the FIRE model. Overall, these results highlight that assuming full information leads to substantially larger estimated contributions of fiscal shocks to inflation, while introducing information frictions significantly attenuates these effects.

The difference in response of inflation lies in how an unfunded shock trans-

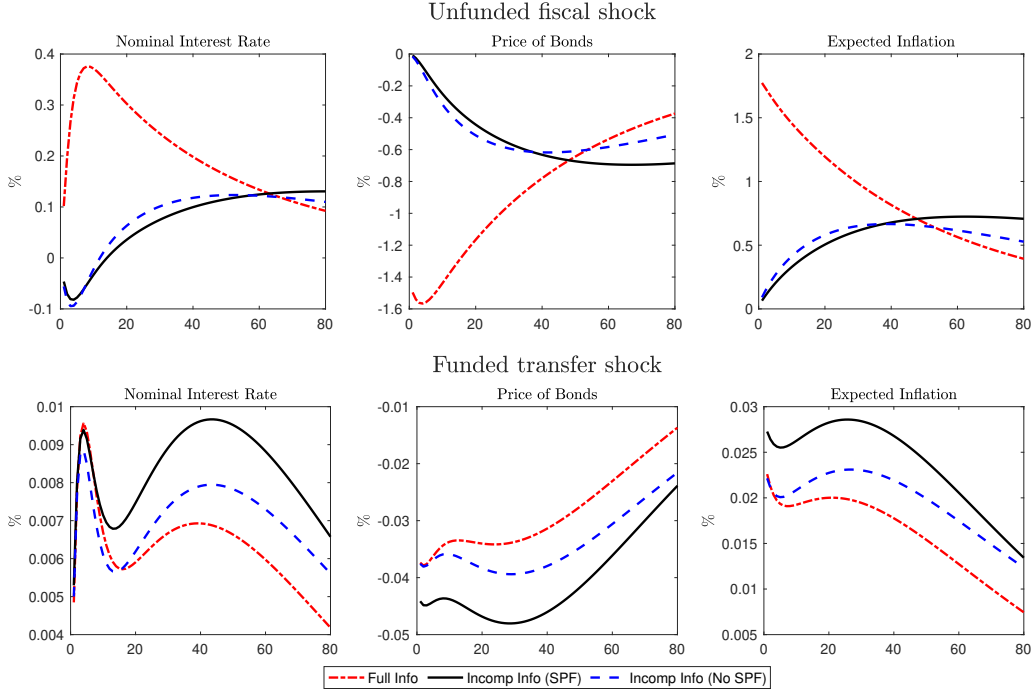


FIGURE 7. IMPULSE RESPONSES TO EXPANSIONARY FUNDED AND UNFUNDED FISCAL SHOCKS

**Notes:** Impulse response to unfunded fiscal shock (top panel) and funded transfer shock (bottom panel) for nominal interest rate (annualized), price of bonds, and expected inflation (annualized) under full information and incomplete information with and without survey expectations. The size of shocks are fixed at one standard deviation expansionary shock, as estimated in the first sample. The units of response are percentage deviations from steady state.

mits under full and incomplete information. Under full information, agents can perfectly separate unfunded shocks from other shocks to the economy. Consequently, inflation rises rapidly to pay for the unfunded shock. This increase in inflation is accommodated by monetary policy in the short run, as captured by the large decrease in real interest rate under FIRE. The interest rate increases sharply as the central bank reacts more aggressively to inflation deviations, with real interest rate reverting to the steady state in approximately two years. Under incomplete information, agents cannot immediately distinguish between a funded and an unfunded shock, attributing some proportion of the observed shock to each type. As in the simple model, a hump-shaped response to inflation once again emerges as agents acquire more information about the nature of the shock. Incorporating survey expectations further reduces the response of inflation in the short and medium run suggesting that professional forecasters attribute larger share of observed shock to funded fiscal shocks.

Figure 7 presents the impulse response functions for nominal interest rate, price

of bonds and expected inflation, to one standard deviation unfunded fiscal and funded transfers shocks. Similarly to Figure 6, the magnitude of the responses of all nominal variables is near 0 for an expansionary funded transfer shock, while the unfunded fiscal shock creates significant effects. The nominal interest rate increases with inflation under both FIRE and IIRE with the persistence of response being higher under IIRE. Combined with higher inflation and interest rate, the decrease in bond prices on impact is nearly four times as significant under FIRE than under IIRE. As the nature of the policy shock becomes clearer under IIRE, bond prices fall further, quantitatively reaching the same levels as FIRE but only after ten additional quarters. An unfunded shock not only affects current inflation but significantly raises five-year-forward inflation expectations. The expected inflation under FIRE is highest on impact and decays with time as the real value of debt decreases. The 5-year forward inflation expectation follows the same hump-shaped pattern as current inflation under incomplete information. The transmission mechanism remains the same. As economic agents learn about the nature of the fiscal shock over time, inflation expectations gradually increase for nearly five years after impact before slowly decaying.

In response to an unfunded fiscal shock, both the FIRE and IIRE models can generate hump-shaped impulse responses of the nominal interest rate; thus, they are equally successful in explaining a Federal Reserve that is seemingly “Behind the Curve”, which is not raising nominal interest rates fast enough to fight against high inflation. The feature is mainly driven by the interest rate inertia parameter  $\rho_r$  in the FIRE model, while information friction plays an additional role in the IIRE model. However, the impulse responses of the bond price and expected inflation provide more direct evidence in favor of the IIRE model. Although the COVID-19 outburst caused initial market turmoil and price drops (more so in the stock market than in the bond market) in early to mid-March 2020, when the fiscal stimulus packages were implemented, bond prices recovered quickly and bond yields stabilized. This movement in bond prices is in contrast to the FIRE model’s implication that there should be a large and immediate drop in bond prices if we assume a large share of unfunded fiscal shock in the CARES Act. Inflation expectations observed through various surveys and derived through market-based financial instruments in 2020 also show little support of the FIRE model and are more in agreement with the IIRE model’s hump-shaped impulse responses.

### C. Determinants of Inflation

We now assess the contribution of the model’s structural shocks to inflation across four sub-periods of the U.S. economy. Table 2 reports the average percentage share of total inflation explained by each shock within a given period. For ease of interpretation, we group all fiscal shocks other than the unfunded fiscal shock into a single category, *Funded Fiscal*, which includes government spending, funded transfers, capital tax, and labor tax shocks.

The first sub-sample, spanning 1968 to 1990, covers the high-inflation episodes of the 1970s and early 1980s, a period characterized by large fiscal expansions and repeated oil price shocks. Column 1 of Table 2 shows that, under the incomplete-information framework, unfunded fiscal shocks account for nearly 39% of observed inflation. Supply-side disturbances, captured by cost-push, price markup, and wage markup shocks, constitute the second-largest source of inflation, explaining approximately 18% of the total. Monetary policy also plays an important role during this period, contributing more than 11% of observed inflation.

Unfunded fiscal shocks remain an important source of inflation in subsequent periods. During the 1991–2007 sub-sample, they account for more than 23% of observed inflation, with their contribution rising to nearly 28% in the period between the Global Financial Crisis and the onset of the COVID recession in 2020. Monetary policy also becomes increasingly important during the post-crisis period, reflecting the prolonged accommodative stance adopted in response to weak economic conditions. According to the model, monetary policy accounts for approximately 20% of inflation between 2008 and early 2020.

|                 | (1)<br>1968:Q3-1990:Q4 | (2)<br>1991:Q1-2007:Q4 | (3)<br>2008:Q1-2020:Q1 | (4)<br>2020:Q2-2022:Q3 |
|-----------------|------------------------|------------------------|------------------------|------------------------|
| Unfunded Fiscal | 38.60                  | 23.41                  | 27.94                  | 27.58                  |
| Monetary Policy | 11.14                  | 6.97                   | 20.49                  | 30.30                  |
| Preference      | 17.39                  | 12.84                  | -9.66                  | 2.98                   |
| Funded Fiscal   | 9.35                   | 8.69                   | 8.34                   | 9.28                   |
| Investment      | -12.08                 | -2.12                  | 7.21                   | 28.57                  |
| Technology      | 8.27                   | 10.19                  | 14.14                  | 11.38                  |
| Cost-Push       | 3.32                   | 4.05                   | -2.55                  | -0.10                  |
| Price Markup    | 6.57                   | 9.61                   | 10.68                  | 6.95                   |
| Wage Markup     | 7.77                   | 9.93                   | 15.62                  | 29.53                  |
| Risk Premium    | 2.98                   | 6.59                   | -2.49                  | -54.67                 |

TABLE 2—AVERAGE CONTRIBUTION TO INFLATION (IN PERCENTAGE)

**Notes:** Table shows the average contribution (in percentage) of each shock to inflation in four sub-samples. Category “Funded fiscal” includes the combined contribution of four funded shocks within the model: government expenditure, funded transfer, capital tax, and labor tax shock. The table is constructed using estimates from the incomplete information model with short-run inflation expectations for sample 1 until 2007:Q4, and for sample 2 starting 2008:Q1.

Under the IIRE framework, a combination of unfunded shocks and supply shocks contributed to the sharp increase in inflation post-pandemic. The IIRE model with short-run inflation expectations identifies unfunded fiscal shock, monetary policy shock, investment adjustment shock, and transitory price and wage markup shocks as the key contributors of post-pandemic inflation. Consistent with the dynamics shown in the IRFs, the information frictions embedded in the model do not allow agents to perfectly distinguish between funded fiscal shocks, unfunded shocks, and monetary policy shock. The estimated model shows that

expansionary monetary policy shocks accounted for over 30% of inflation during the pandemic which is equivalent to the contribution of unfunded fiscal shocks. Transitory supply shocks, i.e., price markup and wage markup shocks, accounted for the largest fraction of inflation at nearly 35%. The global supply chain pressure index built by the New York Fed shows that the supply side pressures, including measures of manufacturing and transportation costs, are highly correlated with post-pandemic inflation, with the index peaking in December 2021. The index also registers a sharp decline in 2022, reaching the average levels by the beginning of 2023, likely easing inflationary pressure.<sup>8</sup> In the next section, we compare the historical decomposition of inflation across different model specifications and highlight the role played by expectations in disciplining the information frictions.

#### *D. Survey Expectations and Information Frictions*

Survey expectations play a central role in identifying the information friction that drives our decomposition of fiscal inflation. In the model, households and firms observe policy outcomes such as inflation, interest rates, transfers, spending, taxes, and debt, but they do not directly observe whether these outcomes are generated by funded fiscal shocks, unfunded fiscal shocks, monetary policy shocks, or supply disturbances. The key latent object is therefore the perceived unfunded component of observed fiscal expansions. This perceived component determines both model-implied inflation expectations and the contribution of unfunded fiscal shocks to realized inflation.

Realized macroeconomic aggregates alone discipline this latent object only imperfectly. The same path of realized inflation can be generated by a large unfunded fiscal shock that agents initially perceive only weakly, or by a smaller unfunded fiscal shock that agents perceive more strongly. Survey inflation expectations help break this near observational equivalence because they provide direct information about the perceived inflation path. In the estimation, we treat survey expectations as proxies for model-implied subjective expectations,

$$\pi_{t+h|t}^{survey} = E_t^{HH} \pi_{t+h} + \nu_t^s.$$

By pinning down  $E_t^{HH} \pi_{t+h}$ , survey data discipline the Kalman-filter inference problem faced by private agents and restrict the weight they assign to the perceived unfunded fiscal shock. This discipline is particularly important for the post-COVID inflation episode. The 2020–2021 fiscal transfers were unprecedented in size, but survey expectations did not immediately rise in proportion to the subsequent inflation surge. Within our model, this pattern implies that agents did not initially interpret the fiscal expansion as fully unfunded. As agents gradually learn about the nature of the underlying shocks, the inflationary effects of un-

<sup>8</sup>The Global Supply Chain Pressure Index can be viewed on the NY Fed’s website: <https://www.newyorkfed.org/research/policy/gscpi#/interactive>.

funded fiscal policy become more persistent but less front-loaded than under full information.

We now examine how this belief discipline affects the historical decomposition of inflation. The comparison shows how incorporating SPF and MSC expectations changes the estimated contribution of unfunded fiscal shocks relative to a specification estimated without survey expectations. If survey expectations are informative about agents' beliefs, then adding them should constrain the model from attributing too much of post-COVID inflation to immediately perceived unfunded fiscal shocks.

Figure 8 illustrates how the estimated contribution of unfunded fiscal shocks to inflation depends on the expectations data used in estimation. To examine the role of expectations in disciplining the model, we re-estimate the incomplete-information framework using one-year-ahead inflation expectations from the Michigan Survey of Consumer Sentiment (MSC).<sup>9</sup> Panels A and B report results for the first sample (1968–2007), while Panels C and D present results for the second sample (2008–2022).

The motivation for this exercise comes from the ongoing debate over the drivers of post-pandemic inflation. Bianchi, Faccini and Melosi (2023) argue that unfunded fiscal shocks account for more than 100% of observed inflation, with non-policy shocks exerting offsetting deflationary pressures. In contrast, Smets and Wouters (2024) emphasize the dominant role of supply shocks and assign only a limited role to unfunded fiscal policy. Our framework suggests that part of this disagreement may stem from differing assumptions about the information available to economic agents. Under full information, agents immediately recognize fiscal transfers as unfunded, generating a sharp inflationary response. Under incomplete information, however, households and firms must infer whether observed fiscal expansions are temporary, funded interventions or persistent unfunded shocks. This distinction becomes especially important during the COVID period, when inflation expectations initially remained relatively subdued despite unprecedented fiscal stimulus.

Panels A and C provide an external validation exercise by comparing the decomposition from the FIRE model in Bianchi, Faccini and Melosi (2023) with the decomposition from our incomplete-information model estimated without survey expectations data. For most of the sample, the two models generate broadly similar decompositions. The differences emerge primarily during the post-COVID period. Whereas Bianchi, Faccini and Melosi (2023) attribute most of the inflation surge to unfunded fiscal transfers, the incomplete-information model assigns

<sup>9</sup>The Michigan Survey data on one-year-ahead household inflation expectations are publicly available beginning in 1978. The survey asks households to forecast CPI inflation, whereas our model uses inflation measured by the GDP deflator. As a result, MSC expectations are not a perfect match for the inflation concept in the model. Given these data limitations, we only incorporate MSC expectations in the second sample, while the first sample is estimated without survey expectations data. Parameter estimates are reported in Appendix Section G. The estimated measurement error for short-run inflation expectations is also substantially larger when using MSC data than when using SPF expectations (0.44 versus 0.25).



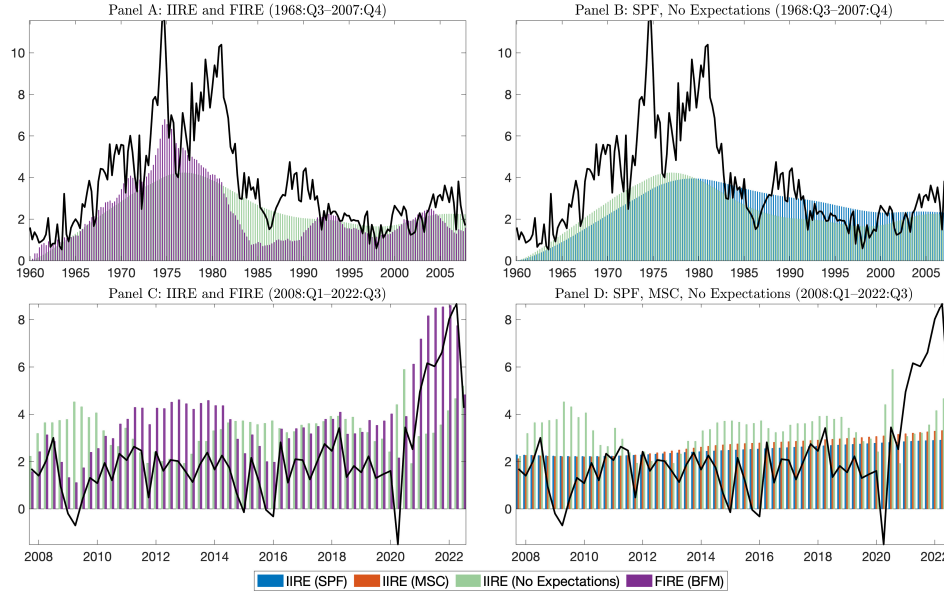


FIGURE 8. FISCAL INFLATION UNDER DIFFERENT MODEL SPECIFICATIONS

**Notes:** This figure compares the contribution of unfunded fiscal shock to inflation under different model specifications. Panels A and C compare the incomplete information model with no short-run expectations with the decomposition in Bianchi, Faccini and Melosi (2023). Panel B and D compare incomplete information model with no short-run expectations to models including survey data from SPF and MSC as observables. Panels A and B show the decomposition in our first sample (1968:Q3 to 2007:Q4) and Panels C and D cover the period between 2008:Q1 and 2022:Q3. The solid black line is the annualized inflation as measured by growth in GDP deflator.

a substantially larger role to supply shocks. Quantitatively, at the inflation peak in 2022, the FIRE model attributes roughly 20% more inflation to unfunded fiscal shocks and approximately 50% less inflation to supply shocks relative to the IIRE specification. These results suggest that the assumption of full information overstates the inflationary effects of unfunded fiscal policy while understating the contribution of supply disruptions. More broadly, the IIRE framework shows that informational frictions are central to reconciling the differing conclusions of Bianchi, Faccini and Melosi (2023) and Smets and Wouters (2024).

Panels B and D then show how introducing survey expectations further disciplines the model's interpretation of inflation dynamics. We compare three incomplete-information specifications: a model estimated without survey expectations data, our baseline model estimated using inflation expectations from the Survey of Professional Forecasters (SPF), and a model estimated using MSC expectations. Incorporating survey data anchors the model-implied path of inflation expectations and therefore constrains how much inflation can be attributed to unfunded fiscal shocks. Both household and professional forecaster surveys indicate

that agents did not initially expect the 2020–2021 stimulus payments to generate a large and immediate increase in inflation (Figure 1). Within the model, this pattern implies that agents initially perceived the fiscal transfers as largely funded or temporary. Additionally, a smaller estimate for  $\Phi$  in the model with SPF expectations suggests that the central bank accommodates a lower fraction of fiscal shock with higher short-run inflation, thus, resulting in lower contribution of unfunded shock to inflation. As a result, models estimated with survey expectations assign a smaller role to unfunded fiscal shocks during the post-pandemic inflation episode.

Without survey expectations data, the IIRE model attributes nearly 60% of observed post-COVID inflation to unfunded fiscal shocks. Incorporating survey-based expectations halves this contribution. Because household inflation expectations in the MSC rose more sharply in 2021 than those of professional forecasters, the MSC-based specification attributes a marginally larger share of inflation to unfunded fiscal shocks relative to the SPF-based model. Nevertheless, both specifications imply a smaller role for unfunded fiscal shocks than the IIRE model estimated without expectations data.

#### *E. Monetary-Fiscal Interaction under Information Frictions*

A novel feature of our model is the interaction between monetary and fiscal policy in the presence of information frictions. The interaction is twofold: First, the unfunded shock drives both the short-term debt and inflation targets. Second, since economic agents cannot perfectly distinguish between funded and unfunded fiscal shocks, an exogenous monetary policy shock (e.g., an interest rate cut) can signal an increase in the central bank’s short-term inflation target, leading to a higher and more persistent inflation response compared to the full-information (FIRE) case.

In response to the COVID pandemic, the Federal Reserve cut its target for the federal funds rate by a total of 1.5 percentage points at its meetings on March 3 and 15, 2020. These cuts reduced the funds rate to a range of 0% to 0.25%. The timing of the interest-rate cuts coincided with the CARES Act, which was signed into law on March 27. We now highlight the point that the interest rate cut may also contribute to higher inflation under incomplete information. Figure 9 shows the impulse response of output, inflation, and interest rate to an exogenous rate cut  $\varepsilon_1^m = -\sigma_m$ . Inflation and output increase, while real interest rates fall under both FIRE and IIRE. However, under IIRE, the effects on inflation are notably stronger. Annualized inflation increases by nearly 0.48 percentage points on impact and remains elevated. Incorporating SPF survey expectations lowers the contribution of monetary policy shock to inflation marginally to 0.42 percentage points. In contrast, under FIRE the inflationary response is more muted, with inflation increasing by only 0.34 percentage points initially. Quantitatively, the peak of the inflation response under FIRE is between 17% and 21% lower than under IIRE at the 5% and 95% credible sets of the parameter estimates. As agents

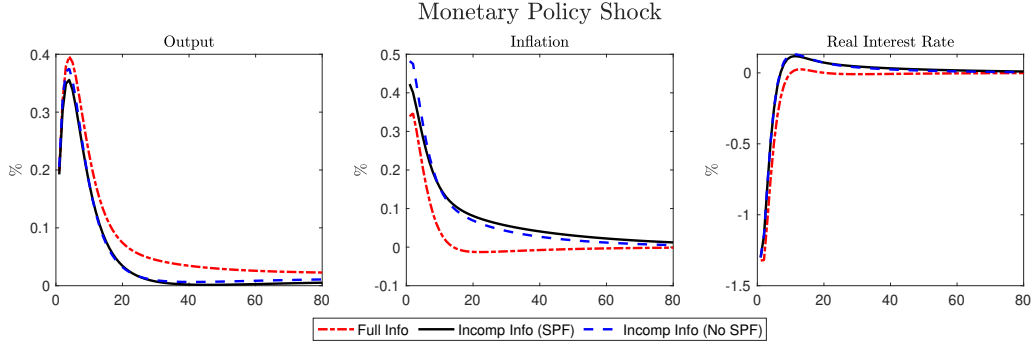


FIGURE 9. IMPULSE RESPONSES TO EXPANSIONARY MONETARY POLICY SHOCK

**Notes:** Impulse response to monetary policy shock for three variables: output, inflation, and real interest rate, under full information and incomplete information estimation with and without survey expectations. The size of shocks are fixed at one standard deviation expansionary shock, as estimated in the first sample. The units are percentage deviations from steady-state.

learn to distinguish between monetary policy shocks and unfunded fiscal shocks, inflation under IIRE gradually declines but remains above FIRE in the long run. Table 2 shows that expansionary monetary policy post-COVID accounted for nearly 30% of inflation under incomplete information. Quantitatively, the response of real interest rate is similar under both models while monetary policy is more expansionary under FIRE.

#### F. Tax multipliers: Funded versus unfunded

We now shift our focus to quantifying the response to tax shocks in our model. Recall that incomplete information affects all policy shocks, including shocks to labor and capital tax rate. Figure 10 reports the impulse responses of output, inflation, and the debt-to-GDP ratio to a one-standard-deviation expansionary capital or labor tax shock (i.e., a tax cut) with incomplete information. For comparison, we also present results for the unfunded fiscal shock. Despite the information frictions, funded capital and labor tax cuts have only negligible effects on inflation compared to the effects under the unfunded shock. When both tax cuts are funded, the real debt-to-GDP ratio increases steadily, stabilizing at about 0.6 percentage points above its steady state. By contrast, in response to an unfunded fiscal shock, output increases persistently and the real interest rate decreases, lowering the borrowing cost (Figure 6), resulting in a persistent decrease in the debt-to-GDP ratio.

In the short run, an expansionary capital tax shock in the model generates a significant increase in output that is 25% higher than that of an unfunded fiscal shock at its peak. To capture the dynamics of output over time in response to capital and labor tax shocks, we compute the present value multipliers (Table

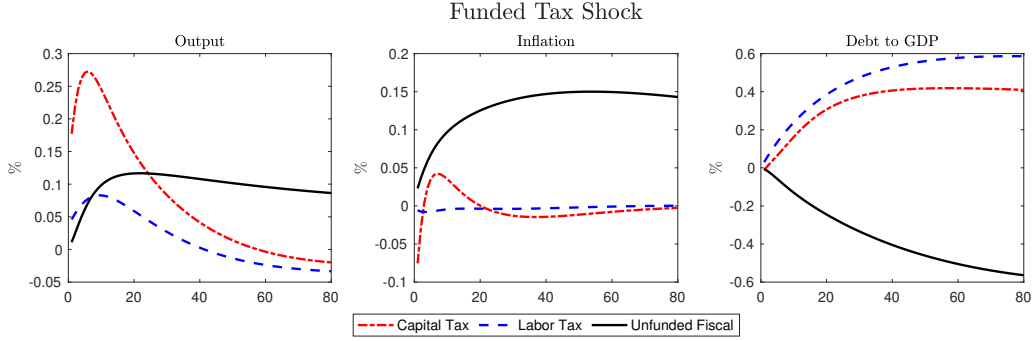


FIGURE 10. IMPULSE RESPONSES TO EXPANSIONARY FUNDED LABOR TAX, CAPITAL TAX, AND UNFUNDED FISCAL SHOCK

**Notes:** Impulse response to funded labor tax, capital tax, and unfunded fiscal shock for three variables: output, inflation, and debt to GDP ratio, under incomplete information estimation (with survey expectations). The size of shocks are fixed at one standard deviation expansionary shock, as estimated in the first sample. The units are percentage deviations from steady-state.

3) following Leeper, Traum and Walker (2011).<sup>10</sup> As funded tax cuts increase output, their multipliers are negative. Capital tax shocks generate an output multiplier of  $-0.5$  on impact and a present value multiplier greater than one after 10 quarters. The multiplier for the labor tax shock is modest:  $-0.15$  on impact and a maximum multiplier of  $0.4$  reached after 25 quarters. Our findings are consistent with the empirical literature on fiscal multipliers, which shows that capital tax cuts are more expansionary than labor tax cuts (Romer and Romer, 2010).

| Variable              | Impact | 5 quarters | 10 quarters | 25 quarters |
|-----------------------|--------|------------|-------------|-------------|
| Labor Tax Shock       | -0.147 | -0.223     | -0.302      | -0.422      |
| Capital Tax Shock     | -0.504 | -0.833     | -1.097      | -1.479      |
| Unfunded Fiscal Shock | 4.524  | 4.632      | 4.764       | 5.067       |

TABLE 3—PRESENT VALUE MULTIPLIERS OF LABOR TAX, CAPITAL TAX, UNFUNDED FISCAL SHOCK

Since the unfunded fiscal shock also enters the tax rules through the debt target, the government can also implement tax cuts supported by the unfunded fiscal shock. A positive unfunded shock reduces tax rates and increases output, leading to positive multipliers. Although the expansionary effect on output is small in the short run, the total tax revenue also responds only weakly to an unfunded

<sup>10</sup>Present value multipliers are calculated as the ratio of the present value of changes in output to the present value of changes in total tax revenue:  $E_t \frac{\sum_{j=0}^k (\Pi_{i=0}^j R_{t+i}^{-1}) \Delta Y_{t+j}}{\sum_{j=0}^k (\Pi_{i=0}^j R_{t+i}^{-1}) \Delta T_{t+j}^{total}}$ .

shock, resulting in a tax multiplier of 4.5 on impact. In the long run, an unfunded tax cut can significantly increase output and reduce the debt burden, leading to a very large multiplier around 5. The size of the unfunded fiscal multiplier is quantitatively similar to the deficit-financed tax multiplier documented in Mountford and Uhlig (2009). Their maximum deficit-financed tax multiplier, peaking at 5.25 in the 12th quarter, is close to our 25-quarter unfunded fiscal shock multiplier, 5.01.

A caveat remains in place: inflation remains persistently high and does not converge to zero, even after 20 years. While higher public debt together with future primary surpluses finance the funded tax cuts, it is certainly not the case for the unfunded tax cuts. Essentially, it is higher inflation, which dilutes real debt, that finances the unfunded tax cuts.

## V. Conclusion

This paper presents a novel informational interaction of monetary and fiscal policy within the larger class of DSGE models to analyze fiscal inflation. We achieve this through the lens of incomplete information between funded and unfunded fiscal shocks. The interaction between fiscal and monetary policy is achieved through augmented monetary and fiscal policy rules with a pair of hidden inflation and debt targets that vary over time. The unfunded fiscal shock drives both short-term targets, altering policy implications.

We embed the augmented policy rules and information frictions into a medium-scale New Keynesian model and estimate it using U.S. macroeconomic data. To discipline information frictions in the model, we incorporate survey measures of short-run inflation expectations and compare the estimated contribution of fiscal policy to inflation across alternative specifications. The inflation decomposition reveals that unfunded fiscal shocks account for more than one-quarter of observed inflation during both the 1970s and post-COVID, while serving as a countervailing force against disinflationary pressures during the 2008 financial crisis. At the same time, incorporating survey expectations data substantially reduces the estimated contribution of unfunded fiscal shocks to inflation. Instead, supply-side disturbances and expansionary monetary policy accounted for a large share of observed inflation, together explaining more than half of the inflation observed post-COVID. These findings suggest that information frictions are crucial for reconciling competing explanations for the surge in inflation observed in the U.S. and globally following the pandemic.

Throughout the paper, we follow the existing literature and utilize a shadow economy to discipline the central bank's accommodation to fiscal inflation. It remains to ask: Given a slow-moving and ever-increasing debt burden, what is the optimal monetary policy to accommodate the unfunded fiscal shocks? We focus on the incomplete information of the private sector and do not model the central bank's expectations to unfunded shocks explicitly. Under what circumstances should the central bank address the issue of fiscal sustainability? Should the

central bank wait and see, or take a preemptive approach? We leave these policy issues for future research.

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## APPENDIX

### RECOVERING AMPF AND PMAF DYNAMICS IN LEEPER (1991)

This section recovers the classic *Active Monetary-Passive Fiscal* (AMPF) and *Passive Monetary-Active Fiscal* (PMAF) inflation dynamics of the simple model in Leeper (1991). The only shock in the economy is the fiscal shock  $\eta_t$ . The equilibrium conditions are

$$\begin{aligned}
 (A1) \quad & \text{Fisher equation:} & i_t &= \mathbb{E}_t^{HH} \pi_{t+1} \\
 (A2) \quad & \text{Monetary policy:} & i_t &= \phi_\pi \pi_t \\
 (A3) \quad & \text{Gov. budget constraint:} & s_{b,t} &= \beta^{-1} [s_{b,t-1} + i_{t-1} - \pi_t - (1 - \beta)s_t] \\
 (A4) \quad & \text{Fiscal policy:} & s_t &= \gamma s_{b,t-1} + \eta_t
 \end{aligned}$$

Denote the inflation dynamics  $\pi_t = \Pi(L)\eta_t = \sum_{j=0}^{\infty} \Pi_j \eta_t$ . We utilize frequency domain techniques (see Kasa, Walker and Whiteman (2014)) to derive the functional form of  $\Pi^{FI}(L)$ . Combining the Fisher equation (A1) and the Taylor rule (A2) and applying the Wiener-Kolmogorov formula lead to

$$L^{-1} [\Pi(L) - \Pi_0] = \phi_\pi \Pi(L).$$

Rearranging the terms yields the following results

$$(A5) \quad \Pi(L) = \frac{\Pi_0}{1 - \phi_\pi L}.$$

If  $\phi_\pi > 1$  (i.e. active monetary policy), the above equation defines a stationary inflation process if and only if  $\Pi_0 = 0$ . It follows  $\Pi(L) = 0$  and  $\pi_t = 0$  under active monetary policy. Combining the government budget constraint (A3) and the fiscal rule (A4) and plugging in  $i_t = 0, \pi_t = 0$  yield

$$(A6) \quad s_{b,t} = \beta^{-1} [1 - (1 - \beta)\gamma] s_{b,t-1} - \beta^{-1} [(1 - \beta)] \eta_t.$$

If  $0 < \beta^{-1} [1 - (1 - \beta)\gamma] < 1$ , or equivalently,  $\gamma > 1$  (that is, passive fiscal policy), then  $s_{b,t}$  defined by (A6) is always a stationary process. We summarize the AMPF inflation and debt dynamics as

$$(A7) \quad \textbf{AMPF:} \quad \pi_t = 0, \quad s_{b,t} = -\frac{\beta^{-1}(1 - \beta)}{1 - \beta^{-1} [1 - (1 - \beta)\gamma] L} \eta_t, \quad \text{when } \phi_\pi > 1, \gamma > 1.$$

If  $0 < \phi_\pi < 1$  (i.e., passive monetary policy), then equation (A5) always defines a stationary AR(1) inflation process, where  $\Pi_0$  is a free parameter. Since  $i_t =$

$\phi_\pi \Pi_0 / (1 - \phi_\pi L) \varepsilon_t$ ,  $\pi_t = \Pi_0 / (1 - \phi_\pi L) \varepsilon_t$ , it follows that

$$i_{t-1} - \pi_t = \frac{\Pi_0 \phi_\pi L}{1 - \phi_\pi L} \eta_t - \frac{\Pi_0}{1 - \phi_\pi L} \eta_t = -\Pi_0 \eta_t.$$

Combining the government budget constraint (A3) and the fiscal rule (A4) and plugging in  $i_{t-1} - \pi_t = \Pi_0 \eta_t$  yield

$$(A8) \quad s_{b,t} = \beta^{-1} [1 - (1 - \beta)\gamma] s_{b,t-1} - \beta^{-1} [\Pi_0 + (1 - \beta)] \eta_t.$$

If  $0 < \gamma < 1$  so that  $\beta^{-1} [1 - (1 - \beta)\gamma] > 1$  (that is, active fiscal policy), we can rewrite equation (A8) using the lag operator  $L$  as

$$(A9) \quad (1 - \beta^{-1} [1 - (1 - \beta)\gamma] L) s_{b,t} = -\beta^{-1} [\Pi_0 + (1 - \beta)] \eta_t$$

Evaluating the above equation at  $L = z_1 = \beta [1 - (1 - \beta)\gamma_\tau]^{-1}$  pins down

$$\Pi_0 = -(1 - \beta),$$

leading to PMAF inflation and debt dynamics as

$$(A10) \quad \textbf{PMAF:} \quad \pi_t = \frac{\beta - 1}{1 - \phi_\pi L} \eta_t, \quad s_{b,t} = 0, \quad \text{when } 0 < \phi_\pi < 1, 0 < \gamma < 1.$$

#### DERIVING THE ANALYTICAL INFLATION PROCESS

The mapping between the household's observables and the underlying shocks is

$$(B1) \quad \begin{bmatrix} f_t^* \\ \pi_t^* \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \frac{1}{1-\rho L} \\ 0 & \mathcal{P}(L) \end{bmatrix}}_{M(L)} \begin{bmatrix} \varepsilon_t^F \\ \varepsilon_t^U \end{bmatrix}$$

We have assumed that  $z = \lambda$  is the only root of  $\mathcal{P}(z)$  inside of  $z \in (-1, 1)$ . Following Rozanov (1967)'s terminology, the existence of such a root implies the representation (B1) is non-fundamental to the household, and a fundamental representation needs to be derived before forming households' expectations. A fundamental representation consists of a fundamental mapping, denoted by  $M^*(L)$ , and its associated fundamental innovations. To derive  $M^*(L)$ , the principle is to utilize the Blaschke factor matrix to flip the root of  $\mathcal{P}(z)$  from  $|z| < 1$  to  $|z| > 1$ . Let  $\mathcal{B}_\lambda(L)$  be

$$(B2) \quad \mathcal{B}_\lambda(L) = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1-\lambda L}{\lambda-L} \end{bmatrix}$$

The fundamental mapping  $M^*(L)$  is given by

$$(B3) \quad M^*(L) = M(L)\mathcal{W}_\lambda\mathcal{B}_\lambda(L),$$

where  $\mathcal{W}_\lambda$  is an orthonormal matrix given by

$$\mathcal{W}_\lambda = \begin{bmatrix} B_1 & A_1 \\ B_2 & A_2 \end{bmatrix} = \frac{1}{\sqrt{\sigma_U^2 + (1 - \rho\lambda)^2\sigma_F^2}} \begin{bmatrix} \sigma_F(1 - \rho\lambda) & -\sigma_U \\ \sigma_U & \sigma_F(1 - \rho\lambda) \end{bmatrix}.$$

Given  $M^*(L)$ , the associated fundamental innovations are

$$(B4) \quad \begin{bmatrix} \tilde{\varepsilon}_t^F \\ \tilde{\varepsilon}_t^U \\ \tilde{\varepsilon}_t \end{bmatrix} = \mathcal{B}_\lambda(L^{-1})\mathcal{W}_\lambda' \begin{bmatrix} \varepsilon_t^F \\ \varepsilon_t^U \\ \varepsilon_t \end{bmatrix}$$

Calculating  $\mathcal{B}_\lambda(L)$  and  $\mathcal{W}_\lambda$  gives

$$(B5) \quad M^*(L) = M(L) \begin{bmatrix} B_1 & A_1 \frac{1-\lambda L}{\lambda-L} \\ B_2 & A_2 \frac{1-\lambda L}{\lambda-L} \end{bmatrix},$$

$$(B6) \quad \begin{bmatrix} \tilde{\varepsilon}_t^F \\ \tilde{\varepsilon}_t^U \\ \tilde{\varepsilon}_t \end{bmatrix} = \begin{bmatrix} B_1 & B_2 \\ A_1 \frac{\lambda-L}{1-\lambda L} & A_2 \frac{\lambda-L}{1-\lambda L} \end{bmatrix} \begin{bmatrix} \varepsilon_t^F \\ \varepsilon_t^U \\ \varepsilon_t \end{bmatrix}$$

Combining the Fisher equation and the monetary policy rule gives a

$$(B7) \quad \pi_t - \pi_t^* = \phi_\pi^{-1} \mathbb{E}_t^{HH} \pi_{t+1}$$

Under FIRE, it can be shown that the inflation process  $\pi_t^{FI}$  is given by a Hansen-Sargent formula

$$(B8) \quad \pi_t^{FI} = \frac{\mathcal{P}(\phi_\pi^{-1}) - \phi_\pi L \mathcal{P}(L)}{1 - \phi_\pi L} \varepsilon_t^U.$$

Denote the incomplete information inflation process

$$(B9) \quad \pi_t^{II} = \Pi_F(L)\varepsilon_t^F + \Pi_U(L)\varepsilon_t^U.$$

We now utilize the fundamental representation to derive  $\mathbb{E}_t^{HH} \pi_{t+1}$ . It follows that

$$\begin{aligned} \mathbb{E}_t^{HH} \pi_{t+1} &= L^{-1} [\Pi_F(L)B_1 + \Pi_U(L)B_2 - (\Pi_F(0)B_1 + \Pi_U(0)B_2)] \tilde{\varepsilon}_t^F \\ &\quad + L^{-1} \left[ \frac{1-\lambda L}{\lambda-L} (\Pi_F(L)A_1 + \Pi_U(L)A_2) - \frac{1}{\lambda} (\Pi_F(0)A_1 + \Pi_U(0)A_2) \right] \tilde{\varepsilon}_t^U \\ &= L^{-1} [\Pi_F(L) - \Pi_F(0)] \varepsilon_t^F + L^{-1} [\Pi_U(L) - \Pi_U(0)] \varepsilon_t^U \\ &\quad + \frac{1-\lambda^2}{\lambda(1-\lambda L)} [\Pi_F(0)A_1 + \Pi_U(0)A_2] (A_1 \varepsilon_t^F + A_2 \varepsilon_t^U) \end{aligned}$$

Plugging the above  $\mathbb{E}_t^{HH} \pi_{t+1}$  into (B7) and matching the coefficients in front of  $\varepsilon_t^F$  lead to

$$L\Pi_F(L) = \phi_\pi^{-1} [\Pi_F(L) - \Pi_F(0)] + \phi_\pi^{-1} \frac{1 - \lambda^2}{\lambda(1 - \lambda L)} [\Pi_F(0)A_1 + \Pi_U(0)A_2] A_1 L$$

Imposing  $L = \phi_\pi^{-1}$  on both sides gives the first restriction

$$(B10) \quad \phi_\pi \Pi_F(0) = \frac{1 - \lambda^2}{\lambda(1 - \lambda \phi_\pi^{-1})} [\Pi_F(0)A_1 + \Pi_U(0)A_2] A_1.$$

It follows that

$$\Pi_F(L) = \frac{\Pi_F(0)}{1 - \lambda}.$$

Similarly, matching the coefficients in front of  $\varepsilon_t^U$  in (B7) yields

$$L\Pi_U(L) - L\mathcal{P}(L) = \phi_\pi^{-1} [\Pi_U(L) - \Pi_U(0)] + \phi_\pi^{-1} \frac{1 - \lambda^2}{\lambda(1 - \lambda L)} [\Pi_F(0)A_1 + \Pi_U(0)A_2] A_2 L$$

Imposing  $L = \phi_\pi^{-1}$  on both sides leads to the second restriction

$$(B11) \quad \phi_\pi [\Pi_U(0) - \mathcal{P}(\phi_\pi^{-1})] = \frac{1 - \lambda^2}{\lambda(1 - \lambda \phi_\pi^{-1})} [\Pi_F(0)A_1 + \Pi_U(0)A_2] A_2$$

It follows that

$$\Pi_U(L) = \frac{\mathcal{P}(\phi_\pi^{-1}) - \phi_\pi L\mathcal{P}(L)}{1 - \phi_\pi L} + [\Pi_U(0) - \mathcal{P}(\phi_\pi^{-1})] \frac{1}{1 - \lambda L}.$$

Finally, it should be noted that  $\Pi_F(0)$  and  $\Pi_U(0)$  are not free parameters. Instead, they are the unique solution to the linear system (B10)-(B11).

#### THE SHADOW ENDOWMENT ECONOMY WITH $S_t = T^* - Z_t$

Consistent with Bianchi, Faccini and Melosi (2023), this section considers a shadow economy in which the only shock is the unfunded transfer shock  $\varepsilon_t^U$ . The equilibrium conditions are

$$\begin{aligned} \text{Fisher equation:} & \quad i_t = \mathbb{E}_t^{HH} \pi_{t+1} \\ \text{Monetary policy:} & \quad i_t = \phi_F \pi_t \\ \text{Gov. budget constraint:} & \quad s_{b,t} = \beta^{-1} [s_{b,t-1} + i_{t-1} - \pi_t - (1 - \beta)s_t] \\ \text{Primary surplus:} & \quad s_t = -\frac{Z^*}{S^*} z_t \\ \text{Unfunded transfers:} & \quad z_t = -\gamma_z s_{b,t-1} + \varepsilon_t^U \end{aligned}$$

Denote  $\pi_t^F$  as the inflation that would arise in the shadow economy in which the fiscal-led regime is always in place (i.e., we restrict  $0 < \frac{Z^*}{S^*}\gamma_z < 1$ ). Repeating the PMAF derivation as in Appendix A implies

$$\pi_t^F = \frac{1 - \beta}{1 - \phi_F L} \frac{Z^*}{S^*} \varepsilon_t^U.$$

Consequently, the inflation target implied by the shadow economy is

$$\pi_t^* = \left(1 - \frac{\phi_F}{\phi_\pi}\right) \pi_t^F = \left(1 - \frac{\phi_F}{\phi_\pi}\right) \frac{1 - \beta}{1 - \phi_F L} \frac{Z^*}{S^*} \varepsilon_t^U.$$

#### THE MEDIUM-SCALE DSGE MODEL ENVIRONMENT

##### D1. Households

There are two types of households in the economy, and their measures sum up to one. Among these households, a fraction of  $\mu$  are hand-to-mouth consumers, and the remaining  $1 - \mu$  are savers.

##### SAVERS

Savers, each indexed by  $j$ , derive utility from the consumption of a composite good  $C_t^{*S}(j)$ , which comprises private consumption  $C_t^S(j)$  and government consumption  $G_t$  such that  $C_t^{*S}(j) = C_t^S(j) + \alpha_G G_t$ . The parameter  $\alpha_G$  governs the substitutability between private and government consumption. When  $\alpha_G$  is negative (positive), these goods are complements (substitutes). External consumption habits imply that the utility is derived relative to the previous period value of aggregate saver consumption of the composite good  $hC_{t-1}^{*S}$ , where  $h \in [0, 1]$  is the habit parameter. Saver households also derive disutility from the supply of differentiated labor services from all their members, indexed by  $l$ ,  $L_t^S(j) = \int_0^1 L_t^S(j, l) dl$ . The period utility function is given by

$$(D1) \quad U_t^S(j) = u_t^d \left( \ln (C_t^{*S}(j) - hC_{t-1}^{*S}) - \frac{L_t^S(j)^{1+\chi}}{1+\chi} \right),$$

where  $u_t^d$  is a discount factor shock, and  $1/\chi$  is the Frisch elasticity of the labor supply.

Savers accumulate wealth in the form of physical capital  $\bar{K}_t^S$ . The law of motion for physical capital is given by

$$(D2) \quad \bar{K}_t^S(j) = (1 - \delta) \bar{K}_{t-1}^S(j) + u_t^i \left[ 1 - s \left( \frac{I_t^S(j)}{I_{t-1}^S(j)} \right) \right] I_t^S(j),$$

where  $\delta$  is the depreciation rate,  $u_t^i$  is a shock to the marginal efficiency of in-

vestment and  $s$  denotes an investment adjustment cost function that satisfies the properties  $s(e^\varkappa) = s'(e^\varkappa) = 0$  and  $s''(e^\varkappa) > 0$ , where  $\varkappa$  is a drift parameter capturing the logarithm of the growth rate of technology in steady state.

Households derive income from renting effective capital  $K_t^S(j)$  to intermediate firms. Effective capital is related to physical capital according to the following law of motion,

$$(D3) \quad K_t^S(j) = \nu_t(j) \bar{K}_{t-1}^S,$$

where  $\nu_t(j)$  is the capital utilization rate. In steady state, the utilization rate  $\nu(j)$  is 1. The cost of utilizing one unit of physical capital is given by the function  $\Psi(\nu_t(j))$  that satisfies the following properties:  $\Psi(1) = 0$  and  $\frac{\Psi''(1)}{\Psi'(1)} = \frac{\psi}{1-\psi}$ , where  $\psi \in [0, 1)$ . We denote the gross capital rental rate as  $R_{K,t}$  and the capital rental income tax rate as  $\tau_{K,t}$ .

The household can also save by purchasing one-period government bonds in zero net supply and a more general portfolio of long-term government bonds in non-zero net supply. The one-period bonds promising a nominal payoff  $B_t$  at time  $t+1$  can be purchased at the present discounted value  $R_{n,t}^{-1}B_t$ , where  $R_{n,t}$  is the gross nominal interest rate set by the central bank. The long-term bond  $B_t^m$  mimics a portfolio of bonds with average maturity  $m$  and duration  $(1 - \beta\rho)^{-1}$ , where  $\rho \in [0, 1]$  is a constant decay rate. This bond can be purchased at a price  $P_t^m$ , determined by the arbitrage condition

$$R_{n,t} = \mathbb{E}_t^{HH}[(1 + P_{t+1}^m)/P_t^m] e^{-u_t^{rp}},$$

where the wedge  $u_t^{rp}$  can be interpreted as a risk premium shock.

In each period, households receive nominal labor income after tax, after-tax revenues from renting capital to firms, lump-sum transfers from the government ( $Z_t^S$ ), and dividends from firms ( $D_t$ ). Households spend to consume and invest in physical capital and bonds. Omitting the index  $j$  to simplify the notation, we can write the nominal budget constraint for the saver household as

$$(D4) \quad \begin{aligned} & P_t(1 + \tau_{C,t})C_t^S + P_t I_t^S + P_t^m B_t^m + R_{n,t}^{-1}B_t \\ &= (1 + \rho P_t^m) B_{t-1}^m + B_{t-1} + (1 - \tau_{L,t}) \int_0^1 W_t(l) L_t^S dl \\ &+ (1 - \tau_{K,t}) R_{K,t} \nu_t \bar{K}_{t-1}^S - \Psi(\nu_t) \bar{K}_{t-1}^S + P_t Z_t^S + D_t, \end{aligned}$$

where  $W_t(l)$  denotes the wage rate faced by all household members, and  $\tau_{C,t}$  and  $\tau_{L,t}$  denote the tax rates on consumption and labor income, respectively. The household maximizes the expected utility  $\mathbb{E}_0^{HH} \sum_{t=0}^{\infty} \beta^t U_t^S$  subject to the sequence of budget constraints in equation (D4) and the law of motion of capital accumulation (D2).

For ease of notation, we drop the index  $j$  in the following. The first-order optimality conditions (FOCs) concerning consumption, labor supply, one-period bond, investment, capital, and capital utilization are

$$(D5) \quad (\partial C_t^{*S}) \quad \Lambda_t^S = u_t^d (C_t^{*S} - hC_{t-1}^{*S})^{-1}$$

$$(D6) \quad (\partial L_t^S) \quad L_t^{S,\chi} = \Lambda_t^S (1 - \tau_{L,t}) \frac{W_t^h}{P_t}$$

$$(D7) \quad (\partial B_t) \quad \Lambda_t^S = \beta R_{nt} \mathbb{E}_t^{HH} \left[ \frac{\Lambda_{t+1}^S}{\pi_{t+1}} \right]$$

$$(D8) \quad (\partial I_t) \quad 1 = Q_t^k \mu_t \left[ 1 - s \left( \frac{I_t}{I_{t-1}} \right) - s' \left( \frac{I_t}{I_{t-1}} \right) \frac{I_t}{I_{t-1}} \right] \\ + \beta \mathbb{E}_t^{HH} \left\{ \frac{\Xi_{t+1}^K}{\Lambda_t^S} \left[ Q_{t+1} u_{t+1}^i s' \left( \frac{I_{t+1}}{I_t} \right) \left( \frac{I_{t+1}}{I_t} \right)^2 \right] \right\}$$

$$(D9) \quad (\partial \bar{K}_t) \quad Q_t = \beta \mathbb{E}_t^{HH} \left\{ \frac{\Xi_{t+1}^K}{\Lambda_t^S} [(1 - \tau_{K,t}) r_{K,t} \nu_{t+1} - \Psi(\nu_{t+1}) + Q_{t+1}(1 - \delta)] \right\}$$

$$(D10) \quad (\partial \nu_t) \quad (1 - \tau_{K,t}) r_{K,t} = \Psi'(\nu_t)$$

where  $\Lambda_t^S$  and  $\Xi_{t+1}^K$  are the Lagrange multipliers associated with the budget and capital accumulation constraints, respectively, and  $Q_t = \frac{\Xi_t^K}{\Lambda_t^S}$  is the Tobin's  $Q$  and equals one in the absence of adjustment costs.

#### HAND-TO-MOUTH HOUSEHOLDS

In every period, hand-to-mouth households derive disposable, after-tax income from labor supply and government transfers and consume all of them. They provide differentiated labor services and set their wage equal to the average wage that is optimally chosen by savers, as described below. Hand-to-mouth households face the same tax rates on consumption and labor income as savers. The specification of period-by-period utility for hand-to-mouth households is the same as that of savers, that is,

$$(D11) \quad U_t^N(j) = u_t^d \left( \ln (C_t^{*N}(j) - hC_{t-1}^{*N}) - \frac{L_t^N(j)^{1+\chi}}{1+\chi} \right).$$

Their budget constraint is

$$(D12) \quad (1 + \tau_{C,t}) P_t C_t^N = (1 - \tau_{L,t}) \int_0^1 W_t(l) L_t^N(l) dl + P_t Z_t^N,$$

where the superscript  $N$  indicates the variables for hand-to-mouth households.

Hand-to-mouth households maximize the discounted utility  $\mathbb{E}_0^{HH} \sum_{t=0}^{\infty} \beta^t U_t^N$

subject to the budget constraint (D12). The F.O.Cs are

$$(D13) \quad (\partial(C_t^{*N})) \quad \Lambda_t^N = u_t^d (C_t^{*N} - hC_{t-1}^{*S})^{-1},$$

$$(D14) \quad (\partial L_t^N) \quad L_t^{S,\chi} = \Lambda_t^N (1 - \tau_{L,t}) \frac{W_t}{P_t}.$$

## D2. Firms

### FINAL GOOD PRODUCERS

There is a perfectly competitive sector of final good firms that produces the homogeneous good  $Y_t$  in time  $t$  by combining a unit measure of intermediate differentiated inputs using aggregation technology

$$(D15) \quad Y_t = \left( \int_0^1 Y_t(i)^{\frac{1}{1+\eta_t^p+u_t^p}} di \right)^{1+\eta_t^p+u_t^p},$$

where  $\eta_t^p$  is the *i.i.d.* price markup shock. The variable  $u_t^p$  is a cost push shock and is assumed to follow a near-unit-root process. The highly persistent cost-push shock captures other external forces, such as international trade, that can generate low-frequency movements of inflation. Profit maximization yields the demand function for intermediate goods as

$$(D16) \quad Y_t(i) = Y_t \left( \frac{P_t(i)}{P_t} \right)^{-\frac{1+\eta_t^p+u_t^p}{\eta_t^p+u_t^p}},$$

where  $P_t(i)$  is the price of the differentiated good  $i$  and  $P_t$  is the price of the final good.

### INTERMEDIATE GOOD PRODUCERS

There is a unit measure of intermediate firms that produce goods according to the production function

$$(D17) \quad Y_t(i) = K_t(i)^\alpha (A_t L_t(i))^{1-\alpha} - A_t \Omega,$$

where  $\Omega$  is a fixed cost of production that grows with the rate of labor-augmenting technological progress  $A_t$ , and  $\alpha \in [0, 1]$  is the capital share. The labor-augmenting technological progress,  $A_t$ , follows an exogenous process that is stationary in its growth rate. Specifically, we assume that  $a_t = \ln(A_t/A_{t-1}) - \varkappa = u_t^a$ . Intermediate firms rent capital and labor from perfectly competitive capital and labor markets, respectively. As described in the following,  $L_t$  is a bundle of all the differentiated labor services supplied in the economy, aggregated into a homogeneous input by a labor agency. Intermediate firms' cost minimization implies the



same nominal marginal cost for all firms

$$(D18) \quad MC_t = (1 - \alpha)^{\alpha-1} \alpha^{-\alpha} (R_{K,t})^\alpha W_t^{1-\alpha} A_t^{-1+\alpha}.$$

Intermediate producers reset prices in the spirit of the Calvo pricing. At time  $t$ , a firm  $i$  can optimally reset its price with probability  $\omega_p$ . Otherwise, it adjusts the price with partial indexation to the previous period's inflation rate according to the rule

$$(D19) \quad P_t(i) = (\Pi_{t-1})^{\xi_p} (\Pi)^{1-\xi_p} P_{t-1}(i),$$

where  $\xi_p \in [0, 1]$  is a parameter,  $\Pi_{t-1} = P_{t-1}/P_{t-2}$ , and  $\Pi$  denotes the aggregate rate of inflation at steady state. Intermediate producers that are allowed to reset their price maximizes the expected discounted stream of nominal profits,

$$(D20) \quad \max \mathbb{E}_t^{Firm} \sum_{s=0}^{\infty} (\beta \omega_p)^s \frac{\Lambda_{t+s}^S}{\Lambda_t^S} \left[ \left( \prod_{k=1}^s \Pi_{t+k-1}^{\xi_p} \Pi^{1-\xi_p} \right) P_t(i) Y_{t+s}(i) - MC_{t+s} Y_{t+s}(i) \right],$$

subject to the demand function (D16), with  $\Lambda_t^S$  denoting the marginal utility of the savers.

The FOC is given by

$$(D21) \quad \max \mathbb{E}_t^{Firm} \sum_{s=0}^{\infty} (\beta \omega_p)^s \frac{\Lambda_{t+s}^S}{\Lambda_t^S} Y_{t+s}(i) \left[ \frac{-1}{\eta_t^p + u_t^p} X_{t,s}^P P_t(i) + \frac{1 + \eta_t^p + u_t^p}{\eta_t^p + u_t^p} MC_{t+s} \right] = 0,$$

where

$$(D22) \quad X_{t,s}^P = \begin{cases} 1 & \text{for } s = 0 \\ \left( \prod_{k=1}^s \Pi_{t+k-1}^{\xi_p} \Pi^{1-\xi_p} \right) & \text{for } s = 1, \dots, \infty \end{cases}$$

### D3. Wage Settings

Both savers and hand-to-mouth households supply a unit measure of differentiated labor service indexed by  $l$ . In each period, a saver household has probability  $\omega_w$  to optimally re-adjust the wage rate that applies to all of its workers,  $W_t(l)$ . If the wage cannot be re-optimized, it will be increased at the geometric average of the steady-state rate of inflation  $\Pi$  and of last period inflation  $\Pi_{t-1}$ , according to the rule

$$(D23) \quad W_t(l) = W_{t-1}(l) (\Pi_{t-1} e^{\varkappa})^{\xi_w} \left( \Pi e^{\xi} \right)^{1-\xi_w},$$

where  $\xi_w \in [0, 1]$  captures the degree of nominal wage indexation. All households, including both savers and non-savers, sell their labor service to a representative,

competitive agency that transforms it into an aggregate labor input, according to the technology

$$(D24) \quad L_t = \left( \int_0^1 L_t(l)^{\frac{1}{1+\eta_t^w}} dl \right)^{1+\eta_t^w},$$

where  $\eta_t^w$  is an *i.i.d.* exogenous wage markup shock. The agency rents labor type  $L_t(l)$  at a price  $W_t(l)$  and sells a homogeneous labor input to the intermediate producers at a price  $W_t$ . The static profit maximization problem yields the labor demand function

$$(D25) \quad L_t(l) = L_t (W_t(l)/W_t)^{-(1+\eta_t^w)/\eta_t^w}.$$

Labor unions use this marginal rate of substitution as the cost of labor services in their negotiations with labor packers. The markup above the marginal disutility is distributed to the households. For those that can adjust, the problem is to choose a wage  $W_t(l)$  that maximizes the discounted total wage income in the future subject to (D23) and (D25),

$$(D26) \quad \max \mathbb{E}_t \sum_{s=0}^{\infty} \omega_w^s \beta^s \frac{\Lambda_{t+s}^S P_t}{\Lambda_t^S P_{t+s}} \left[ W_{t+s}(l) - W_{t+s}^h \right] L_{t+s}(l).$$

The FOC becomes

$$(D27) \quad \mathbb{E}_t \sum_{s=0}^{\infty} \omega_w^s \beta^s \frac{\Lambda_{t+s}^S P_t}{\Lambda_t^S P_{t+s}} L_{t+s}(l) \left[ \left( X_{t,s}^W W_t(l) - W_{t+s}^h \right) \left( -\frac{1 + \eta_{w,t+s}}{\eta_{w,t+s}} \right) - X_{t,s}^W W_t(l) \right] = 0,$$

where

$$(D28) \quad X_{t,s}^W = \begin{cases} 1 & \text{for } s = 0 \\ \left( \prod_{k=1}^s \Pi_{t+k-1}^{\xi^w} \Pi^{1-\xi^w} \right) & \text{for } s = 1, \dots, \infty \end{cases}$$

#### D4. Monetary and Fiscal Policy

We have assumed that the government supplies one-period bonds in zero net supply, and both types of households receive the same amount of transfers. It follows that the government's nominal budget constraint is

$$(D29) \quad P_t^m B_t^m + \tau_{K,t} R_{K,t} K_t + \tau_{L,t} W_t L_t + \tau_{C,t} P_t C_t = (1 + \rho P_t^m) B_{t-1}^m + P_t G_t + P_t Z_t,$$

where  $C_t$  and  $Z_t$  denote aggregate consumption and total transfers, respectively. Their expressions are the following

$$(D30) \quad C_t = \mu C_t^N + (1 - \mu) C_t^S,$$

$$(D31) \quad Z_t = \int_0^1 Z_t(j) dj.$$

The budget constraint (D29) implies that the fiscal authority finances government expenditures, transfers, and the rollover of expiring long-term debt by issuing new long-term debt obligations as well as by raising labor, capital, and consumption taxes.

The aggregate resource constraint is given by

$$(D32) \quad Y_t = C_t + I_t + G_t + \Psi_t(\nu_t) \bar{K}_{t-1}.$$

We rescale the variables entering the fiscal rules as  $g_t = G_t/A_t$  and  $z_t = Z_t/A_t$ . For each variable  $x_t$ ,  $\hat{x}_t$  denotes the percentage deviation from its balanced-growth steady state.

Let  $s_{b,t} = (P_t^m B_t^m)/(P_t Y_t)$  denote the real market debt-to-GDP ratio. The fiscal authority adjusts government spending  $\hat{g}_t$ , transfers  $\hat{z}_t$ , and tax rates on capital income, labor income, and consumption  $\hat{\tau}_J$ ,  $J \in \{k, l, c\}$  as follows:

$$(D33) \quad \hat{g}_t = \rho_G \hat{g}_{t-1} - (1 - \rho_G) [\gamma_G (\hat{s}_{b,t-1} - \hat{s}_{b,t}^*) + \phi_{g,y} \hat{y}_t] + \varepsilon_t^g,$$

$$(D34) \quad \hat{z}_t = \rho_Z \hat{z}_{t-1} - (1 - \rho_Z) [\gamma_Z (\hat{s}_{b,t-1} - \hat{s}_{b,t}^*) + \phi_{z,y} \hat{y}_t] + \varepsilon_t^z,$$

$$(D35) \quad \hat{\tau}_{J,t} = \rho_J \hat{\tau}_{J,t-1} + (1 - \rho_J) \gamma_J [\hat{s}_{b,t-1} - \hat{s}_{b,t}^*] + u_t^J, \text{ for } J \in \{k, l\};$$

$$(D36) \quad \hat{\tau}_{c,t} = \rho_c \hat{\tau}_{c,t-1};$$

where  $\gamma_G$ ,  $\gamma_Z$ , and  $\gamma_J > 0$  are large enough to guarantee that the debt remains on a stable path. The time-varying debt target,  $\hat{s}_{b,t}^*$ , follows a stationary AR(1) process

$$\hat{s}_{b,t}^* = \rho_s \hat{s}_{b,t-1}^* + \varepsilon_t^U, \quad \varepsilon_t^U \sim N(0, \sigma_U^2),$$

with  $\rho_s \in (0, 1)$ .  $\varepsilon_t^U$  is the unfunded fiscal shock that increases the government's debt target.

Finally, the central bank follows the Taylor rule and adjusts the short-term interest rate  $R_t^n$  in response to inflation deviations and the output gap. The linearized monetary policy rule is the following:

$$(D37) \quad \hat{r}_t^n = \rho_r \hat{r}_{t-1}^n + (1 - \rho_r) [\phi_\pi (\hat{\pi}_t - \hat{\pi}_t^*) + \phi_y \hat{y}_t] + u_t^m$$

where  $u_t^m$  is the monetary policy shock and  $\phi_\pi > 1$  satisfies the Taylor principle. We introduce a time-varying inflation target which follows a stationary AR(1) process, that is

$$\hat{\pi}_t^* = \phi_F \hat{\pi}_{t-1}^* + \Phi \varepsilon_t^U, \quad \Phi > 0$$

### D5. Specifying Shocks and Introducing Incomplete Information

Throughout the paper, we use the notation  $u_t^x$  to denote a persistent shock and  $\eta_t^x$  to denote a transitory shock. For each  $x \in \{d, i, rp, p, a, g, z, m\}$ , we specify a stationary AR(1) process for the persistent shock  $u_t^x$  as

$$u_t^x = \rho_x u_{t-1}^x + \varepsilon_t^x, \quad \varepsilon_t^x \sim N(0, \sigma_x^2)$$

with  $\rho_x \in (0, 1)$ . We let the transitory shock  $\eta_t^p \sim N(0, \sigma_{\eta,p}^2)$  and  $\epsilon_t^U \sim N(0, \sigma_U^2)$  follow an *i.i.d.* process.

We now introduce incomplete information by assuming all households, intermediate firms, and final good firms share the same information set, denoted by  $\mathcal{I}_t^{HH}$ . We assume agents can observe the entire histories of nominal interest rate  $\{r_{t-k} | k \geq 0\}$  and inflation  $\{\pi_{t-k} | k \geq 0\}$ . The households can also observe history of surpluses  $\{\tau_{t-k} | k \geq 0\}$ , the realized real market debt  $\{s_{b,t-k} | k \geq 0\}$ , and the histories of labor, capital, and consumption tax rates  $\{\tau_{J,t-k} | k \geq 0\}$ . However, households cannot distinguish between the exogenous shock to fiscal and monetary policy and the shock to time-varying debt and inflation targets.

There are now four shocks to fiscal and monetary policy rules, but only three signals and incomplete information arise naturally in the model. The monetary policy rule (D37) and the fiscal rules (D33) and (D34) indicate that policy variables can also serve as signals to households. For instance, rewriting the monetary rule (D37) as

$$\hat{r}_t^n - \rho_r \hat{r}_{t-1}^n - (1 - \rho_r) (\phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t) = \underbrace{-(1 - \rho_r) \phi_\pi \pi_t^* + u_t^m}_{s_t^m}$$

indicates the history of the right side variables,  $s_t^m$ , is also known to households. Similarly, rewriting the fiscal rules (D33) and (D34) as

$$\begin{aligned} \hat{g}_t - \rho_G \hat{g}_{t-1} + (1 - \rho_G) [\gamma_G \hat{s}_{b,t-1} + \phi_{g,y} \hat{y}_t] &= \underbrace{(1 - \rho_G) \gamma_G \hat{s}_{b,t}^* + \varepsilon_t^g}_{s_t^g} \\ \hat{z}_t - \rho_Z \hat{z}_{t-1} + (1 - \rho_Z) [\gamma_Z \hat{s}_{b,t-1} + \phi_{z,y} \hat{y}_t] &= \underbrace{(1 - \rho_Z) \gamma_Z \hat{s}_{b,t}^* + \varepsilon_t^z}_{s_t^z} \end{aligned}$$

suggests that the two right side variables,  $s_t^g$  and  $s_t^z$ , are also in the household's information set. Additionally, agents cannot distinguish between shocks to time-varying debt targets and shocks to the tax rules. The tax rules (D35) can also be rewritten with signals where  $s_t^J$  are in the household's information set. We do not include shock to consumption tax rate in our estimation.

$$\hat{\tau}_{J,t} - \rho_J \hat{\tau}_{J,t-1} - (1 - \rho_J) \gamma_J \hat{s}_{b,t-1} = \underbrace{-(1 - \rho_J) \gamma_J \hat{s}_{b,t}^* + \varepsilon_t^j}_{s_t^J} \quad \forall J \in \{k, l\}$$

Finally, the cost-push shock signal is a composite of transitory and persistent components:

$$\hat{t}p_t = \underbrace{\hat{u}_t^p + \hat{\eta}_t^p}_{s_t^p}$$

We do not introduce incomplete information to any other shocks in the model and assume that they can be observed perfectly by households. These shocks are intended to improve the empirical fit of the medium-scale DSGE model. Formally, we define the incomplete information set  $\mathcal{I}_t^{HH}$  as

$$\mathcal{I}_t^{HH} = \{s_{t-k}^m, s_{t-k}^g, s_{t-k}^z, s_{t-k}^k, s_{t-k}^l, s_{t-k}^p, \mathcal{M} | k \geq 0\}.$$

#### D6. Deriving the log-linearized equilibrium conditions

To make the model stationary, we de-trend the non-stationary variables, accounting for the unit root in the labor-augmenting technology process. We define the following variables:  $y_t = \frac{Y_t}{A_t}$ ,  $c_t^* = \frac{C_t^{*S}}{A_t}$ ,  $c_t^S = \frac{C_t^S}{A_t}$ ,  $c_t^N = \frac{C_t^N}{A_t}$ ,  $k_t = \frac{K_t}{A_t}$ ,  $g_t = \frac{G_t}{A_t}$ ,  $z_t = \frac{Z_t}{A_t}$ ,  $b_t = \frac{P_t^m B_t^m}{P_t A_t}$ ,  $s_{b,t} = \frac{P_t^m B_t^m}{P_t Y_t}$ ,  $w_t = \frac{W_t}{P_t A_t}$ , and  $\lambda_t^S = \Lambda_t^S A_t$ . In what follows,  $e^\pi$  denotes the steady-state growth of the technology process. That is,  $e^a = e^\pi$ .

Production function:

$$(D38) \quad \hat{y}_t = \frac{y + \Omega}{y} \left[ \alpha \hat{k}_t + (1 - \alpha) \hat{L}_t \right].$$

Capital-labor ratio:

$$(D39) \quad \hat{r}_{K,t} - \hat{w}_t = \hat{L}_t - \hat{k}_t.$$

Marginal cost:

$$(D40) \quad \hat{m}_t = \alpha \hat{r}_{K,t} + (1 - \alpha) \hat{w}_t.$$

Phillips curve:

$$(D41) \quad \hat{\pi}_t = \frac{\beta}{1 + \xi_p \beta} \mathbb{E}_t \hat{\pi}_{t+1} + \frac{\xi_p}{1 + \xi_p \beta} \hat{\pi}_{t-1} + \kappa_p \hat{m}_t + \kappa_p \hat{t}p_t,$$

where  $\kappa_p = [(1 - \beta\omega_p)(1 - \omega_p)] / [\omega_p(1 + \beta\xi_p)]$ .

Saver household's FOC for consumption:

$$(D42) \quad \hat{\lambda}_t^S = \hat{u}_t^d - \frac{h}{e^z - h} \hat{u}_t^a - \frac{e^z}{e^z - h} \hat{c}_t^{*S} + \frac{h}{e^z - h} \hat{c}_{t-1}^{*S} - \frac{\tau^C}{1 + \tau^C} \hat{\tau}_t^C$$

Cost-Push shock Process:

$$(D43) \quad \hat{t}p_t = \hat{u}_t^p + \hat{\eta}_t^p$$

Public/private consumption in utility:

$$(D44) \quad \hat{c}_t^* = \frac{c^S}{c^S + \alpha_G g} \hat{c}_t^S + \frac{\alpha_G g}{c^S + \alpha_G g} \hat{g}_t$$

Euler equation:

$$(D45) \quad \hat{\lambda}_t^S = \hat{r}_{n,t} + \mathbb{E}_t \hat{\lambda}_{t+1}^S - \mathbb{E}_t \hat{\pi}_{t+1} - \mathbb{E}_t \hat{u}_{t+1}^a + \hat{u}_t^{rp}$$

The maturity structure of debt is

$$(D46) \quad \hat{r}_{n,t} + \hat{P}_t^m = \frac{P_m}{1 + P_m} \mathbb{E}_t \hat{P}_{t+1}^m - \hat{u}_t^{rp}$$

where  $P_m = \frac{\rho}{R_n - \rho}$ .

Saver household's FOC for capacity utilization:

$$(D47) \quad \hat{r}_{K,t} - \frac{\tau_K}{1 - \tau_K} \hat{r}_{K,t} = \frac{\psi}{1 - \psi} \hat{v}_t.$$

Saver household's FOC for capital:

$$(D48) \quad \hat{q}_t = \mathbb{E}_t \hat{\pi}_{t+1} - \hat{r}_{n,t} + \beta e^{-\varkappa} (1 - \tau_K) r_K \mathbb{E}_t \hat{r}_{K,t+1} - \beta e^{-\varkappa} \tau_K r_K \mathbb{E}_t \hat{r}_{K,t+1} + \beta e^{-\varkappa} (1 - \delta) \mathbb{E}_t \hat{q}_{t+1} - \hat{u}_t^{rp}$$

Saver household's FOC for investment:

$$(D49) \quad \hat{i}_t = \frac{1}{(1 + \beta) s e^{2\varkappa}} \hat{q}_t + \hat{u}_t^i + \frac{\beta}{1 + \beta} \mathbb{E}_t \hat{i}_{t+1} - \frac{1}{1 + \beta} \hat{u}_t^a + \frac{\beta}{1 + \beta} \mathbb{E}_t \hat{u}_{t+1}^a + \frac{1}{1 + \beta} \hat{i}_{t-1}$$

Effective capital:

$$(D50) \quad \hat{k}_t = \hat{v}_t + \hat{k}_{t-1} - \hat{u}_t^a.$$

Law of motion for capital:

$$(D51) \quad \hat{\hat{k}}_t = (1 - \delta) e^{-\varkappa} \left( \hat{\hat{k}}_{t-1} - \hat{u}_t^a \right) + \left[ 1 - (1 - \delta) e^{-\varkappa} \right] \left[ (1 + \beta) s e^{2\varkappa} \hat{u}_t^i + \hat{i}_t \right].$$

Hand-to-mouth household's budget constraint:

$$(D52) \quad \tau_C C_N \hat{\tau}_{C,t} + (1 + \tau_C) C_N \hat{C}_t^N = (1 - \tau_L) w L (\hat{w}_t + \hat{L}_t) - \tau_L w L \hat{\tau}_{L,t} + z \hat{z}_t.$$

Wage equation:

$$(D53) \quad \begin{aligned} \hat{w}_t &= \frac{1}{1 + \beta} \hat{w}_{t-1} + \frac{\beta}{1 + \beta} \mathbb{E}_t \hat{w}_{t+1} - \kappa_w \left[ \hat{w}_t - \chi \hat{L}_t - \hat{d}_t + \lambda_t^S - \frac{\tau_L}{1 - \tau_L} \hat{\tau}_{L,t} \right] + \frac{\chi w}{1 + \beta} \hat{\pi}_{t-1} \\ &- \frac{1 + \beta \chi w}{1 + \beta} \hat{\pi}_t + \frac{\beta}{1 + \beta} \mathbb{E}_t \hat{\pi}_{t+1} + \frac{\chi w}{1 + \beta} \hat{u}_{t-1}^a - \frac{1 + \beta \chi w - \rho_a \beta}{1 + \beta} \hat{u}_t^a + \kappa_w \eta_t^w \end{aligned}$$

$$\text{where } \kappa_w \equiv [(1 - \beta \omega_w)(1 - \omega_w)] / \left[ (1 + \beta) \omega_w \left( 1 + \frac{(1 + n^w) \xi}{\eta^w} \right) \right].$$

Aggregate households' consumption:

$$(D54) \quad c \hat{c}_t = c^S (1 - \mu) \hat{c}_t^S + c^N \mu \hat{c}_t^N.$$

Aggregate resource constraint:

$$(D55) \quad y \hat{y}_t = c \hat{c}_t + i \hat{i}_t + g \hat{g}_t + \psi'(1) k \hat{k}_t.$$

Government budget constraint:

$$(D56) \quad \begin{aligned} &\frac{b}{y} \hat{b}_t + \tau_K r_K \frac{k}{y} [\hat{\tau}_{K,t} + \hat{r}_{K,t} + \hat{k}_t] + \tau_L w \frac{L}{y} [\hat{\tau}_{L,t} + \hat{w}_t + \hat{L}_t] + \tau_C \frac{C}{y} (\hat{\tau}_{C,t} + \hat{c}_t) \\ &= \frac{1}{\beta} \frac{b}{y} [\hat{b}_{t-1} - \hat{\pi}_t - \hat{P}_{t-1}^m - \hat{u}_t^a] + \frac{b}{y} \frac{\rho}{y e^\pi} \hat{P}_t^m + \frac{g}{y} \hat{g}_t + \frac{z}{y} \hat{z}_t, \end{aligned}$$

where we define  $b_t = \frac{P_t^m B_t}{P_t A_t}$  so that  $s_{b,t} = \frac{P_t^m B_t}{P_t Y_t} = \frac{b_t}{y_t}$ . It follows that

$$(D57) \quad s_{b,t} = \hat{b}_t - \hat{y}_t.$$

Fiscal rules:

$$(D58) \quad \hat{g}_t = \rho_G \hat{g}_{t-1} - (1 - \rho_G) [\gamma_G (\hat{s}_{b,t-1} - \hat{s}_{b,t}^*) + \phi_{g,y} \hat{y}_t] + \hat{u}_t^g$$

$$(D59) \quad \hat{z}_t = \rho_Z \hat{z}_{t-1} - (1 - \rho_Z) [\gamma_Z (\hat{s}_{b,t-1} - \hat{s}_{b,t}^*) + \phi_{z,y} \hat{y}_t] + \hat{u}_t^z$$

$$(D60) \quad \hat{\tau}_{L,t} = \rho_L \hat{\tau}_{L,t-1} + (1 - \rho_L) \gamma_L [\hat{s}_{b,t-1} - \hat{s}_{b,t}^*] + \hat{u}_t^l$$

$$(D61) \quad \hat{\tau}_{K,t} = \rho_K \hat{\tau}_{K,t-1} + (1 - \rho_K) \gamma_K [\hat{s}_{b,t-1} - \hat{s}_{b,t}^*] + \hat{u}_t^k$$

where the time-varying debt target follows an AR(1) process

$$(D62) \quad s_{b,t}^* = \rho_s s_{b,t-1}^* + \epsilon_t^U$$

Monetary Rule:

$$(D63) \quad \hat{r}_{n,t} = \rho_r \hat{r}_{n,t-1} + (1 - \rho_r) [\phi_\pi (\hat{\pi}_t - \hat{\pi}_t^*) - \phi_y \hat{y}_t] + u_t^m$$

Time-varying inflation target:

$$(D64) \quad \hat{\pi}_t^* = \phi_F \hat{\pi}_{t-1}^* + \Phi \epsilon_t^U$$

#### THE JUSTIFICATION OF THE INFLATION TARGET

The short-term inflation target in the DSGE model follows an AR(1) process with unfunded shock. In this section we show that this specification of the inflation target is equivalent to the solution derived from analytically solving a shadow economy with a fiscal led regime.

##### *E1. The Analytical Solutions to The Shadow Economy*

Suppose that the shadow economy is a stylized 3-equation New Keynesian model augmented with a government budget constraint and a fiscal rule. The linearized equilibrium conditions are as follows.

$$(E1) \quad i_t = \phi_F \pi_t, \quad 0 < \phi_F < 1,$$

$$(E2) \quad \pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t,$$

$$(E3) \quad x_t = \mathbb{E}_t x_{t+1} - \sigma (i_t - \mathbb{E}_t \pi_{t+1}),$$

$$(E4) \quad \hat{z}_t = \gamma_z s b_{t-1} + \eta_t,$$

$$(E5) \quad s b_t = \beta^{-1} (s b_{t-1} + i_t - \pi_t + (1 - \beta) \lambda \hat{z}_t),$$

where  $z_t$  denotes the transfer,  $s_t$  denotes the primary surplus, and  $\lambda = \frac{Z_{ss}}{S_{ss}}$ , the ratio of steady-state transfer to steady-state primary surplus. Substituting (E4) into (E5), we have

$$(E6) \quad \begin{aligned} s b_t &= \beta^{-1} (s b_{t-1} + i_t - \pi_t + (1 - \beta) \lambda \hat{z}_t) \\ &= \beta^{-1} (s b_{t-1} + i_t - \pi_t + (1 - \beta) \lambda (\gamma_z s b_{t-1} + \eta_t)). \end{aligned}$$

Using the monetary rule  $i_t = \phi_F \pi_t$ :

$$(E7) \quad s b_t = \beta^{-1} \left( \underbrace{[1 + (1 - \beta) \lambda \gamma_z]}_{A_z} s b_{t-1} - \underbrace{(1 - \phi_F)}_{\text{since } \phi_F \pi_t - \pi_t} \pi_t + \underbrace{(1 - \beta) \lambda}_{C_z} \eta_t \right).$$

Define  $A_z \equiv 1 + (1 - \beta) \lambda \gamma_z$  and  $C_z \equiv (1 - \beta) \lambda$ . Multiplying by  $\beta$  on both sides of the linearized budget constraint yields:

$$(E8) \quad \beta s b_t = A_z s b_{t-1} - (1 - \phi_F) \pi_t + C_z \eta_t.$$



E2. *Guess-and-verify Solutions*

To obtain analytical solutions for the shadow economy, we use the guess-and-verify solution. Guess that the equilibrium variables are linear in lagged debt and the transfer shock:

$$(E9) \quad \pi_t = \pi_s sb_{t-1} + \pi_\eta \eta_t,$$

$$(E10) \quad x_t = x_s sb_{t-1} + x_\eta \eta_t,$$

$$(E11) \quad sb_t = s_s sb_{t-1} + s_\eta \eta_t.$$

Because  $\eta_t$  is i.i.d., we have  $\mathbb{E}_t sb_{t+1} = s_s sb_t$ ,  $\mathbb{E}_t \pi_{t+1} = \pi_s sb_t$ , and  $\mathbb{E}_t x_{t+1} = x_s sb_t$ .

Substitute (E9)–(E11) into (E8):

$$(E12) \quad \beta(s_s sb_{t-1} + s_\eta \eta_t) = A_z sb_{t-1} - (1 - \phi_F)(\pi_s sb_{t-1} + \pi_\eta \eta_t) + C_z \eta_t.$$

Matching coefficients gives

$$(E13) \quad \beta s_s = A_z - (1 - \phi_F)\pi_s,$$

$$(E14) \quad \beta s_\eta = -(1 - \phi_F)\pi_\eta + C_z.$$

Similarly, by substituting (E9) and (E10) into (E2), we have:

$$(E15) \quad \pi_s sb_{t-1} + \pi_\eta \eta_t = \beta \pi_s sb_t + \kappa(x_s sb_{t-1} + x_\eta \eta_t).$$

Using  $sb_t = s_s sb_{t-1} + s_\eta \eta_t$  and matching coefficients yield

$$(E16) \quad \pi_s = \beta \pi_s s_s + \kappa x_s,$$

$$(E17) \quad \pi_\eta = \beta \pi_s s_\eta + \kappa x_\eta.$$

Finally, combining (E3), and (E1) and substituting (E9) and (E11) yields:

$$(E18) \quad (1 - s_s)x_s = \sigma \pi_s (s_s - \phi_F),$$

$$(E19) \quad \phi_F \pi_\eta \sigma - \pi_s s_\eta \sigma - s_\eta x_s + x_\eta = 0.$$

From (E18), we have

$$(E20) \quad x_s = \frac{\sigma \pi_s (s_s - \phi_F)}{1 - s_s}, \quad s_s \neq 1.$$

E3. *Solving for The Debt Dynamics*

Since  $sb_{t-1}$  is a state variable, we need to solve the law of motion for  $sb_t$  before we proceed to the analytical solutions of  $\pi_t$ . Combine (E16) with (E20) and

assuming  $\pi_s \neq 0$  we have

$$(E21) \quad (1 - s_s)(1 - \beta s_s) = \kappa \sigma (s_s - \phi_F).$$

This yields the quadratic

$$(E22) \quad \beta s_s^2 - (1 + \beta + \kappa \sigma) s_s + (1 + \kappa \sigma \phi_F) = 0,$$

where the two roots are

$$(E23) \quad s_s^{(\pm)} = \frac{1 + \beta + \kappa \sigma \pm \sqrt{(1 + \beta + \kappa \sigma)^2 - 4\beta(1 + \kappa \sigma \phi_F)}}{2\beta}.$$

To avoid explosive debt dynamics requires the root to be stable and hence we have:

$$(E24) \quad s_s = \frac{1 + \beta + \kappa \sigma - \sqrt{(1 + \beta + \kappa \sigma)^2 - 4\beta(1 + \kappa \sigma \phi_F)}}{2\beta}.$$

With  $s_s$ , we can solve for  $\pi_s$  and  $x_s$  from (E13) and (E16), respectively:

$$(E25) \quad \pi_s = \frac{A_z - \beta s_s}{1 - \phi_F}$$

and from (E16) we can get,

$$(E26) \quad x_s = \frac{1 - \beta s_s}{\kappa} \pi_s.$$

Define  $r \equiv \frac{C_z}{A_z} = \frac{(1-\beta)\lambda}{1+(1-\beta)\lambda\gamma_z}$ , where  $A_z \neq 0$ . Guess that  $\pi_\eta = r\pi_s$  and  $s_\eta = rs_s$ , we solve  $\pi_\eta$  and  $s_\eta$  from (E13), which yields

$$(E27) \quad \pi_\eta = \frac{C_z}{A_z} \pi_s = \frac{(1 - \beta)\lambda}{1 + (1 - \beta)\lambda\gamma_z} \cdot \frac{A_z - \beta s_s}{1 - \phi_F},$$

and

$$(E28) \quad s_\eta = \frac{C_z}{A_z} s_s = \frac{(1 - \beta)\lambda}{1 + (1 - \beta)\lambda\gamma_z} s_s.$$

Then, we have

$$(E29) \quad sb_t = s_s sb_{t-1} + s_\eta \eta_t = s_s \frac{\pi_t - \pi_\eta \eta_t}{\pi_s} + s_\eta \eta_t.$$

The one-period-ahead inflation is

$$\begin{aligned}
 \pi_{t+1} &= \pi_s s b_t + \pi_\eta \eta_{t+1} \\
 (E30) \quad &= s_s \pi_t + (\pi_s s_\eta - s_s \pi_\eta) \eta_t + \pi_\eta \eta_{t+1}.
 \end{aligned}$$

Since  $\pi_s s_\eta - s_s \pi_\eta = 0$ , we have

$$(E31) \quad \pi_{t+1} = s_s \pi_t + \pi_\eta \eta_{t+1},$$

which is AR(1) consistent with specification of equation (31) in the quantitative model.

## STATE-SPACE REPRESENTATION FOR ESTIMATION

This section establishes the incomplete information model solution as a state-space representation used in the estimation. We follow the notation and algorithm of Blanchard, L'Huillier and Lorenzoni (2013) closely. The signal extraction problem is defined by a pair of equations

$$(F1) \quad \mathbf{x}_t = \mathbf{A}\mathbf{x}_{t-1} + \mathbf{B}\nu_t,$$

$$(F2) \quad \mathbf{s}_t = \mathbf{C}\mathbf{x}_t + \mathbf{D}\nu_t,$$

where  $\nu_t$  is an  $n_\nu$ -dimensional vector of mutually independent *i.i.d.* shocks. The dimensions of exogenous variables  $\mathbf{x}_t$  and signals  $\mathbf{s}_t$  are  $n_x$  and  $n_s$ . Let  $\mathbf{y}_t$  denote a vector of endogenous state variables of size  $n_y$ . The economic model can be described in terms of the stochastic difference equation

$$(F3) \quad \mathbf{F}\mathbb{E}_t[\mathbf{y}_{t+1}] + \mathbf{G}\mathbf{y}_t + \mathbf{H}\mathbf{y}_{t-1} + \mathbf{M}\mathbf{s}_t + \mathbf{N}\mathbb{E}_t[\mathbf{s}_{t+1}] = \mathbf{0},$$

where  $\mathbb{E}_t(\cdot)$  is the rational expectations operator under incomplete information and  $\mathbf{F}, \mathbf{G}, \mathbf{H}, \mathbf{M}, \mathbf{N}$  are matrices of parameters. The solution of the model can be described as

$$(F4) \quad \mathbf{y}_t = \mathbf{P}\mathbf{y}_{t-1} + \mathbf{Q}\mathbf{s}_t + \mathbf{R}\mathbf{x}_{t|t},$$

where the matrices  $\mathbf{P}, \mathbf{Q}, \mathbf{R}$  can be found by solving the three matrix equations

$$\begin{aligned} \mathbf{F}\mathbf{P}^2 + \mathbf{G}\mathbf{P} + \mathbf{H} &= \mathbf{0}, & (\mathbf{F}\mathbf{P} + \mathbf{G})\mathbf{Q} + \mathbf{M} &= \mathbf{0}, \\ (\mathbf{F}\mathbf{P} + \mathbf{G})\mathbf{R} + [\mathbf{F}(\mathbf{Q}\mathbf{C} + \mathbf{R}) + \mathbf{N}\mathbf{C}]\mathbf{A} &= \mathbf{0}. \end{aligned}$$

We know from the Kalman recursion that the law of motion of  $\mathbf{x}_{t|t}$  can be written as

$$\mathbf{x}_{t|t} = \mathbf{K}\mathbf{C}\mathbf{A}\mathbf{x}_{t-1} + (\mathbf{I} - \mathbf{K}\mathbf{C})\mathbf{A}\mathbf{x}_{t-1|t-1} + \mathbf{K}(\mathbf{C}\mathbf{B} + \mathbf{D})\nu_t,$$

where  $\mathbf{K}$  is the Kalman gain matrix of size  $n_x \times n_s$ . It follows that we can write the model solution  $\mathbf{y}_t$  as

$$\mathbf{y}_t = \mathbf{P}\mathbf{y}_{t-1} + (\mathbf{Q} + \mathbf{R}\mathbf{K})\mathbf{C}\mathbf{A}\mathbf{x}_{t-1} + \mathbf{R}(\mathbf{I} - \mathbf{K}\mathbf{C})\mathbf{A}\mathbf{x}_{t-1|t-1} + (\mathbf{Q} + \mathbf{R}\mathbf{K})(\mathbf{C}\mathbf{B} + \mathbf{D})\nu_t$$

Define the extended state variables as  $[\mathbf{y}_t, \mathbf{x}_t, \mathbf{x}_{t|t}]'$ . The state-space representation of the incomplete information model solution can be written as

$$\begin{bmatrix} \mathbf{y}_t \\ \mathbf{x}_t \\ \mathbf{x}_{t|t} \end{bmatrix} = \begin{bmatrix} \mathbf{P} & (\mathbf{Q} + \mathbf{R}\mathbf{K})\mathbf{C}\mathbf{A} & \mathbf{R}(\mathbf{I} - \mathbf{K}\mathbf{C})\mathbf{A} \\ \mathbf{0} & \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}\mathbf{C}\mathbf{A} & (\mathbf{I} - \mathbf{K}\mathbf{C})\mathbf{A} \end{bmatrix} \begin{bmatrix} \mathbf{y}_{t-1} \\ \mathbf{x}_{t-1} \\ \mathbf{x}_{t-1|t-1} \end{bmatrix} + \begin{bmatrix} (\mathbf{Q} + \mathbf{R}\mathbf{K})(\mathbf{C}\mathbf{B} + \mathbf{D}) \\ \mathbf{B} \\ \mathbf{K}(\mathbf{C}\mathbf{B} + \mathbf{D}) \end{bmatrix} \nu_t.$$

## RESULTS FROM ESTIMATION

| Results from First Sample               |        |        |       |       |                                  |       |        |       |       |
|-----------------------------------------|--------|--------|-------|-------|----------------------------------|-------|--------|-------|-------|
| Parameter                               | Mode   | Median | 5%    | 95%   | Parameter                        | Mode  | Median | 5%    | 95%   |
| Debt to GDP $s_b$                       | 2.391  | 2.405  | 2.331 | 2.486 | AR(1) MP-rule $\rho_r$           | 0.701 | 0.711  | 0.681 | 0.737 |
| SS growth $100\gamma$                   | 0.437  | 0.421  | 0.402 | 0.465 | AR(1) G-rule $\rho_g$            | 0.970 | 0.968  | 0.960 | 0.975 |
| SS inflation $100\Pi$                   | 0.525  | 0.484  | 0.426 | 0.569 | AR(1) Z-rule $\rho_z$            | 0.959 | 0.961  | 0.950 | 0.978 |
| Inverse Frisch $\xi$                    | 2.295  | 2.253  | 2.239 | 2.309 | AR(1) TK-rule $\rho_{TK}$        | 0.928 | 0.932  | 0.913 | 0.947 |
| Non-savers $\mu$                        | 0.090  | 0.094  | 0.081 | 0.111 | AR(1) TL-rule $\rho_{TL}$        | 0.934 | 0.941  | 0.924 | 0.955 |
| Wage Calvo $\omega_w$                   | 0.710  | 0.710  | 0.692 | 0.727 | AR(1) Infl. target $\phi_F$      | 0.993 | 0.989  | 0.983 | 0.993 |
| Price Calvo $\omega_p$                  | 0.706  | 0.711  | 0.689 | 0.730 | Technology $\rho_a$              | 0.135 | 0.126  | 0.111 | 0.156 |
| Cap util cost $\psi$                    | 0.736  | 0.673  | 0.647 | 0.694 | Preference $\rho_b$              | 0.684 | 0.727  | 0.692 | 0.747 |
| Invest adj cost                         | 5.185  | 5.067  | 5.045 | 5.160 | Monetary $\rho_m$                | 0.357 | 0.345  | 0.318 | 0.361 |
| Wage index $\chi_w$                     | 0.064  | 0.059  | 0.053 | 0.064 | Investment $\rho_i$              | 0.881 | 0.870  | 0.855 | 0.885 |
| Price index $\chi_p$                    | 0.060  | 0.066  | 0.062 | 0.070 | Risk Prem. $\rho_{rp}$           | 0.820 | 0.811  | 0.791 | 0.833 |
| Habits $\theta$                         | 0.915  | 0.911  | 0.900 | 0.921 | Cost-push $\rho_{cp}$            | 0.998 | 0.997  | 0.995 | 0.999 |
| C/G sub $\alpha_G$                      | -0.014 | 0.032  | 0.004 | 0.077 | Gov spending $\sigma_g$          | 1.695 | 1.787  | 1.716 | 1.888 |
| Taylor rule coeff $\phi_\pi$            | 1.701  | 1.743  | 1.690 | 1.822 | Technology $\sigma_a$            | 1.055 | 1.175  | 1.117 | 1.222 |
| TR coeff on Y $\phi_y$                  | 0.001  | 0.001  | 0.000 | 0.006 | Preference $\sigma_b$            | 4.940 | 4.967  | 4.904 | 4.996 |
| Debt response $\gamma_G$                | 0.274  | 0.252  | 0.240 | 0.265 | Monetary $\sigma_m$              | 0.276 | 0.263  | 0.247 | 0.285 |
| Debt response $\gamma_{TK}$             | 0.011  | 0.012  | 0.011 | 0.013 | Investment $\sigma_i$            | 0.601 | 0.629  | 0.574 | 0.699 |
| Debt response $\gamma_{TL}$             | 0.004  | 0.003  | 0.001 | 0.005 | Wage $\sigma_w$                  | 0.338 | 0.344  | 0.308 | 0.371 |
| Debt response $\gamma_z$                | 0.013  | 0.014  | 0.003 | 0.026 | Price markup $\sigma_p$          | 0.170 | 0.157  | 0.136 | 0.177 |
| Output response $\phi_{z,y}$            | 0.183  | 0.189  | 0.181 | 0.198 | Risk prem. $\sigma_{rp}$         | 0.504 | 0.498  | 0.473 | 0.520 |
| Output response $\phi_{g,y}$            | 0.190  | 0.206  | 0.199 | 0.210 | Cost-push $\sigma_{cp}$          | 0.150 | 0.140  | 0.132 | 0.160 |
| Infl. target strength $\Phi$            | 0.087  | 0.090  | 0.078 | 0.106 | Transfer $\sigma_z$              | 3.079 | 3.057  | 3.023 | 3.085 |
|                                         |        |        |       |       | Unfunded $\sigma_U$              | 0.378 | 0.402  | 0.375 | 0.436 |
|                                         |        |        |       |       | Capital tax $\sigma_{tk}$        | 4.980 | 4.978  | 4.943 | 4.997 |
|                                         |        |        |       |       | Labor tax $\sigma_{tl}$          | 2.669 | 2.628  | 2.514 | 2.674 |
|                                         |        |        |       |       | GDP M.E. $\sigma_y^{ME}$         | 0.434 | 0.433  | 0.398 | 0.458 |
|                                         |        |        |       |       | Debt to GDP M.E. $\sigma_b^{ME}$ | 0.339 | 0.385  | 0.353 | 0.407 |
| Results from Second Sample              |        |        |       |       |                                  |       |        |       |       |
| Parameter                               | Mode   | Median | 5%    | 95%   |                                  |       |        |       |       |
| Gov spending $\sigma_g$                 | 2.359  | 2.360  | 2.356 | 2.363 |                                  |       |        |       |       |
| Technology $\sigma_a$                   | 3.291  | 3.293  | 3.290 | 3.294 |                                  |       |        |       |       |
| Preference $\sigma_b$                   | 4.996  | 4.994  | 4.992 | 4.995 |                                  |       |        |       |       |
| Monetary $\sigma_m$                     | 0.156  | 0.156  | 0.150 | 0.164 |                                  |       |        |       |       |
| Investment $\sigma_i$                   | 2.463  | 2.460  | 2.456 | 2.463 |                                  |       |        |       |       |
| Wage $\sigma_w$                         | 0.493  | 0.493  | 0.489 | 0.494 |                                  |       |        |       |       |
| Price markup $\sigma_p$                 | 0.111  | 0.110  | 0.108 | 0.111 |                                  |       |        |       |       |
| Risk prem. $\sigma_{rp}$                | 4.881  | 4.888  | 4.886 | 4.889 |                                  |       |        |       |       |
| Cost-push $\sigma_{cp}$                 | 0.558  | 0.544  | 0.542 | 0.547 |                                  |       |        |       |       |
| Transfer $\sigma_z$                     | 4.995  | 4.995  | 4.994 | 4.997 |                                  |       |        |       |       |
| Unfunded $\sigma_U$                     | 2.937  | 2.936  | 2.933 | 2.940 |                                  |       |        |       |       |
| Capital tax $\sigma_{tk}$               | 4.981  | 4.982  | 4.981 | 4.983 |                                  |       |        |       |       |
| Labor tax $\sigma_{tl}$                 | 2.664  | 2.651  | 2.649 | 2.654 |                                  |       |        |       |       |
| GDP M.E. $\sigma_y^{ME}$                | 1.745  | 1.744  | 1.741 | 1.751 |                                  |       |        |       |       |
| Debt to GDP M.E. $\sigma_b^{ME}$        | 4.980  | 4.982  | 4.981 | 4.983 |                                  |       |        |       |       |
| Break-even inflation $\sigma_\pi^{5yr}$ | 0.269  | 0.271  | 0.268 | 0.273 |                                  |       |        |       |       |

TABLE G1—PARAMETER ESTIMATES FROM FIRST SAMPLE IIRE ESTIMATION WITHOUT EXPECTATIONS

**Notes:** Table reports the mode, median, and confidence intervals from first and second sample estimation under incomplete information without inflation expectation as an observable. Prior Type indicates the prior density function where N, G, B, and I stand for Normal, Gamma, Beta, and Inverse-Gamma respectively. The posterior median and confidence intervals are computed using MCMC with 100,000 simulations and 50% burn-in.

| Parameter                              | IIRE (SPF) |        |       |       | IIRE (MSC) |        |       |       |
|----------------------------------------|------------|--------|-------|-------|------------|--------|-------|-------|
|                                        | Mode       | Median | 5%    | 95%   | Mode       | Median | 5%    | 95%   |
| Gov spending $\sigma_g$                | 2.422      | 2.405  | 2.401 | 2.412 | 2.388      | 2.368  | 2.349 | 2.398 |
| Technology $\sigma_a$                  | 3.373      | 3.380  | 3.374 | 3.385 | 3.359      | 3.327  | 3.275 | 3.394 |
| Preference $\sigma_b$                  | 4.993      | 4.992  | 4.991 | 4.996 | 4.997      | 4.995  | 4.989 | 5.000 |
| Monetary $\sigma_m$                    | 0.188      | 0.189  | 0.185 | 0.192 | 0.183      | 0.208  | 0.196 | 0.224 |
| Investment $\sigma_i$                  | 2.496      | 2.572  | 2.536 | 2.580 | 2.601      | 2.613  | 2.598 | 2.647 |
| Wage $\sigma_w$                        | 0.566      | 0.549  | 0.542 | 0.563 | 0.574      | 0.512  | 0.497 | 0.531 |
| Price markup $\sigma_p$                | 0.081      | 0.084  | 0.082 | 0.085 | 0.073      | 0.068  | 0.058 | 0.087 |
| Risk prem. $\sigma_{rp}$               | 4.407      | 4.424  | 4.416 | 4.432 | 4.906      | 4.934  | 4.890 | 4.976 |
| Cost-push $\sigma_{cp}$                | 0.442      | 0.427  | 0.421 | 0.431 | 0.503      | 0.461  | 0.442 | 0.480 |
| Transfer $\sigma_z$                    | 4.996      | 4.998  | 4.995 | 5.000 | 4.995      | 4.998  | 4.995 | 5.000 |
| Unfunded $\sigma_U$                    | 0.205      | 0.200  | 0.197 | 0.202 | 0.273      | 0.287  | 0.279 | 0.293 |
| Capital tax $\sigma_{tk}$              | 4.986      | 4.987  | 4.985 | 4.989 | 4.995      | 4.991  | 4.989 | 4.994 |
| Labor tax $\sigma_{tl}$                | 2.670      | 2.687  | 2.680 | 2.690 | 2.665      | 2.680  | 2.650 | 2.733 |
| GDP M.E. $\sigma_y^{ME}$               | 1.715      | 1.721  | 1.710 | 1.735 | 1.746      | 1.628  | 1.608 | 1.661 |
| Debt to GDP M.E. $\sigma_b^{ME}$       | 4.989      | 4.987  | 4.984 | 4.988 | 4.995      | 4.995  | 4.978 | 5.000 |
| Inf expectation M.E. $\pi_{ME}^e$      | 0.248      | 0.258  | 0.255 | 0.266 | 0.442      | 0.410  | 0.391 | 0.433 |
| Breakeven inflation $\sigma_\pi^{5yr}$ | 0.234      | 0.252  | 0.247 | 0.262 | 0.229      | 0.245  | 0.237 | 0.274 |

TABLE G2—PARAMETER ESTIMATES FROM SECOND SAMPLE FOR IIRE MODEL WITH SHORT-RUN SPF AND MSC INFLATION EXPECTATIONS (2008:Q1-2022:Q3)

**Notes:** Table reports the mode, median, and confidence intervals from second sample estimation under incomplete information with short-run inflation expectations data from SPF and MSC as observables. Prior Type indicate the prior density function where N, G, B, and I stand for Normal, Gamma, Beta, and Inverse-Gamma respectively. The posterior median and confidence intervals are computed using MCMC with 100,000 simulations and 50% burn-in.

## G1. Impulse Response for all other shocks

This section presents the impulse response for the remaining shocks in the model. Each figure presents the IRFs for six variables- output, inflation, nominal interest rate, real interest rate, expected inflation, and real marginal cost, under full information and incomplete information. Figure G1 shows the impulse response for funded fiscal shock. A decrease in fiscal spending decreases production and causes deflation. The presence of hand-to-mouth agents in our model allows us to have small inflation effects from funded fiscal spending under both FIRE and incomplete information (IIRE).

We also present results for tax shocks in our fiscal block, i.e. an expansionary labor-tax shock and capital-tax shock. A one-standard-deviation negative (expansionary) labor tax shock is deflationary and reduces expected inflation under both FIRE and incomplete information (Figure G2). Cut in capital tax rate expands output and increases inflation, both of which effects are more prominent under IIRE. Output increases due to the fall in real marginal cost, which expands investment.

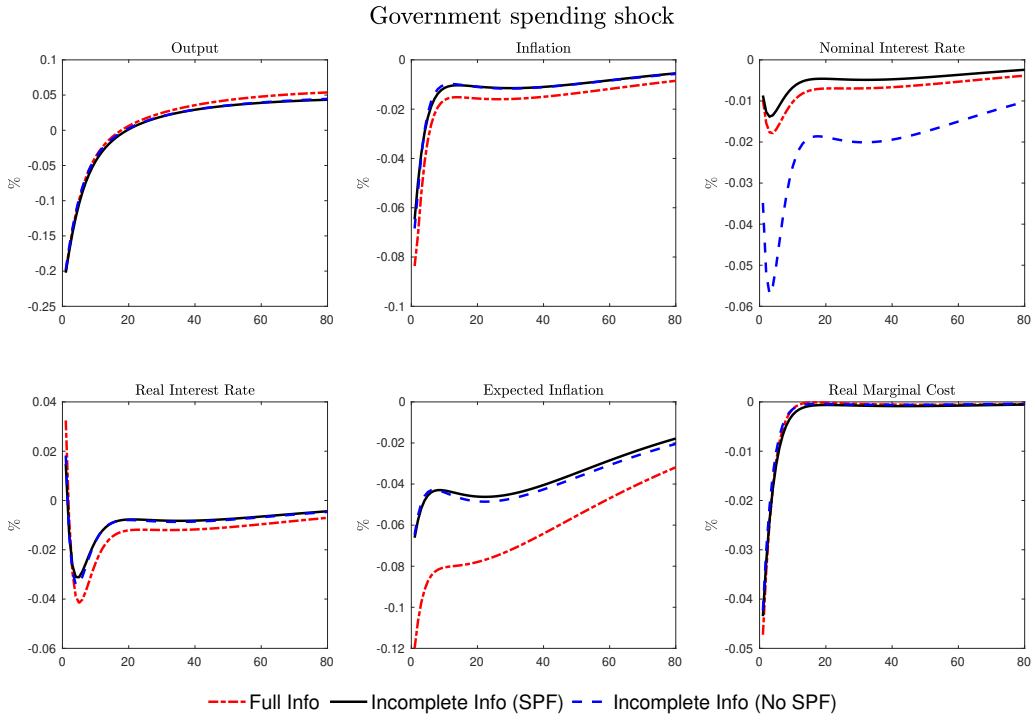


FIGURE G1. IMPULSE RESPONSES TO A ONE STANDARD DEVIATION GOVERNMENT SPENDING SHOCK

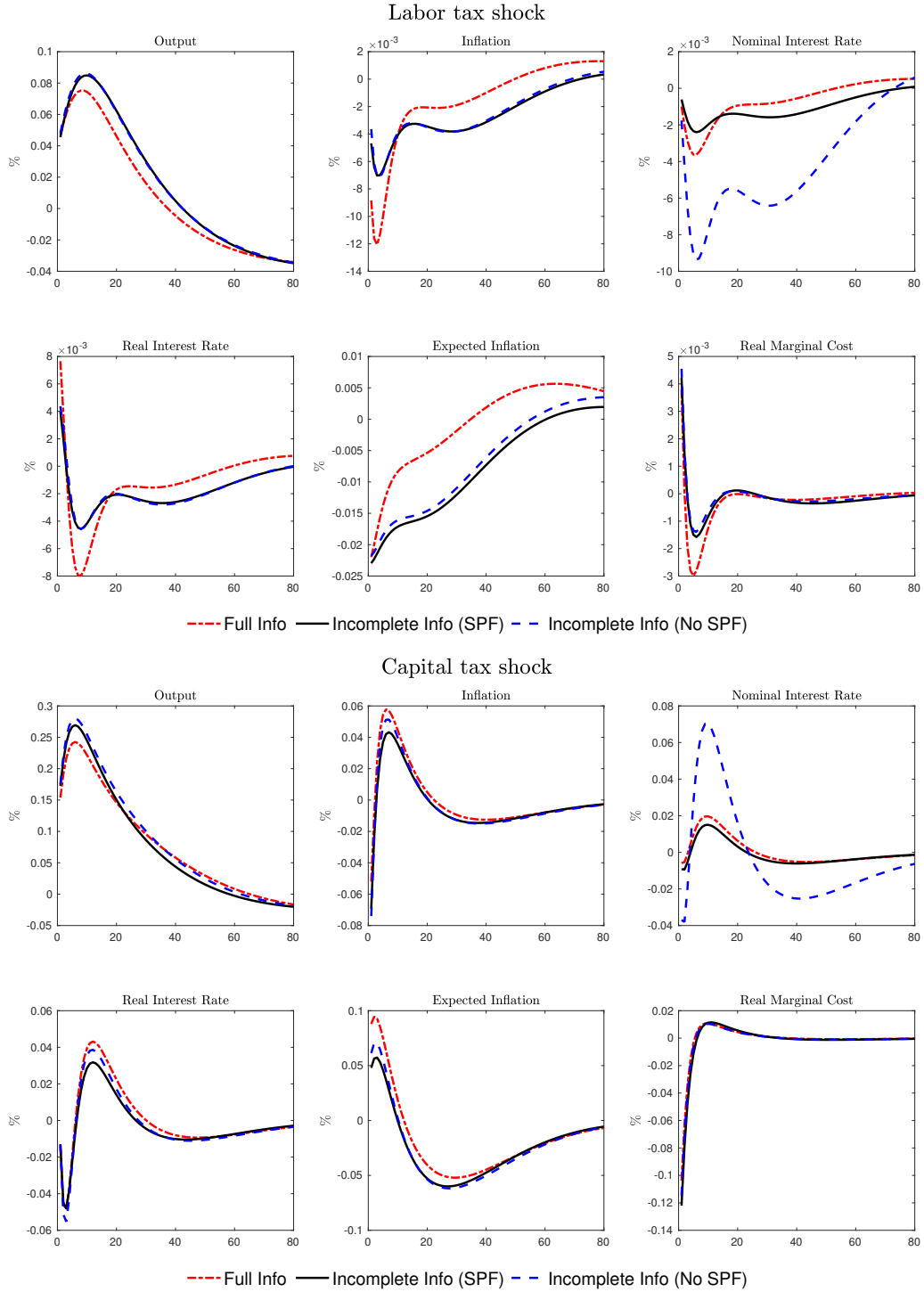


FIGURE G2. IMPULSE RESPONSES TO A ONE STANDARD DEVIATION LABOR AND CAPITAL TAX SHOCK



We do not introduce incomplete information with respect to any other shocks in our model. The response of macro variables to a one standard deviation risk premium shock (G3), technology shock, investment-specific technology shock (Figure G4), preference shock, wage markup shock (Figure G5), cost push shock, and price markup shock (G6) are largely similar under full information and incomplete information models.

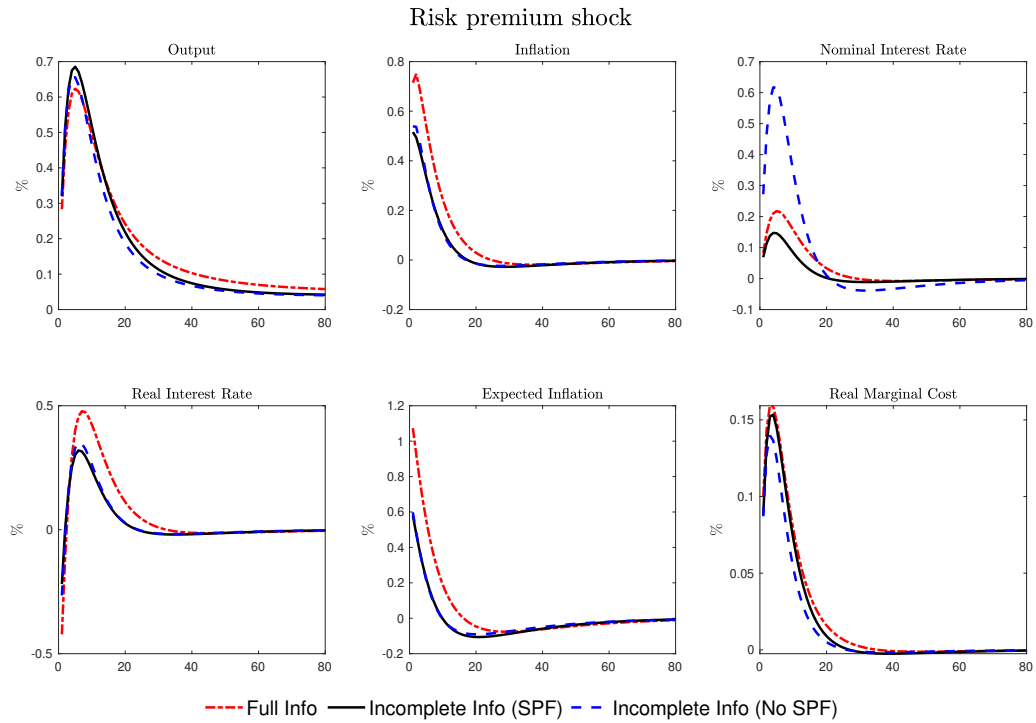


FIGURE G3. IMPULSE RESPONSES TO A ONE STANDARD DEVIATION RISK PREMIUM AND WAGE MARKUP SHOCK

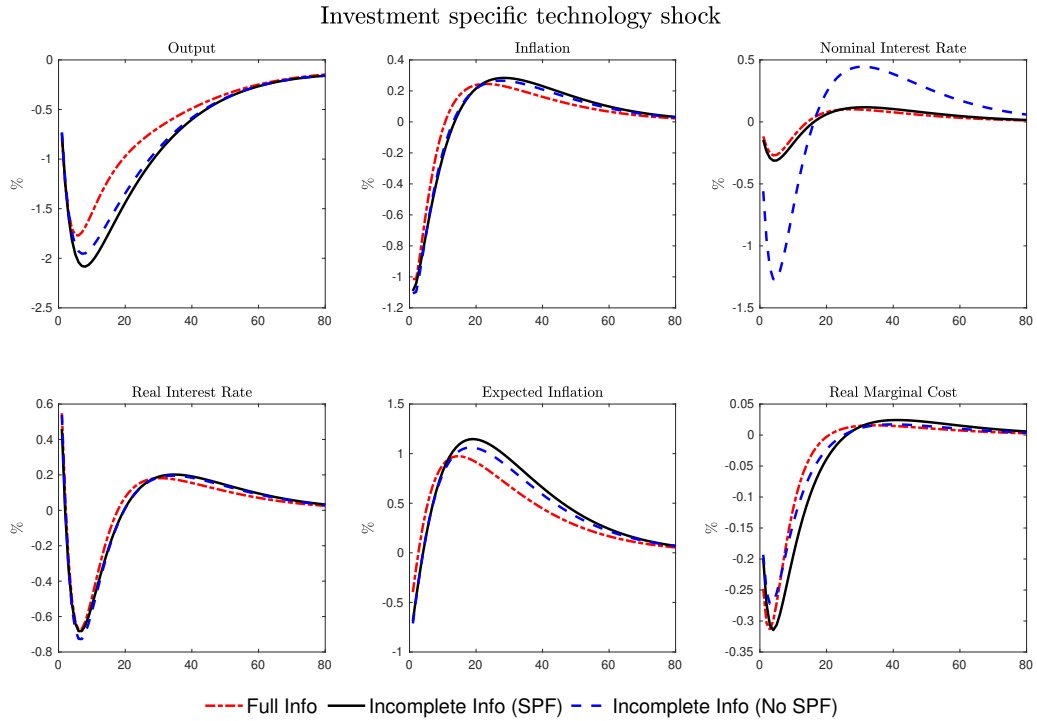
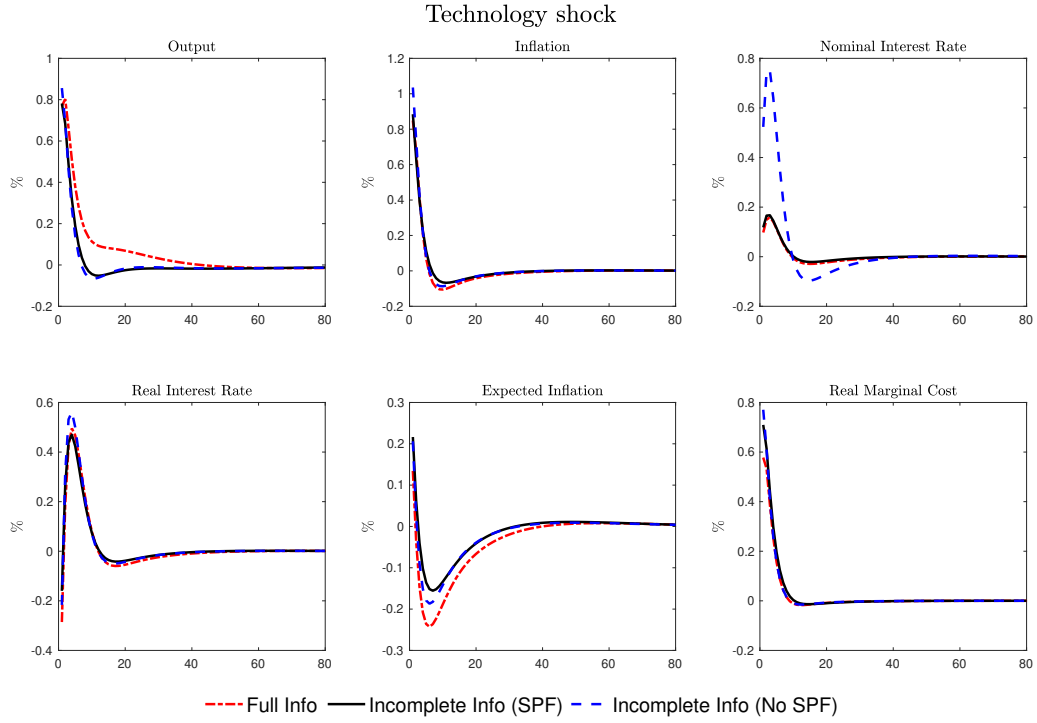


FIGURE G4. IMPULSE RESPONSES TO A ONE STANDARD DEVIATION TECHNOLOGY AND INVESTMENT SPECIFIC TECHNOLOGY SHOCK

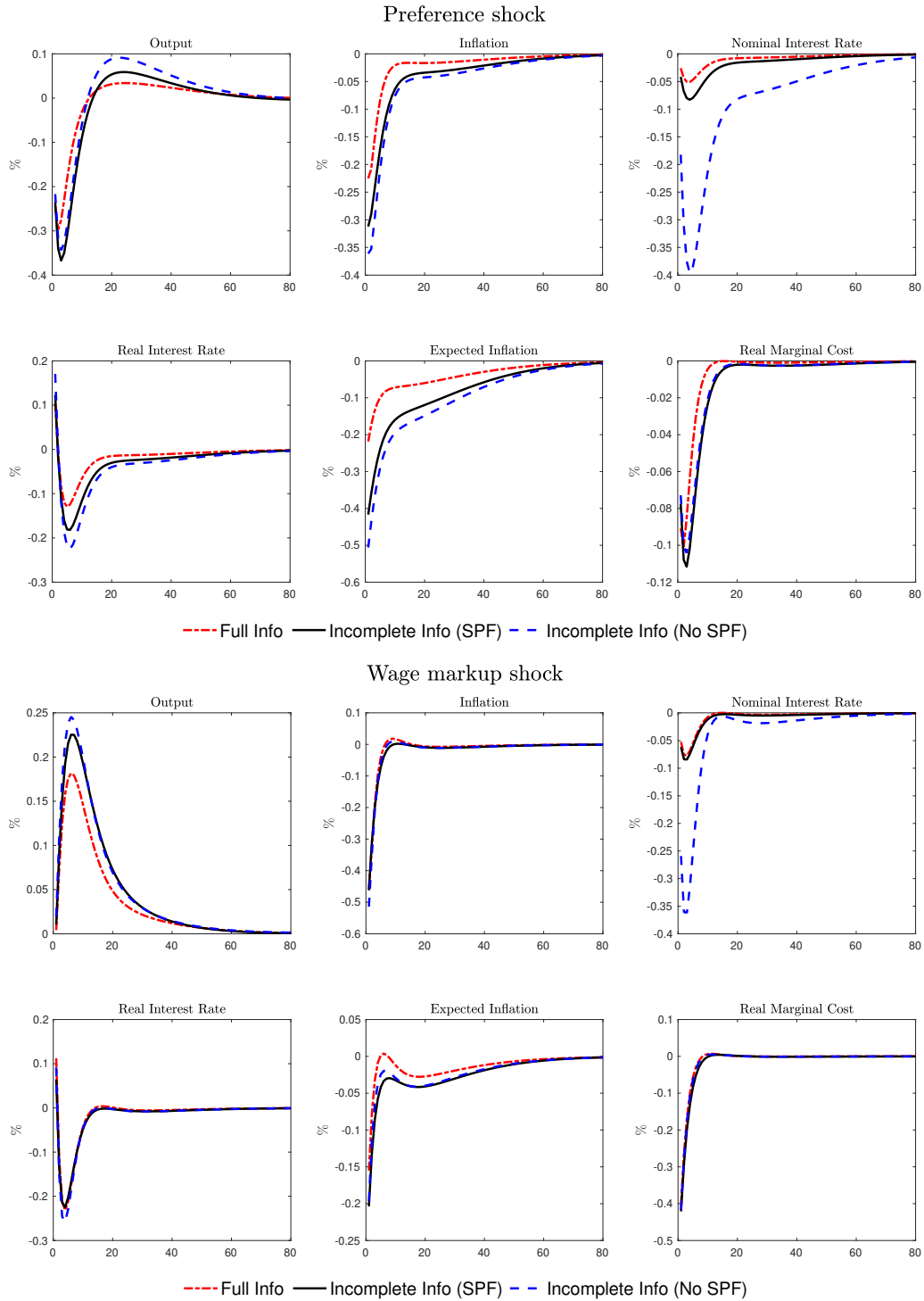


FIGURE G5. IMPULSE RESPONSES TO A ONE STANDARD DEVIATION PREFERENCE AND WAGE MARKUP SHOCK

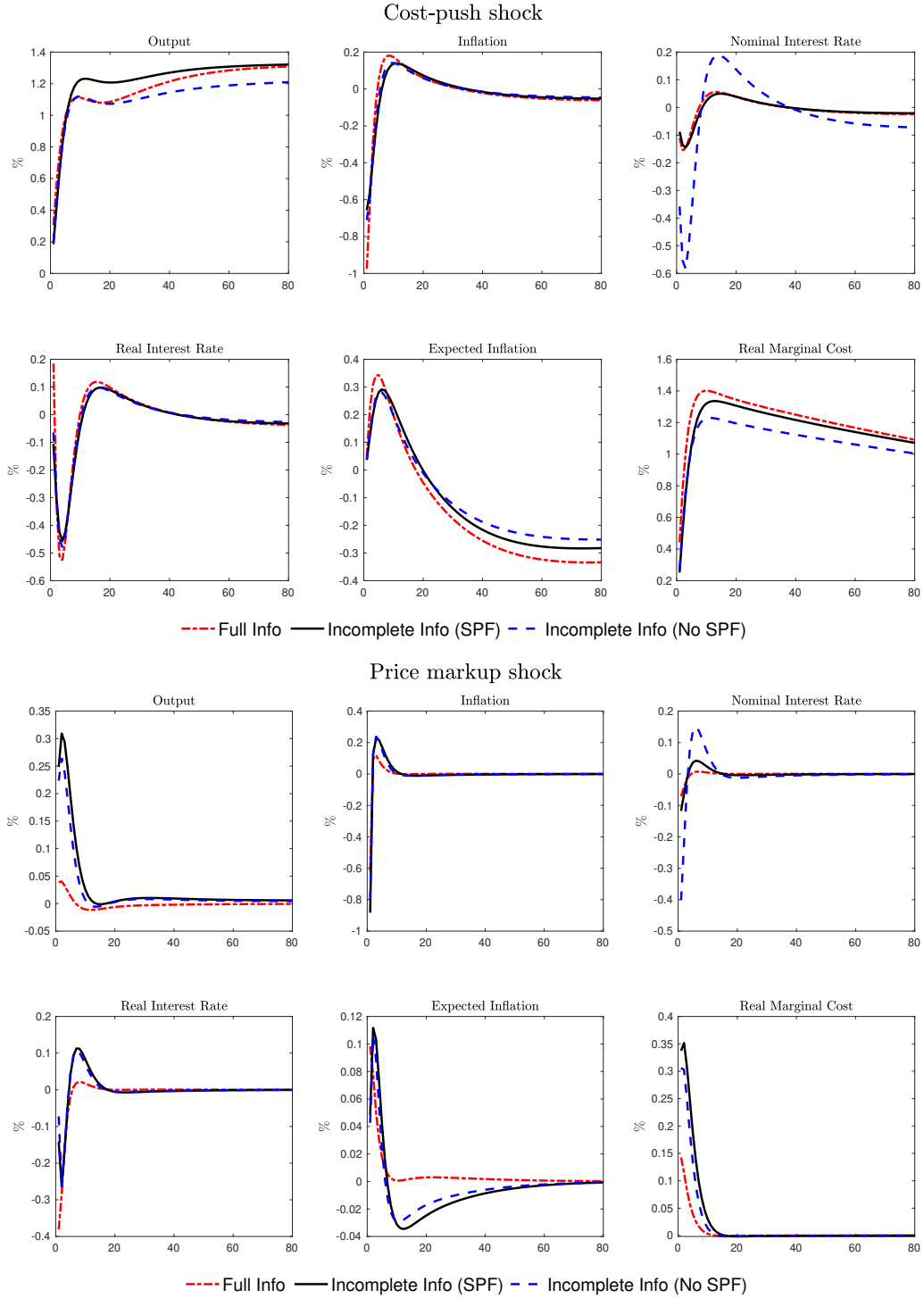


FIGURE G6. IMPULSE RESPONSES TO A ONE STANDARD DEVIATION COST PUSH AND PRICE MARKUP SHOCK

## DATA CONSTRUCTION

**GDP, Consumption, and Investment:** All national accounting variables are obtained from National Income and Product Account (NIPA) tables and converted to real values by dividing with the respective price indices with base year 2012:Q3. Consumption is the sum of personal expenditure in non-durable goods and services. Investment is the sub of gross private domestic investment and personal consumption expenditure in durable goods. We scale the real variables with population and convert into growth rates using the formula in Section H.H1.

**Hours worked and the average weekly earning:** We follow Bianchi, Facchini and Melosi (2023) in constructing the hours gap. We use average weekly hours of all non-farm workers, convert it to per capita using the population index (base year 2012:Q3) and take a difference of hours per capita from its trend which is computed as a fourth degree polynomial. Average weekly wages are constructed by dividing wage and salary accrual with average hours of non-farm workers. Wages are then converted to index with base year 2012:Q3, divided by GDP deflator, and converted to growth rates.

**Interest rate and inflation:** Interest rates are measured as the effective federal funds rate, inflation is computed as growth rate of GDP deflator index (BY 2012:Q3). Short-term inflation expectations are measured as one-quarter ahead inflation expectations from the Survey of Professional Forecasters, and one-year ahead inflation expectations from Michigan Survey of Consumer Sentiments, while long-term inflation expectations are measured using the 5-year breakeven inflation rate. In our second sample estimation, we also incorporate interest rate expectations measured by overnight index swaps for one to ten quarters ahead federal funds rate.

**Government expenditure and transfers:** We construct the government expenditure variable as a sum of gross government consumption expenditure and gross government investment expenditure. The nominal variables are converted to real value by dividing with the government expenditure implicit price deflators. To construct an index of government transfers we add together government social benefits, current transfer payments (incl. grants-in-aid to state and local government) and divide by GDP deflator. All real quantities are converted to growth rate after being normalized by the population index.

**Revenue from labor and capital tax:** Data to calculate revenue from labor and capital tax are obtained from NIPA tables. Following Leeper, Plante and Traum (2010) we first calculate the personal tax rate as:

$$\tau_p = \frac{IT}{WS + PRI/2 + CI}$$

where  $IT$  is the personal current tax revenues (Table 3.2, line 3),  $WS$  is the wage and salary accrual (Table 1.12, line 3),  $PRI$  is the proprietors' income (Table 1.12, line 9), and  $CI$  is the capital income defined as the sum of rental income (Table 1.12, line 12), corporate profits (Table 1.12, line 13), interest income (Table 1.12, line 18), and  $PRI/2$ . Using the personal income tax rate, the labor and capital tax revenue is calculated as:

$$T_L = \tau^p(WS + PRI/2) + CSI$$

$$T_K = \tau^p CI + CT$$

where  $CSI$  are contributions to government social insurance (NIPA Table 3.2, line 10) and  $CT$  are taxes on corporate income (Table 3.2, line 8).

#### *H1. Observables*

All nominal series are converted to real values using GDP deflator indexed to 100 in 2012:Q3 and normalized using civilian non-institutional population (obtained from BLS) indexed to 100 in 2012:Q1 such that:

$$x = \ln \left( \frac{x}{\text{Popindex}} \right) \times 100$$

where  $x$  is the GDP, consumption, investment, average weekly earnings, government transfers, government consumption and investment expenditure, labor tax revenue, and capital tax revenue. All above series are then converted to growth rates.

The measurement equations are then defined as

$$X_t = C + HY_t + K\Xi_t$$

where  $\Xi$  is the vector of measurement errors. The observables are related to the

model variables through the following equations:

$$\begin{bmatrix}
 \text{dlOutput}_t \\
 \text{dlConst}_t \\
 \text{dlInv}_t \\
 \text{dlWage}_t \\
 \text{dlGovSpend}_t \\
 \text{dlTransfers}_t \\
 \text{lGovDebt}_t \\
 \text{lHours}_t \\
 \text{lInflation}_t \\
 \text{lFedFunds}_t \\
 \text{dlLaborTaxRevenue}_t \\
 \text{dlCapitalTaxRevenue}_t
 \end{bmatrix}
 =
 \begin{bmatrix}
 100e^\gamma \\
 100e^\gamma \\
 100e^\gamma \\
 100e^\gamma \\
 100e^\gamma \\
 100e^\gamma \\
 \log(s_b) \times 100 \\
 0 \\
 \pi^* \\
 \log(R_s) \times 100 \\
 100e^\gamma \\
 100e^\gamma
 \end{bmatrix}
 +
 \begin{bmatrix}
 \hat{y}_t - \hat{y}_{t-1} + \hat{u}_t^a \\
 \hat{c}_t - \hat{c}_{t-1} + \hat{u}_t^a \\
 \hat{i}_t - \hat{i}_{t-1} + \hat{u}_t^a \\
 \hat{w}_t - \hat{w}_{t-1} + \hat{u}_t^a \\
 \hat{g}_t - \hat{g}_{t-1} + \hat{u}_t^a \\
 \hat{z}_t - \hat{z}_{t-1} + \hat{u}_t^a \\
 \hat{b}_t \\
 \hat{L}_t \\
 \hat{\pi}_t \\
 \hat{R}_t \\
 \hat{T}_t^L - \hat{T}_{t-1}^L + \hat{u}_t^a \\
 \hat{T}_t^K - \hat{T}_{t-1}^K + \hat{u}_t^a
 \end{bmatrix}$$

Both short-term and long-term break-even inflation expectations and interest rate expectations are incorporated using the following measurement equation:

$$\pi_t^f = \bar{\pi} + E_t^{HH} \pi_{t+1} + \zeta_t^\pi$$

where  $\zeta_t$  is an i.i.d. measurement error between the observed inflation/interest rate forecast and model-generated forecast.